UNIT- I

BASIC CIRCUITS

Voltage: Voltage is nothing but potential difference between two points also called as Electro motive force (EMF).

Potential difference :- In Electrical terminology called as Voltage and is donated with "p". It is Expressed in terms of energy (W). per unit charge (Q)

$$\boxed{V = \frac{W}{Q}}$$

W (Energy) is Expressed in Joules

(charge) Q is Expressed in Culombs

(Voltage) V is Expressed in Volts.

Current :

Movement of free Electrons in any Semiconductive are Conductive materials is called Current.

Current is defined as rate of flow of Electrons in a Conductive or semiconductive materials.

It is measured by the no of electrons that flows to a Point in unit time.

Mathematically it is Expressed as $\boxed{I = \frac{Q}{t}}$

i.e, I is Current in Ampheres

c is charge in Cullombs

t is time in Seconds.

Power and Energy :

Energy is the Capacity of doing Work i.e. Energy is nothing but stored Work.

Power is also Expressed as $\boxed{P = V \times I}$

Energy may Exist in many forms Such as Mechanical, chemical, Electrical etc...

Power is the rate of change of Energy and it is denoted with "p".

If certain amount of Energy is used over a certain length of time. than

Power $\boxed{P = \frac{\text{Energy}}{\text{time}}}$ $\boxed{P = \frac{W}{t}}$

Where Energy is in Joules

time is in Seconds

Power is in Watts

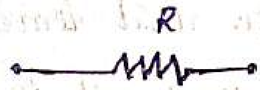
<u>circuit</u> :

A closed circuit is defined as a circuit in which the current has a complete path to flow.

<u>Basic circuit Elements</u>

<u>Resistance</u> : It opposes the flow of Electrons of current.

It is denoted with "R".

Simbol of the resistance is 

The unit of resistance is ohm (Ω)

ohm is defined as the resistance offered by the material when a Current of one Amphere flows between two terminals with one volt Applied across it.

$$\boxed{R = \frac{V}{I}}$$

according to ohms law the Current is directly proportional to voltange and inversely proportional to total resistance of the circuit.

Power observed by the resistance is

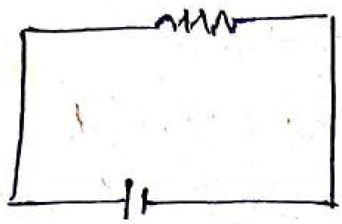
$$P = V \cdot I \qquad\qquad P = V\left(\frac{V}{R}\right)$$

$$P = (IR)\, I \qquad\qquad \boxed{P = \frac{V^2}{R}}$$

$$\boxed{P = I^2 R}$$

1. Problem

In the following circuit find out the Power



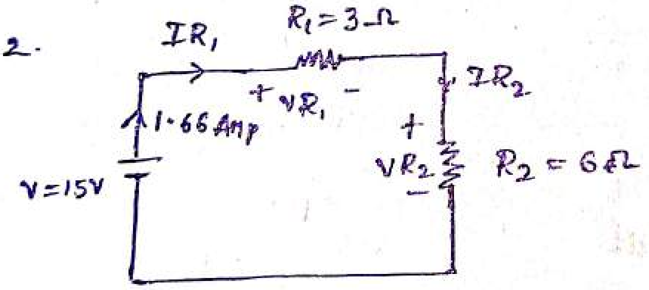
$$I_R = \frac{V}{R} = \frac{12}{3} = 4 \text{ Amps.}$$

$\therefore$ Power ; $P = V \cdot I = (12)\,4 = 48$ Watts.

$$P = I^2 R = (4)^2\, 3 = 48 \text{ Watts.}$$

$$P = \frac{V^2}{R} = \frac{(12)^2}{3} = 48 \text{ Watts.}$$

2.



In the circuit Calculate IR_1, IR_2, VR_1, VR_2, PR_1, PR_2

Equivalent resistance

$$R_{eq} = R_1 + R_2$$

$$R_{eq} = 3 + 6 = 9\,\Omega \quad \boxed{R_{eq} = 9\,\Omega}$$

$$I_T = \frac{V}{R_{eq}} = \frac{15}{9} = 1{\cdot}66 \text{ Amp.}$$

Energy may Exist in many forms such as Mechanical, chemical, electrical etc...

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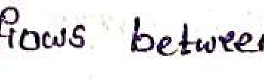
Simbol of the resistance is 

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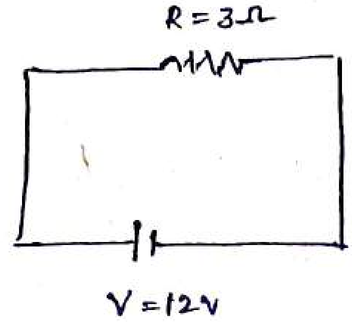

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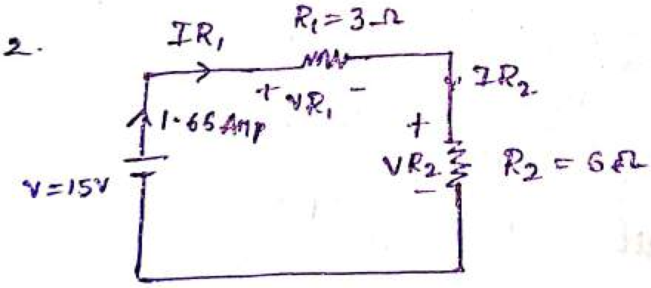


In the circuit Calculate I_{R_1}, I_{R_2}, V_{R_1}, V_{R_2}, P_{R_1}, P_{R_2}

Equivalent resistance $\quad R_{eq} = R_1 + R_2$

$R_{eq} = 3 + 6 = 9\,\Omega \quad \boxed{R_{eq} = 9\,\Omega}$

$$I_T = \frac{V}{R_{eq}} = \frac{15}{9} = 1.66 \text{ Amp.}$$

Power at R_1,

$$PR_1 = VR_1 \, IR_1 = (7.5)(2.5)$$

$$\boxed{PR_1 = 18.75 \text{ Watts}}$$

Power at R_2,

$$PR_2 = VR_2 \, IR_2 = (7.5)(1.25)$$

$$\boxed{PR_2 = 9.375 \text{ Watts}}$$

Power at R_3,

$$PR_3 = VR_3 \, IR_3$$

$$= (7.5)(1.25)$$

$$\boxed{PR_3 = 9.375 \text{ watts}}$$

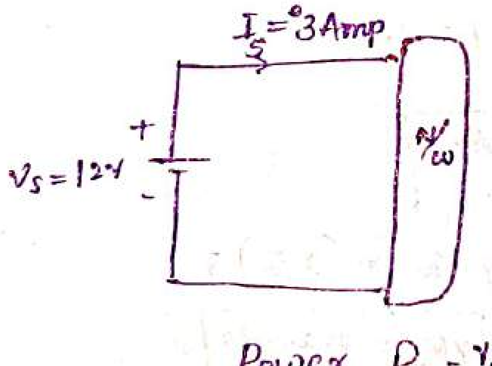
Total power in the circuit,

$$P_T = V \, I_T$$

$$P_T = (15)(2.5)$$

$$\boxed{P_T = 37.5 \text{ Watts}}$$

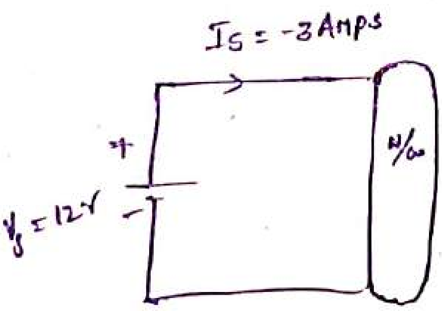
<u>Examples for power</u>



Power, $P_S = V_S I_S$

$P_S = (12)3$

$P_S = 36$ Watts. (Delivering Power)

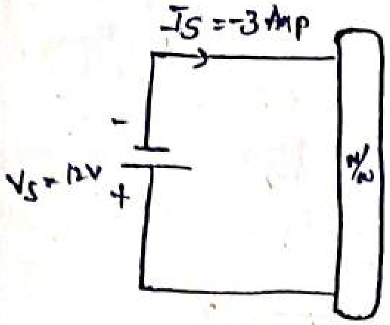


Power, $P_S = V_S I_S$

$P_S = (12)(-3)$

$P_S = -36$ Watts

(obsorbing power)



Power, $P_S = V_S I_S$

$P_S = (-12)(-3)$

$P_S = 36$ Watts (Delivering Power)

Energy

Capacity of doing Work is Called Energy.
it is represented with "E".

$$E = P \times t$$

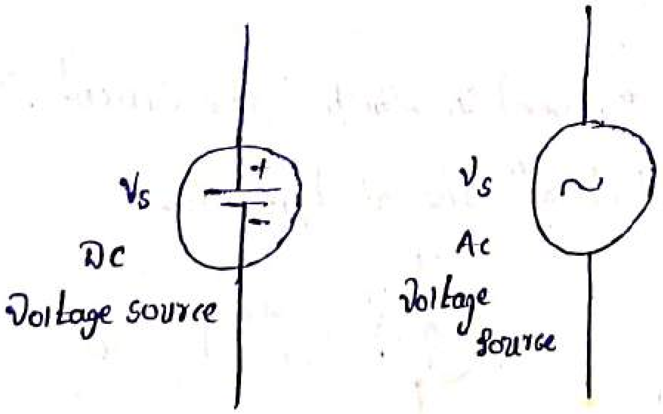
$$E = \int_{t_0}^{t} P\,dt$$

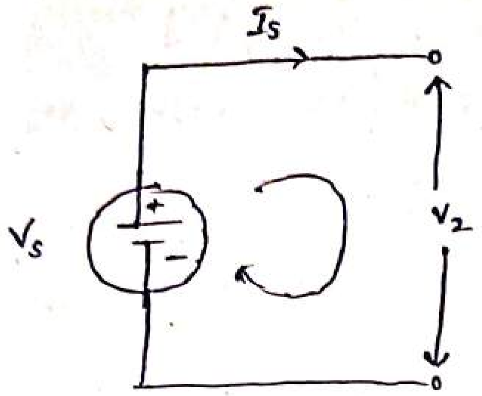
Energy sources

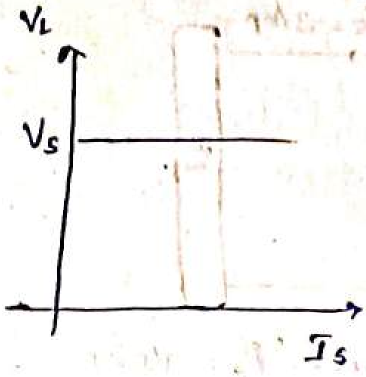
Electrical Energy Sources are Catagorised into ideal Voltage Sources and ideal Current sources.

Ideal Voltage Source

An ideal voltage source is a two terminal Element in Which the voltage V_s is Completely independent of current I_s through its terminals.





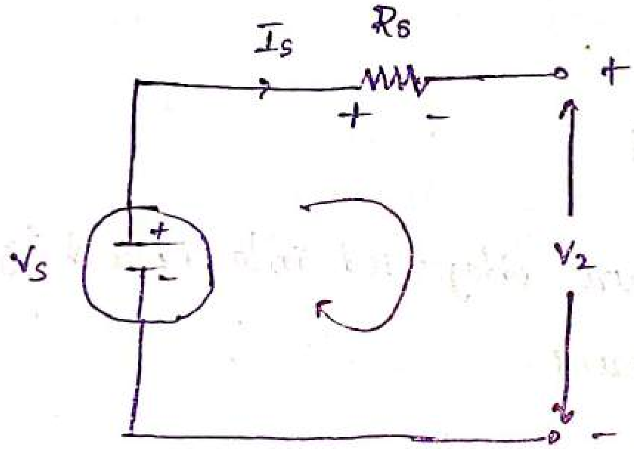


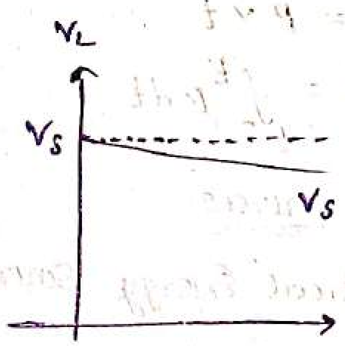
$V_S = V_L$

$V_S - V_L = 0$

Practical Voltage Source

In many practical voltage sources the internal resistance (R_S) is represented in series with the Source





$V_S - V_{R_1} - V_L = 0$

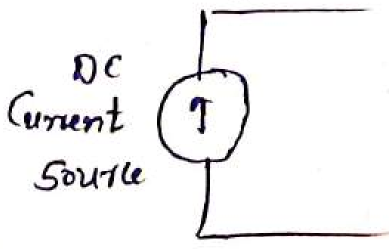
$V_S = V_1 + V_R$

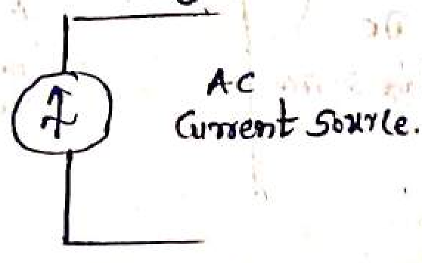
$V_S \neq V_L$

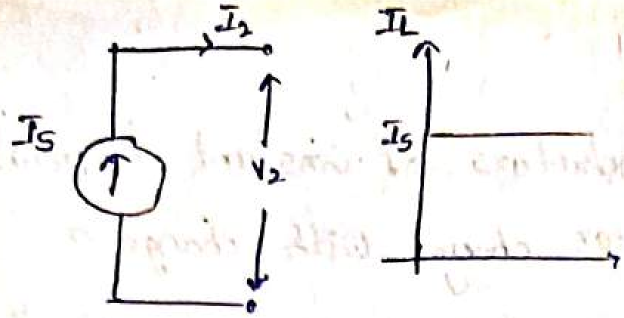
$V_L << V_S$

Ideal Current Source

It is two terminal Element in which the current I_S is completely independent of the voltage V_S.



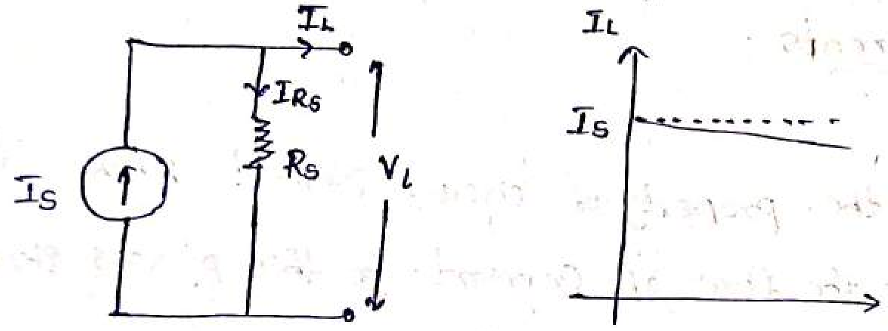




$I_S = I_L$

Practicle Current Source

In practicle Current Source the internal resistance r_s is Connected in parallel with I_S.



$I_S \neq I_L$

$I_L << I_S$

$I_S = I_{RS} + I_L$

Ohms law :

ohms law states that at const temperature the current flowing through an Element is directly proportional to Voltage across that Element.

$$\boxed{I \propto V}$$

$$I = kV$$

$$k = \frac{1}{R}$$

$$\boxed{I = \frac{V}{R}}$$

$$\boxed{V = IR}$$

limitations

1. This law is valid for conductors of constant temperature. The resistance of conductor changes with change in temperature.

2. incase of insulators ohms law is not followed. insulators donot conduct electricity.

3. ohms law not followed by semi conductors

Network Elements :

1. Resistor

resistance is the property of circuit element which offers oposition to the flow of current. in this process electrical energy is converted into heat energy.

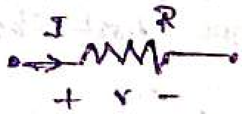
ii) The resistance of an element is represented as

$$R = \frac{\rho l}{a} \text{ ohms.}$$

where, ρ = Resistivity of material (Ω-M)

l = length of the conductor (m)

a = Area of crosssection of conductor (m^2)

resistance is represented as 

Power , $P = VI$

$P = I^2 R$ $\quad P = \frac{V^2}{R}$.

Energy , $E = P \times t$ Joules

$E = VIt$ Joules.

$1\,kwh = 1\,unit = 3.6 \times 10^6 J$

1. Find the resistance, Power and Energy in a standard copper wire of 200 meters long and 25 mm^2 in cross-section and resistivity of copper is 1.72×10^{-8} ohmmeter. Voltage is 12 volts and time is 20 sec.

(A)

length of wire = 200 m

Area $a = 25\ mm^2$.

$= 25 \times (10^{-3})^2\ m^2$

$= 25 \times 10^{-6}\ m^2$

$\rho = 1.72 \times 10^{-8}\ \Omega m$

$V = 12\ V$

$t = 20$ sec.

i) Resistance

$$R = \frac{\rho l}{a}$$

$$R = \frac{(1.72 \times 10^{-8})\,200}{25 \times 10^{-6}}$$

$$\boxed{R = 13.76 \times 10^{-2}\ \text{ohms.}}$$

ii) Power, $P = VI$

$$P = \frac{V^2}{R} = \frac{(12)^2}{13.76 \times 10^{-2}}$$

$$\boxed{P = 1046.5\ W}$$

iii) Energy $= (1046.5)\ 20$.

$$\boxed{E = 20930\ \text{joules}}$$

$t = 20\,sec = \frac{20}{60}$ min

$t = \frac{20}{60 \times 60}$ hrs.

$$\boxed{E = 5.81\ Wh}$$

Energy, $E = (1046.5)\ \frac{20}{60 \times 60}$

2. The resistance of wire is "R" Ω when length is l and area is "A". Find out the resistance of the same wire in terms of R when the length of the conductor is double and area of cross section is ~~halfed~~ halved?

A. $R_1 = R$

$R_2 = ?$

$l_1 = l$

$l_2 = 2l$

$a_1 = a$

$a_2 = a/2$

$$R = \frac{\rho l}{a}$$

$$R \propto \frac{l}{a}$$

$$\frac{R_2}{R_1} = \frac{l_2}{l_1} \times \frac{a_1}{a_2}$$

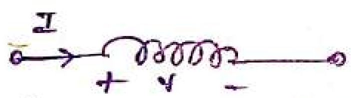
$$\frac{R_2}{R} = \frac{2l}{l} \times \frac{a}{a/2}$$

$$\frac{R_2}{R} = 4$$

$$\boxed{R_2 = 4R.}$$

Inductor:

The Electrical Element which stores the Energy in association with current is called an inductor.

inductor 

inductor is notated with "L" and units are henry's (H)

$$L = \frac{\mu N^2 l}{a}$$

where

μ = permiability

N = Number of turns of the coil.

l = length of the coil.

a = Area of cross section.

The voltange induced in a inductor is Directly proportional to rate of change of current to the Inductor

$$v \propto \frac{di}{dt}$$ [inductor does not allow the sudden changes in current]

$$V = L \frac{di}{dt}$$

Where L is called inductance of the coil.

Voltage current relation is.

$$V = L \frac{dI}{dt}$$

Rewriting the Equation,

$$dI = \frac{1}{L} V \cdot dt$$

Integerate on both sides,

$$\int_0^t dI = \frac{1}{L} \int_0^t V dt$$

$$i(t) - i(0) = \frac{1}{L} \int_0^t V dt$$

$$\boxed{i(t) = \frac{1}{L} \int_0^t V dt + i(0)}$$

Power absorbed by inductor

$$P = VI$$

$$P = L \frac{dI}{dt} I$$

Energy started by inductor is

$$E = \int P dt$$

$$E = \int L \frac{dI}{\cancel{dt}} I \cancel{dt}$$

$$E = L \int I dI$$

$$E = L \frac{I^2}{2}$$

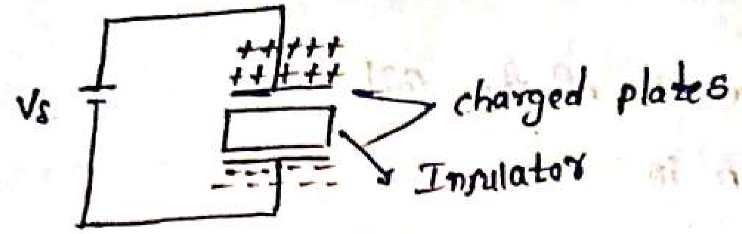
$$\boxed{E = \frac{1}{2} L I^2} \text{ Joules.}$$

Capacitor:

The Electrical Element stores the Energy in association With voltage is called a Capacitor.

It is notated "C"

units are Farads (F).



Capacitance $C = \frac{\epsilon A}{d}$

Where,

ϵ = Permitivity of the material.

$\epsilon = \epsilon_0 \epsilon_r$.

ϵ_0 = permitivity of air

ϵ_r = Relative permitivity of Insulator.

A = Area of crossection of Plates.

d = Distance b/w the Plates.

⇒ assume a Capacitor is charged with q columbs and voltage across it is v volts i.e, Capacitor C is

$$\boxed{C = \frac{q}{v}} \quad (or) \quad \boxed{q = Cv}$$

We know that Current charged relation is $I = \frac{dq}{dt}$

$$I = \frac{d}{dt}(Cv)$$

$$I = C\frac{dv}{dt}$$ [Capacitor does not allows sudden changes in Voltage]

Power :

Power abbsorbed by Capacitor is $P = vI$

$$P = vC\frac{dv}{dt}.$$

Energy stored by capacitor is

$$E = \int P\,dt$$

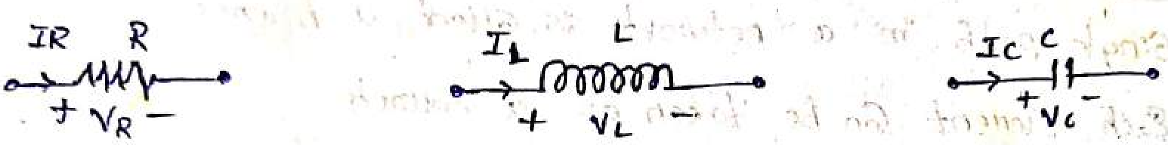
$$E = \int vC\frac{dv}{\cancel{dt}}\,\cancel{dt}$$

$$E = C\int v\,dv$$

$$E = C\frac{V^2}{2}$$

$$\boxed{E = \frac{1}{2}CV^2}$$ Joules.

Resistor (R) Ω	Inductor (L) Henry	Capacitor (C) Farady's
$R = \frac{\rho l}{a}$ ohms	$L = \frac{\mu N^2 l}{a}$ Henry	$C = \frac{\epsilon A}{d}$ Farady's.

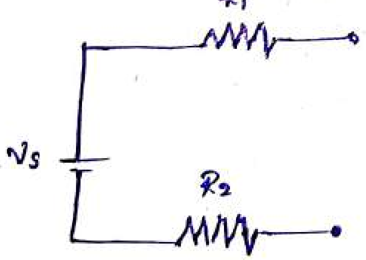


$V_R = I_R R$	$V_L = L\frac{dI}{dt}$	$V_C = \frac{1}{C}\int I\,dt$
$I_R = \frac{V_R}{R}$	$I_L = \frac{1}{L}\int V_L\,dt$	$I = C\frac{dV_C}{dt}$
Energy $E = 0\,J$	inductor $E = \frac{1}{2}LI^2$ Joules.	Energy $E = \frac{1}{2}CV^2$ Joules.

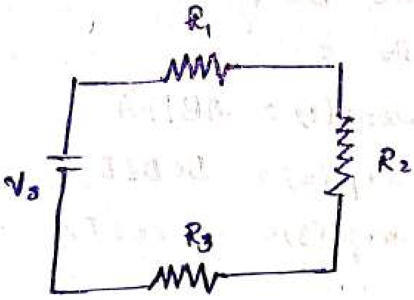
Network and circuit :

The interconnection of two or more circuit elements is called a network.

if the network contains atleast one closed path. is called a circuit.



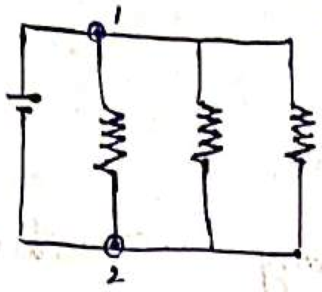
Network.



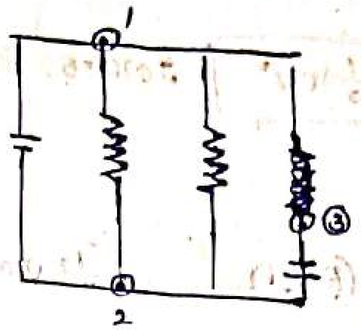
circuit

Node

~~required at to~~ the point at which two or more elements have common connection is called a node.



$n = 2$



$n = 3$.

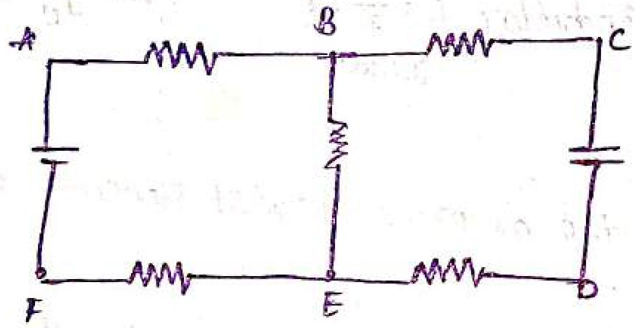
Branch

Single path in a network is called a Branch.

Each Element Can be taken as a Branch.

loop

If the node at which we started is the same as the node at which we ended. Without processing any node twice. is called a loop.



nodes, $n = 6$

Branches $b = 7$

loops $l = 3$.

loop(1) = ABEFA

loop(2) = BCDEB

loop(3) = ABCDEFA

kirchoff's law

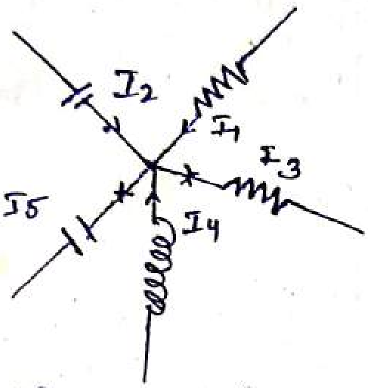
kirchoff proposed two laws one is kirchoffs Current law (kcl and kirchoff's voltage law (kvl).

1. kirchoff's current law (kcl)

It states that Algebraic sum of Currents at a node in a network at any instant of time is Zero.

(or)

The sum of the Currents Coming towards a node is Equal to sum of the Currents leaving from that node.



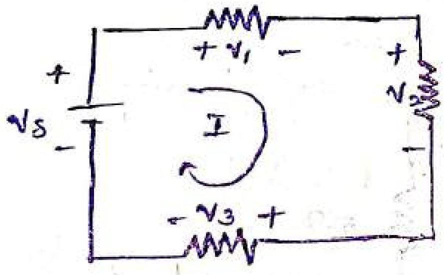
KCL Equations

$I_1 + I_1 - I_3 + I_4 - I_5 = 0$

$I_1 + I_2 + I_4 = I_3 + I_5$

Kirchoff's Voltage law:

It states that any instant of time the algebraic sum of Voltages in a closed Path is zero.

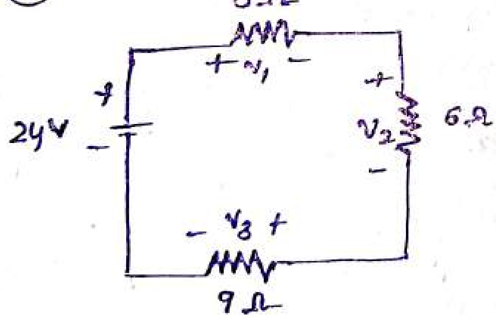


KVL Equations.

$-V_S + V_1 + V_2 + V_3 = 0.$

$V_1 + V_2 + V_3 = V_S.$

(1)



find V_1, V_2, V_3.

Equivalent Resistance

$R_{eq} = R_1 + R_2 + R_3$

$= 3+6+9 = 18\Omega$

$\boxed{R_{eq} = 18\Omega}$

Current through the circuit

$$I = \frac{V}{R_{eq}} = \frac{24}{18}.$$

$$\boxed{I = 1.33\,A}$$

Voltage across $3\,\Omega$

$$V_1 = 3I$$
$$= 3(1.33)$$
$$\boxed{V_1 = 3.99\,V}$$

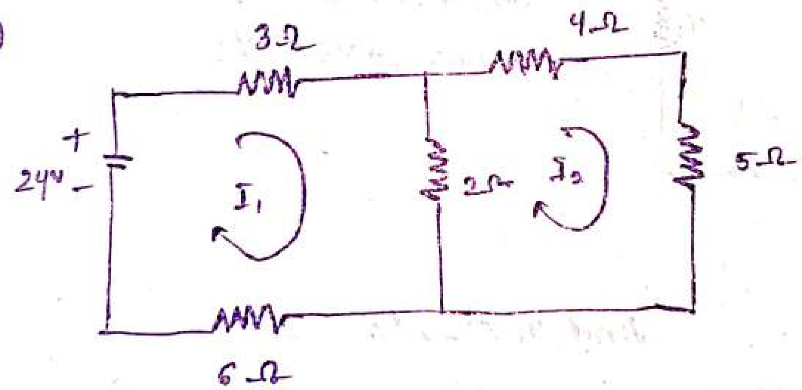
Voltage across $6\,\Omega$

$$V_2 = 6I$$
$$= 6(1.33)$$
$$\boxed{V_2 = 7.98\,V}$$

Voltage across $9\,\Omega$

$$V_3 = 9I$$
$$= 9(1.33)$$
$$\boxed{V_3 = 11.97\,V}$$

(2)



Find voltage across Each resistor.

Apply KVL in loop (1)

$$-24 + 3I_1 + 2(I_1 - I_2) + 6I_1 = 0.$$
$$-24 + 3I_1 + 2I_1 - 2I_2 + 6I_1 = 0.$$
$$11I_1 - 2I_2 = 24 \quad \text{— (1)}.$$

Similarly

Apply KVL in loop (2).

$4I_2 + 5I_2 + 2(I_2 - I_1) = 0.$

$4I_2 + 5I_2 + 2I_2 - 2I_1 = 0$

$-2I_1 + 11I_2 = 0. \quad — (2).$

Solving Eqn (1) & (2).

$$\boxed{\begin{aligned} I_1 &= 2.25 A \\ I_2 &= 0.41 A. \end{aligned}}$$

Voltage across 3Ω

$V_{3\Omega} = 3I_1$

$= 3(2.25)$

$$\boxed{V_{3\Omega} = 6.75V}$$

Voltage across 2Ω

$V_{2\Omega} = 2(I_1 - I_2)$

$= 2(2.25 - 0.41)$

$$\boxed{V_{2\Omega} = 3.68V}$$

Voltage across 6Ω

$V_{6\Omega} = 6I$

$= 6(2.25)$

$$\boxed{V_{6\Omega} = 13.5V}$$

Voltage across 4Ω

$V_{4\Omega} = 4I_2.$

$= 4(0.41)$

$$\boxed{V_{4\Omega} = 1.64V}$$

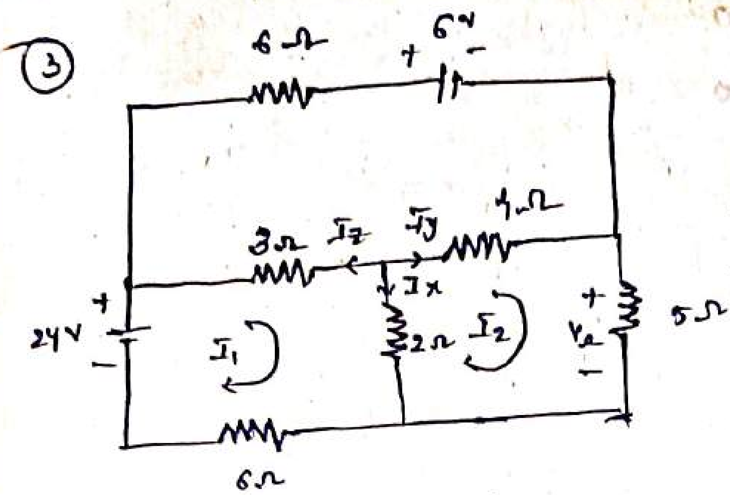
Voltage across 5Ω

$V_{5\Omega} = 5I_2$

$= 5(0.41)$

$$\boxed{V_{5\Omega} = 2.05V}$$

(3)



Apply kvl in loop (1)

$-24 + 3(I_1 - I_3) + 2(I_1 - I_2) + 6I_1 = 0$

$-24 + 3I_1 - 3I_3 + 2I_1 - 2I_2 + 6I_1 = 0$

$11I_1 - 2I_2 - 3I_3 = 24 \rightarrow (1)$

Apply kvl in loop (2)

$4(I_2 - I_3) + 5I_2 + 2(I_2 - I_1) = 0$

$4I_2 - 4I_3 + 5I_2 + 2I_2 - 2I_1 = 0$

$-2I_1 + 11I_2 - 4I_3 = 0 \rightarrow (2)$

Apply kvl in loop (3)

$6I_3 + 6 + 4(I_3 - I_2) + 3(I_3 - I_1) = 0.$

$6I_3 + 6 + 4I_3 - 4I_2 + 3I_3 - 3I_1 = 0$

$-3I_1 - 4I_2 + 13I_3 = -6$

$3I_1 + 4I_2 - 13I_3 = 6 \rightarrow (3).$

Solve Eqn (1) (2) & (3).

$I_1 = 2.339\,A$

$I_2 = 0.51\,A.$

$I_3 = 0.235\,A.$

$I_x = I_1 - I_2$

$= 2.339 - 0.51$

$\boxed{I_x = 1.82\,A}$

$I_y = I_2 - I_3$

$= 0.51 - 0.235$

$\boxed{I_y = 0.275\,A}$

$I_z = I_3 - I_1$

$= 0.235 - 2.329$

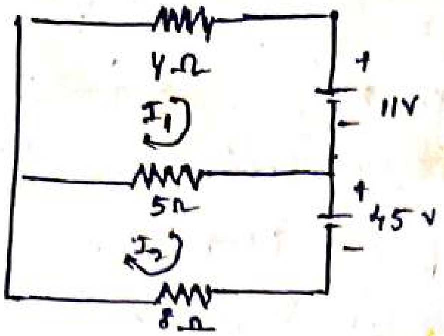
$\boxed{I_z = -2.104\,A}$

$V_o = 5I_2$

$= 5(0.51)$

$\boxed{V_o = 2.55\,V}$

4



Apply KVL in loop ①

$$4I_1 + 11 + 5(I_1 - I_2) = 0$$

$$4I_1 + 11 + 5I_1 - 5I_2 = 0$$

$$9I_1 - 5I_2 = -11 \longrightarrow ①.$$

Apply KVL in loop (2)

$$5(I_2 - I_1) + 45 + 8I_2 = 0$$

$$5I_2 - 5I_1 + 45 + 8I_2 = 0.$$

$$-5I_1 + 13I_2 = -45 \longrightarrow ②.$$

Matrix form of Equ ① & ②.

$$\begin{bmatrix} 9 & -5 \\ -5 & 13 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -11 \\ -45 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 9 & -5 \\ -5 & 13 \end{bmatrix}^{-1} \begin{bmatrix} -11 \\ -45 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{\mathrm{Adj}\begin{bmatrix} 9 & -5 \\ -5 & 13 \end{bmatrix}}{\begin{vmatrix} 9 & -5 \\ -5 & 13 \end{vmatrix}} \begin{bmatrix} -11 \\ -45 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{\begin{bmatrix} 13 & 5 \\ 5 & 9 \end{bmatrix}}{9(13) - 25} \begin{bmatrix} -11 \\ -45 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{1}{92} \underset{2\times2}{\begin{bmatrix} 13 & 5 \\ 5 & 9 \end{bmatrix}} \underset{2\times1}{\begin{bmatrix} -11 \\ -45 \end{bmatrix}}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -4 \\ -5 \end{bmatrix}$$

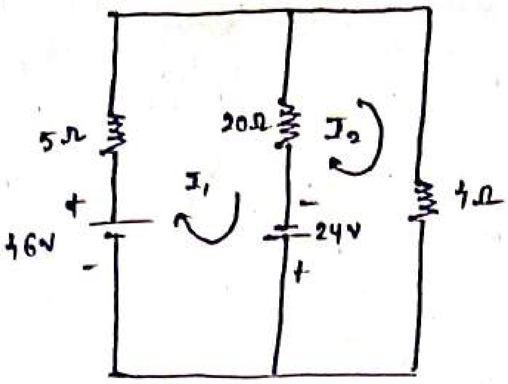
$$\begin{bmatrix} 13 \times -11 + 5(-45) \\ 5 \times -11 + 9(-45) \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{1}{92} \begin{bmatrix} -143 + -225 \\ -55 + -405 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{1}{92} \begin{bmatrix} -368 \\ -460 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -4 \\ -5 \end{bmatrix}$$

(5)



find I_1, I_2 ?

Apply k.v.L in loop 1

$$-46 + 5I_1 + 20(I_1 - I_2) - 24 = 0.$$

$$-46 + 5I_1 + 20I_1 - 20I_2 - 24 = 0.$$

$$25I_1 - 20I_2 = 70 \longrightarrow (1)$$

Apply kvL in loop 2.

$$24 + 20(I_2 - I_1) - 4I_2 = 0.$$

$$24 + 20I_2 - 20I_1 - 4I_2 = 0.$$

$$-20I_1 + 24I_2 = -24 \quad - (2).$$

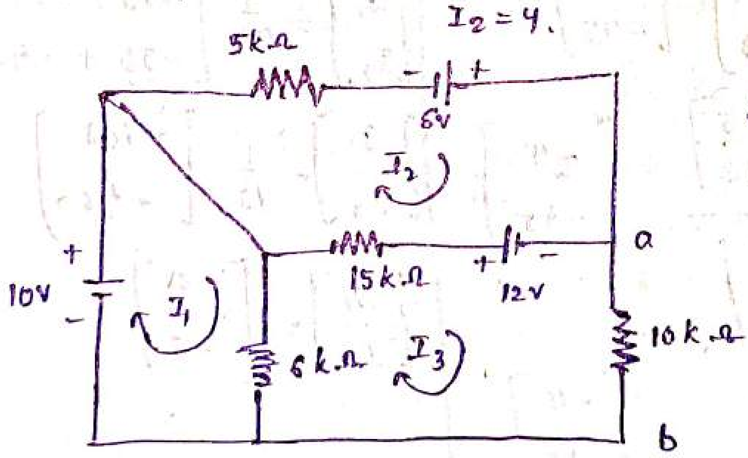
~~$I_1 = 1008$~~ solve Equ (1) & (2).

~~$I_2 = 18$.~~

$$I_1 = 6$$

$$I_2 = 4.$$

(6)



find I_1, I_2, I_3 & V_{ab}.

$$-10 + 6(I_1 - I_3) = 0.$$

$$-10 + 6I_1 - 6I_3 = 0.$$

$$6I_1 - 6I_3 = 10. \quad - (1).$$

$$-6 - 12 + 15(I_2 - I_3) + 5I_2 = 0$$

$$-18 + 15I_2 - 15I_3 + 5I_2 = 0$$

$$20I_2 - 15I_3 = 18. \quad - (2).$$

$12 - 10I_3 - 6(I_3 - I_1) - 15(I_3 - I_2)$

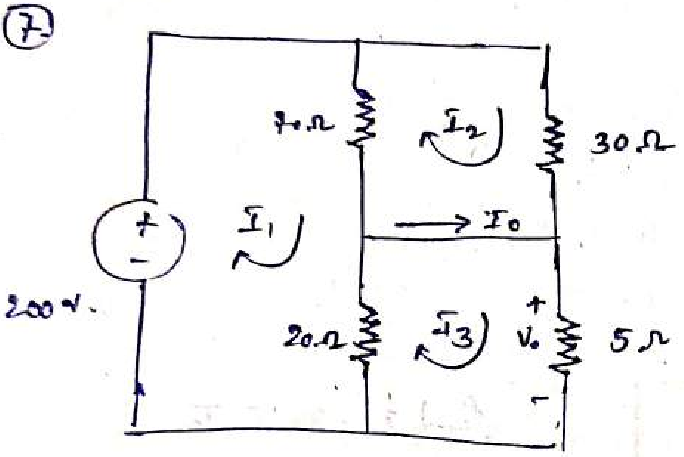
$12 - 10I_3 - 6I_3 + 6I_1 - 15I_3 + 15I_2 = 0.$

$-31I_3 + 15I_2 + 6I_1 = -12 \rightarrow$

$6I_1 + 15I_2 - 31I_3 = -12. \rightarrow (3).$

$x = \frac{701}{165} \quad y = \frac{156}{55}, \quad z = \frac{142}{55}.$

(7)



$-200 + 70(I_1 - I_2) + 20(I_1 - I_3) = 0.$

$-200 + 70I_1 - 70I_2 + 20I_1 - 20I_3 = 0.$

$-200 + 90I_1 - 70I_2 - 20I_3 = 0.$

$90I_1 - 70I_2 - 20I_3 = 200 \quad - (10).$

$9I_1 - 7I_2 - 2I_3 = 20 \quad - (1)$

In loop (2).

$30I_2 + 70(I_2 - I_1) = 0.$

$30I_2 + 70I_2 - 70I_1 = 0$

$100I_2 - 70I_1 = 0.$

$-7I_1 + 10I_2 = 0. \quad - (2)$

In loop (3).

$5I_3 + 20(I_3 - I_1) = 0.$

$5I_3 + 20I_3 - 20I_1 = 0.$

$25I_3 - 20I_1 = 0.$

$-20I_1 + 25I_3 = 0 - (3). \Rightarrow -4I_1 + 5I_3 = 0 \rightarrow (3)$

$x = 8, y = 5.6, z = 6.4.$

Solving eqn ① ② & ③.

$I_1 = 8A$

$I_2 = 5.6A$

$I_3 = 6.4A$.

$I_0 = I_3 - I_2$

$I_0 = 6.4 - 5.6$

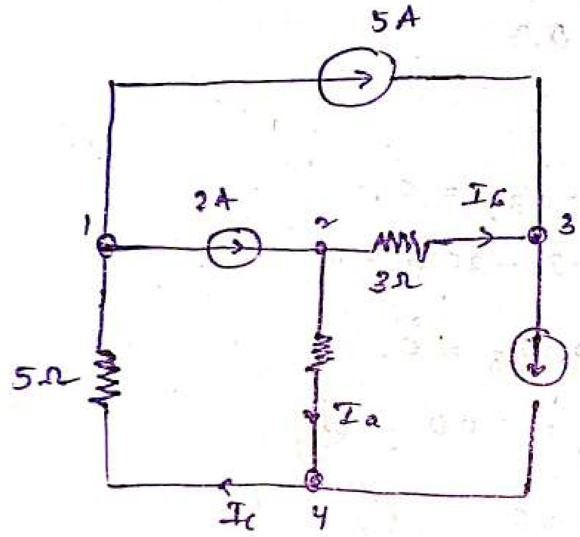
$\boxed{I_0 = 0.8A}$

$V_0 = 5I_3$

$V_0 = 5(6.4)$

$\boxed{V_0 = 32V}$

⑧



find I_a, I_b & I_c

Apply kcl at node ③

$I_b + 5 - 8 = 0$.

$\boxed{I_b = 3A}$

Apply kcl at node ②,

$2 - I_a - I_b = 0$

$-I_a = I_b - 2$

$I_a = 3 - 2$

$\boxed{I_a = -1A}$

Apply kcl at node 4

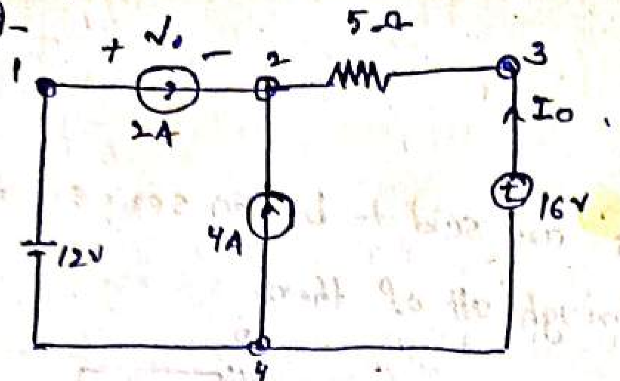
$I_a + 8 - I_c = 0$

$I_c = I_a + 8$.

$I_c = -1 + 8$.

$\boxed{I_c = 7A}$

(9)-



find I_0, V_0.

Apply kcl at node (2).

$$2+4+I_0=0.$$

$$\boxed{I_0=-6A}$$

~~at node~~ Apply kvl at outer loop.

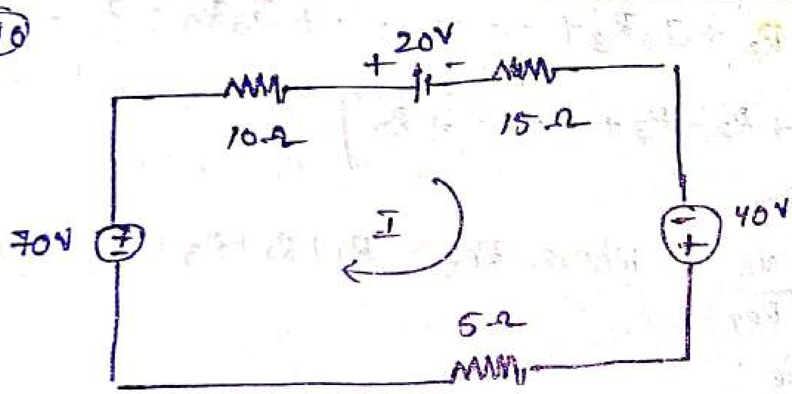
$$-12+V_0+5I_0+16=0.$$

$$-12+V_0+5(+6)+16=0.$$

$$-12+V_0+30+16=0.$$

$$\boxed{V_0=-34V}$$

(10)



find I.

$$-40+5I+10I+20+15I=0.$$

$$30I-20-70=0.$$

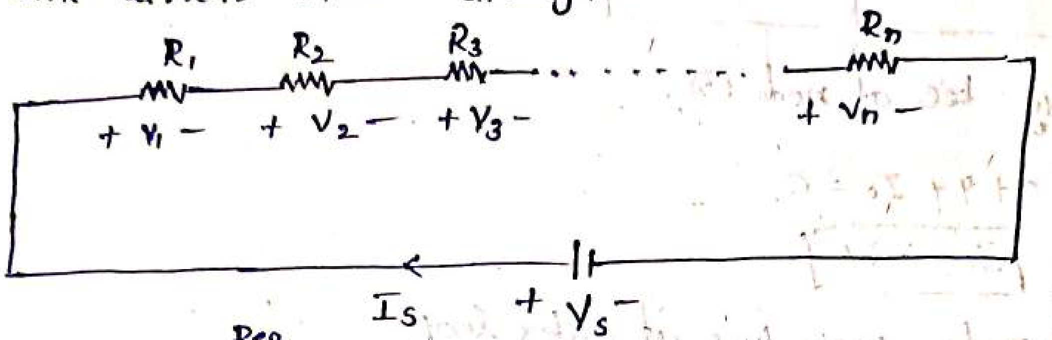
$$30I=20+70.$$

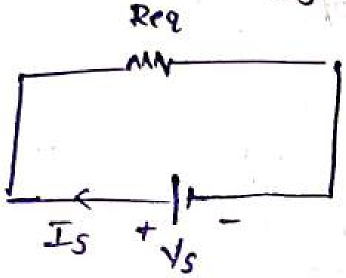
$$I=\frac{90}{30} \Rightarrow 3A.$$

Resistive Networks

1. Series Connected

Two or more resistances are said to be in series if the same current flows through all of them





Apply KVL in loop

$$-V_s + I_s R_1 + I_s R_2 + I_s R_3 + \cdots + I_s R_n = 0$$

$$V_s = I_s \left[R_1 + R_2 + R_3 + \cdots + R_n \right]$$

$$V_s = I_s \; R_{eq}$$

$$\Rightarrow I_s = \frac{V_s}{R_{eq}} \qquad \text{where, } R_{eq} = R_1 + R_2 + R_3 + \cdots + R_n$$

Voltage division Rule

Voltage across R_1, $V_1 = I_s R_1$

$$V_1 = \left(\frac{V_s}{R_{eq}}\right) R_1$$

$$V_1 = \frac{R_1}{R_{eq}} V_s.$$

Voltage across R_2, $V_2 = I_s R_2$

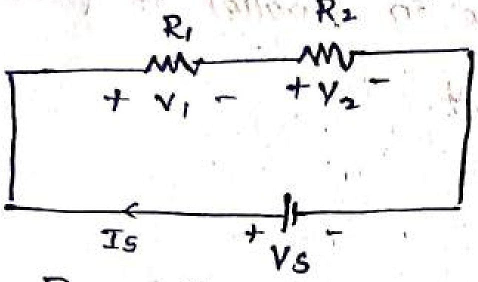
$$V_2 = \left(\frac{V_s}{R_{eq}}\right) R_2$$

$$V_2 = \left(\frac{R_2}{R_{eq}}\right) V_s.$$

$$V_3 = I_S R_3$$
$$= \left(\frac{V_S}{R_{eq}}\right) R_3.$$
$$V_3 = \left(\frac{R_3}{R_{eq}}\right) V_S.$$

Simply voltage across any conductor: $\boxed{V_n = \left(\frac{R_n}{R_{eq}}\right) V_S}$

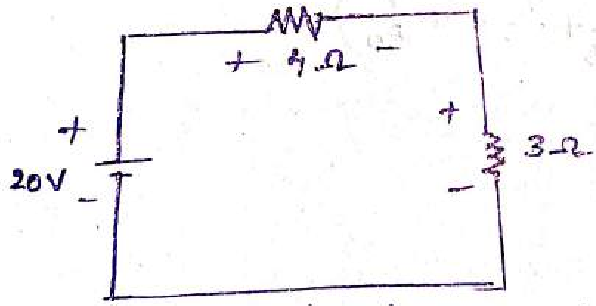
If $n = 2$



$$R_{eq} = R_1 + R_2$$
$$I_S = \frac{V_S}{R_{eq}}$$

Voltage across R_1

$$V_1 = \left(\frac{R_1}{R_1 + R_2}\right) V_S$$
$$V_2 = \left(\frac{R_2}{R_1 + R_2}\right) V_S$$



Find vatage across each resistance.

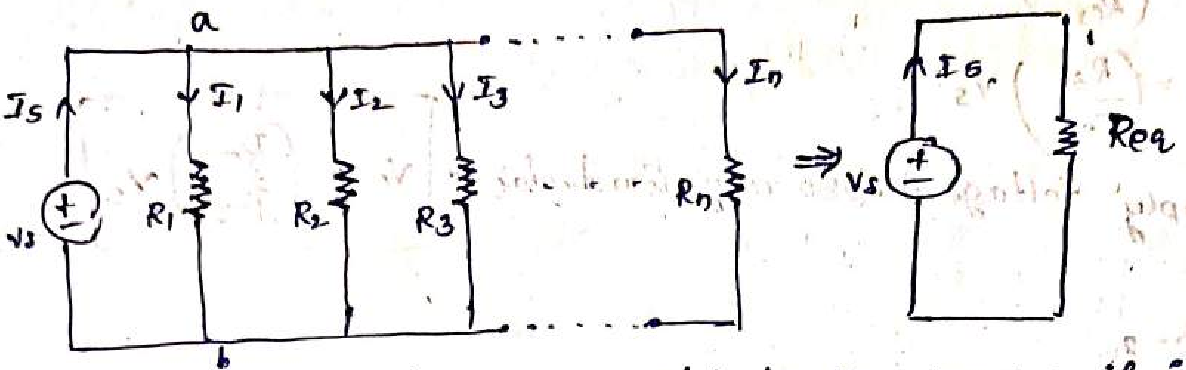
voltage across 4Ω = $V_{4\Omega} = \left(\frac{4}{7}\right) 20$

$$V_{4\Omega} = 11.4 \text{ volts}$$

voltage across 3Ω = $V_{3\Omega} = \left(\frac{3}{7}\right) 20$

$$V_{3\Omega} = 8.5 \text{ volts.}$$

(ii) <u>Parallel Connected</u>



Two or more resistances are said to be in parallel if identical voltages appears across each element.

Apply KCL at node a

$$I_S = I_1 + I_2 + I_3$$

$$\frac{V}{R_{eq}} = \frac{V_s}{R_1} + \frac{V_s}{R_2} + \frac{V_s}{R_3} + \cdots + \frac{V_s}{R_n}.$$

$$\frac{V}{R_{eq}} = V_s\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}\right]$$

$$\frac{1}{R_{eq}} = \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}\right]$$

$$R_{eq} = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}\right)}$$

If $n = 2$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\boxed{R_{eq} = \frac{R_1 R_2}{R_1 + R_2}}$$

<u>Current division Rule</u>

Current through R_1,

$$I_1 = \frac{V_s}{R_1}$$

$$I_1 = \frac{I_s R_{eq}}{R_1}$$

$$I_2 = \left(\frac{R_{eq}}{R_1}\right) I_s$$

Current through R_2,

$$I_2 = \frac{V_s}{R_2}$$

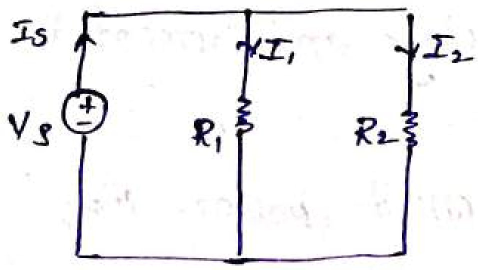
$$= \frac{I_s R_{eq}}{R_2}$$

$$= \left(\frac{R_{eq}}{R_2}\right) I_s.$$

Similarly I_n

$$\boxed{I_n = \left(\frac{R_{eq}}{R_n}\right) I_s}$$

If $n=2$,



$$I_1 = \left(\frac{R_{eq}}{R_1}\right) I_s$$

$$I_1 = \frac{\left(\frac{R_1 R_2}{R_1+R_2}\right)}{R_1} \times I_s$$

$$I_1 = \left(\frac{R_2}{R_1+R_2}\right) I_s.$$

$$V_1 = \left(\frac{R_1}{R_1+R_2}\right) V_s$$

$$V_2 = \left(\frac{R_2}{R_1+R_2}\right) V_s.$$

$$I_2 = \left(\frac{R_{eq}}{R_2}\right) I_s.$$

$$\left(\frac{\frac{R_1 \cancel{R_2}}{R_1+R_2}}{\cancel{R_2}}\right) I_s$$

$$I_2 = \left(\frac{R_1}{R_1+R_2}\right) I_s.$$

* The amount of work done to bring a negative charged Particle from one point to another point is called as voltage.
* It is notated with V or v
* units :- volts

<u>Current</u> :- The flow of electrons in a conductor is called current

(or)

The rate of charge is called current. i.e, $I = \frac{dq}{dt}$

It is denoted with I (or) i

units :- Ampeares.

<u>Power</u> :- The product of voltage and current is called Power.

(or)

The rate of workdone is called power. i.e.,

$$P = \frac{dw}{dt} = \frac{E}{t}$$ Here E = Energy, T = time

units : watts (W).

Classification of Elements:

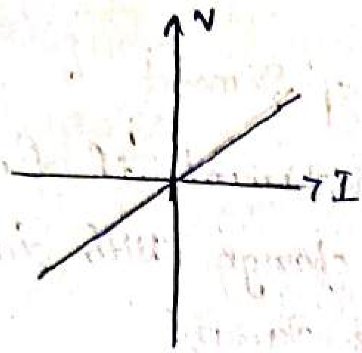
1. linear & non-linear Elements
2. Active & passive Elements.
3. unilateral & Bilateral Elements.
4. Time variant and time invariant Elements.
5. lumped & Distributed.

Linear and non linear Elements.

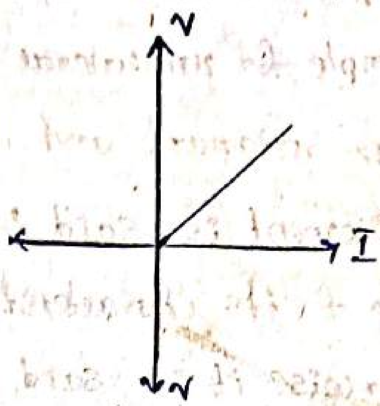
The element is said to be linear where the v-I characterstics follows only one Equation of a straight line passing through origin for all the time is called as linear.

P

~~Watts~~ H



linear



Non-linear

Example for linear Elements: Resistor

Example for Non-linear elements: Diode

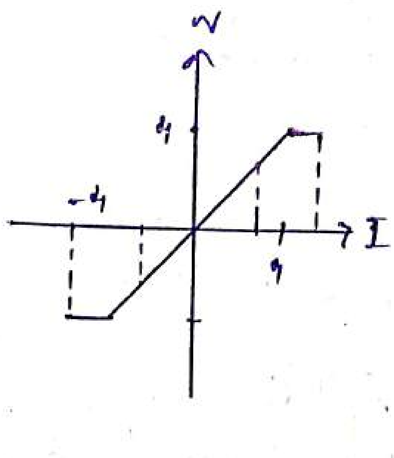
2. Active and passive.

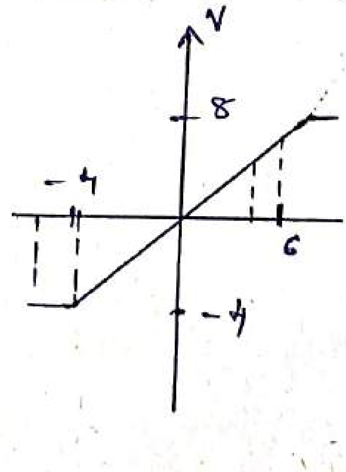
An element is said to be active if it delivers an net amount of Energy to the outside ~~volt~~ world otherwise it is said to be passive.

* Example for active elements are voltage sources, Current sources, Battery
* Example for passive Resistor, Inductor, capacitor.

3. Bilateral & unilateral Elements.

An element is said to be bilateral if it offers same impedence for the different directions of same current flow., otherwise it is said to be unilateral.



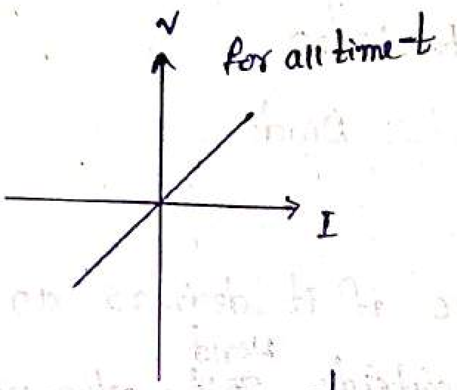


Example for Bilateral Resistor

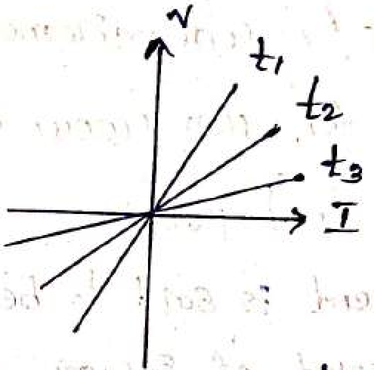
Example for unilateral Diode.

Time invariant and time varying Element.

An element is said to be time invariant if for all time t its characteristics donot change with time. otherwise it is said to be time variant.



Time invariant



Time variant.

Examples of Time invariant: Resistor.

Example for Time variant: Electrical load variation in a day.

Lumped & Distributed type Elements.

An element is said to be lumped if it is separable or replaceble with other element in the network. otherwise it is said to be Distributed.

Classification of sources:

Sources

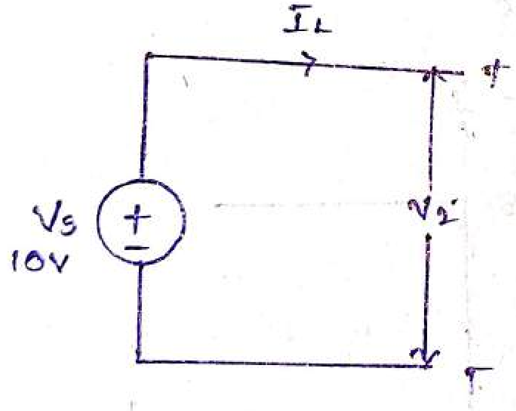
- Independent Sources
  - Independent Voltage Source
    - ideal Voltage Source
    - Practical Voltage Source
  - Independent Current Source
    - ideal Current Source
    - Practical Current Source
- Dependent Sources
  - VDVS
  - CDVS
  - VDCS
  - CDCS.

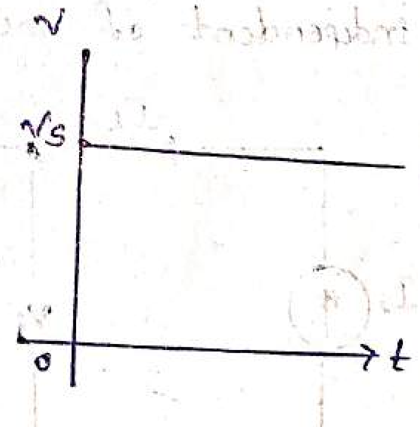
Independent sources:

i) ideal Voltage Source.

The source having zero internal resistance is called ideal voltage source.

In an ideal voltage source the load voltage is independent of the load current ~~drop~~. drawn



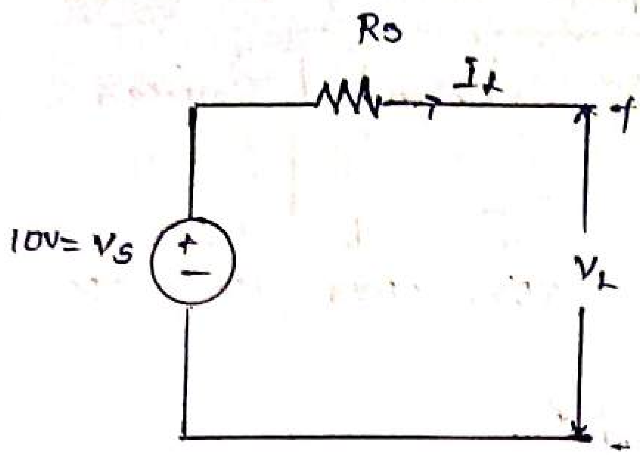


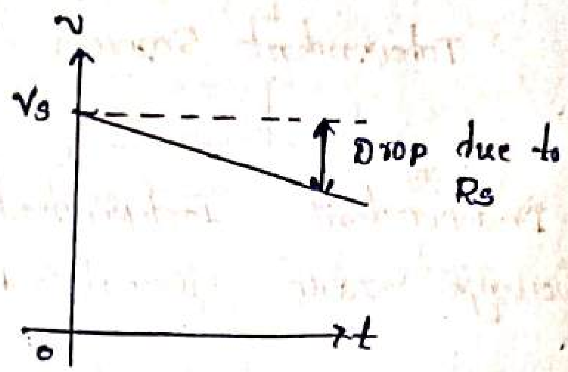
$V_S = V_L = 10V$

Practical voltage source:

The source having a finite internal resistance is called practical voltage source.

In a practical voltage source load voltage is a function of load current drawn.





$R_S \rightarrow$ Internal risistana of source

$V_S = V_L +$ Drop in R_S

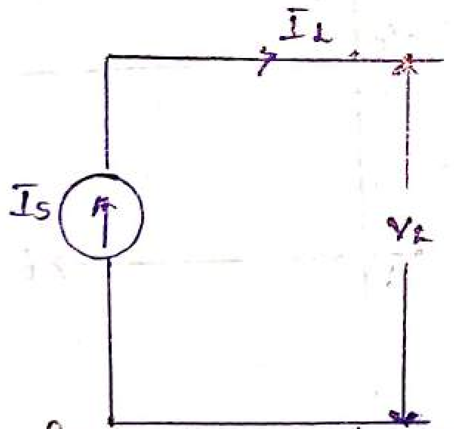
$V_S = V_L + (I_L R_S)$

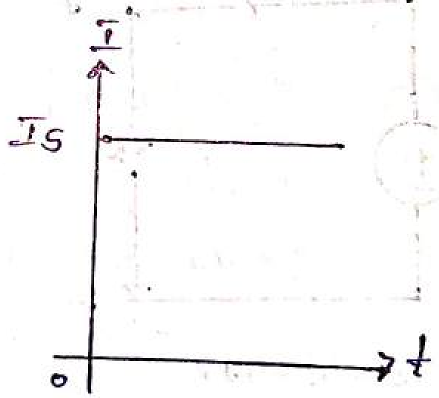
$V_L = V_S - (I_L R_S)$

ideal. Current Source:

* The Current does not have the effect of internal resistance is Called as Ideal Current source.

* In an ideal Current Source load Current is independent of low voltage.

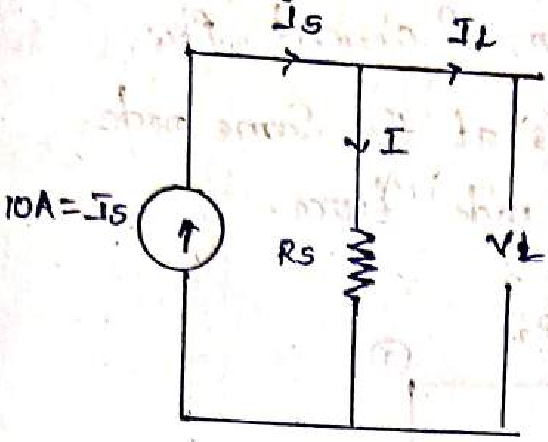


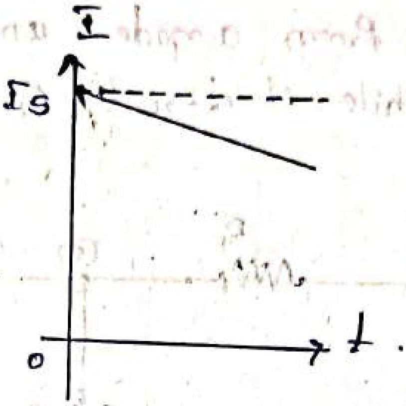


Practical Current source

The current Source have an effect of internal resistance is called practical Current Source.

In a practical Current Source the load Current is a function of load Voltage.



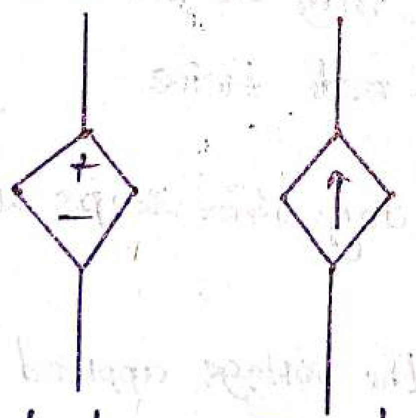


$$I_S = I_L + I_{R_S}$$

$$I_S = I_L + \frac{V_L}{R_S}$$

$$I_L = I_S - \frac{V_L}{R_S}.$$

The value of the source depends on are controlled by Some other parameter in the same circuit.
The dependent sources are represented as 



Dependent Voltage Source

Dependent Current Source.

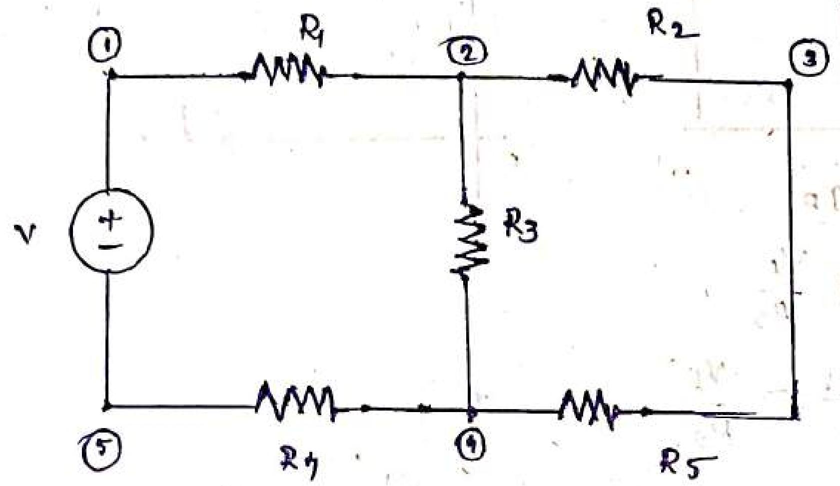
Circuit terminologys:

1) Branch : A single path Containing one single element which connects one node to any other node.

2) Node: A point at which to or more elements have Common Connection is Called a node.

Loop :

A path containing the direction of current flow starting from a node and ends at the same node. Mean while it doesn't cross a node twice.



Branches $b = 6$

nodes $n = 5$

loops $l = 3$

loop 1 1 − 2 − 4 − 5 − 1

loop 2 1 − 2 − 3 − 4 − 5 − 1

loop 3 2 − 3 − 4 − 2

Path:

A set of elements that may be traversed in order without passing through the node twice

Mesh:

A loop does not contain any other loops with in it.

Imp

Ohm's law

At constant temperature the ~~voltage applied across~~ current flowing through a conductor is directly proportional to voltage across that conductor.

$$\boxed{I \propto V}$$

$$I = kV$$

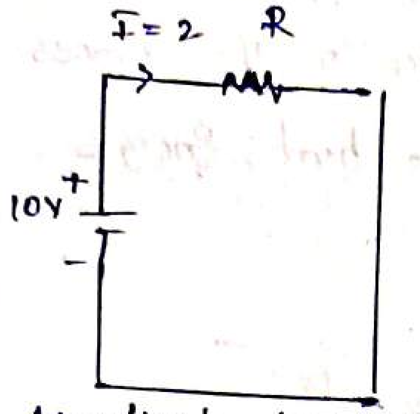
where $k = \frac{1}{R}$

$$\boxed{I = \frac{V}{R}}$$

$$\boxed{V = IR}$$

$$\boxed{V_{RS} = I_L R_S}$$

① find the resistance of the circuit show in the below figure.



According to ohms law

$$R = \frac{V}{I} = \frac{10}{2} = 5$$

$$R = 5\Omega.$$

②. Calculate the power where the voltage applied across resistor $R = 10\Omega$ is 20 Voltage

$\boxed{P = VI}$ $\boxed{P = \frac{V^2}{R}}$

$$I = \frac{V}{R} \Rightarrow \frac{20}{10} = 2.$$

$\boxed{P = I^2 R}$

$$P = 20(2) = 40 \text{ Watt}.$$

③. Calculate the resistance of the conductor where the voltage applied across the conductor is 30 V and the power supplied by the source is 50 W.

$$P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P}$$

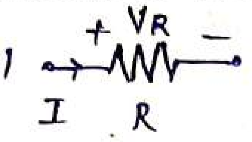
$$= \frac{(30)^2}{50}$$

$\boxed{R = 18\Omega}$

Network elements

Resistor:- The Element which offers oposition to the flow of current is called resistance in this process the Electrical Energy is converted into heat Energy.

The resistance is notated by R

Symbol 

The resistance is Expressed as $R = \frac{\rho l}{a}$

Where $\rho \rightarrow$ Resistivity of material (Ω-m)

$l \rightarrow$ length of the material (m)

$a \rightarrow$ Area of Crosssection (m^2).

→ units of Resistance is ohm's (Ω)

→ voltage accross resistance, $V_R = IR$

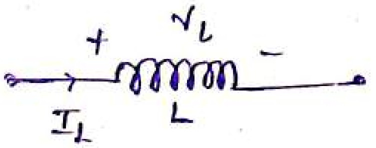
→ Current through resistance, $I = \frac{V_R}{R}$

Inductor: It is an Element that stores the Energy in association with the current flowing through it.

Voltage across an inductor is directly proportional to rate of change of Current flowing through it.

→ Induction does not allows sudden changes in Current.

→ It is notated with 'L' and is represented as



→ $V_L \propto \frac{dI_L}{dt}$

$$\boxed{V_L = L\frac{dI_L}{dt}}$$ → Voltage accross inductor

$$L = \frac{V_L \times dt}{dI_L} = \frac{\text{Volt-Sec}}{\text{Amp}} \approx \text{Henry (H)}$$

units of Inductor is Henry. $L = \frac{\mu N^2 l}{a}$

Energy stored by Inductor:

$\mu \rightarrow$ Permiability

$N \rightarrow$ No of turns

$l \rightarrow$ length

$a \rightarrow$ area

Voltage $V_L = L \frac{dI}{dt}$

W.K.T, Power, $P = V_L I$

Energy stored, $E = \int P\, dt$

$$E = \int V_L I$$

$$E = \int L \left(\frac{dI}{dt} \times I\right) dt$$

$$E = L \int I\, dI$$

$$E = L\left(\frac{I^2}{2}\right)$$

$$\boxed{E = \frac{1}{2} L I^2}$$ Joules (J)

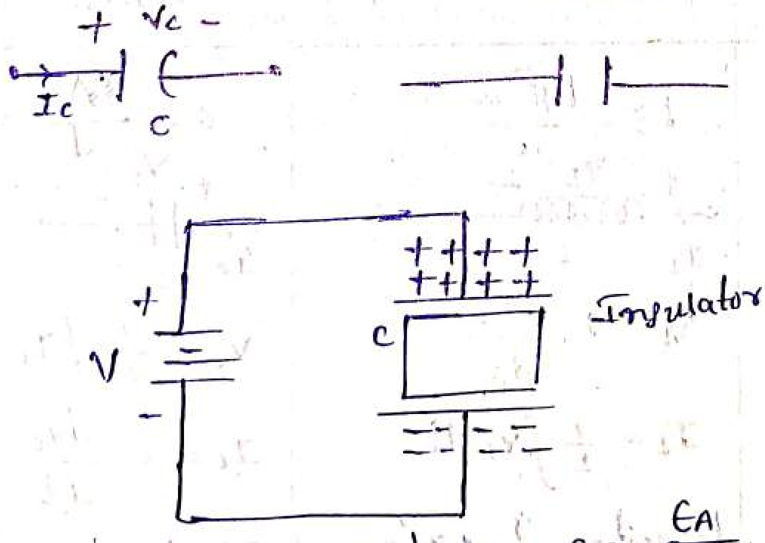
→ Voltage across an Inductor, $V_L = L \frac{dI_L}{dt}$

→ Current through an Inductor $I_L = \frac{1}{L}\int V_L\, dt$.

Capacitor:

The Element which stores the Energy in association with voltage across it is called a capacitor.

→ It is notated with "C" and is represented as



Capacitance is expressed as $C = \frac{\epsilon A}{d}$

Where $\epsilon \rightarrow$ permitivity of the material.

$\varepsilon = \varepsilon_0 \varepsilon_r$

$\varepsilon_0 \rightarrow$ permitivity of air.

$\varepsilon_r \rightarrow$ Relative permetivity

$A \rightarrow$ Area of Crossection of plates.

$d \rightarrow$ distance between plates.

units $\rightarrow$ Farads (F).

$\rightarrow$ current through Capacitance is directly proportional to rate of change of voltage across it.

Energy stored by Capacitor :-

We know that power $P = V I_c$

$\therefore$ Energy stored $E = \int P\, dt$

$$E = \int V \cdot I_c\, dt$$

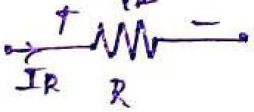
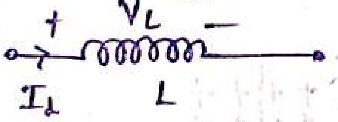
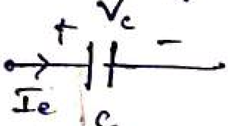
$$E = \int V \cdot C\left(\frac{dV}{dt}\right) dt$$

$$E = C\int V\, dV$$

$$E = C\frac{V^2}{2}$$

$$\boxed{E = \frac{1}{2} CV^2} \text{ Joules}$$

Comparision between R & L & C

Resistor (R)	Inductor (L)	Capacitor (C)
$R = \frac{\rho l}{a}$	$L = \frac{\mu N^2 a}{l}$	$C = \frac{\varepsilon A}{d}$
 $+\ V_R\ -$, I_R, R	 $+\ V_L\ -$, I_L, L	 $+\ V_c\ -$, I_c, C
$V_R = I_R R$	$V_L = L\frac{dI_L}{dt}$	$V_c = \frac{1}{C}\int I_c\, dt$
$I_R = \frac{V_R}{R}$	$I_L = \frac{1}{L}\int V_L\, dt$	$I_c = C\frac{dV_c}{dt}$
$E_{stored} = 0$ Joules	$E_{stored} = \frac{1}{2} L I_L^2$ J	$E_{stored} = \frac{1}{2} C V_c^2$ J
units : ohms (Ω)	units :- Henries (H)	units :- Farads (F).

kirchoff's laws

kirchoff proposed two laws

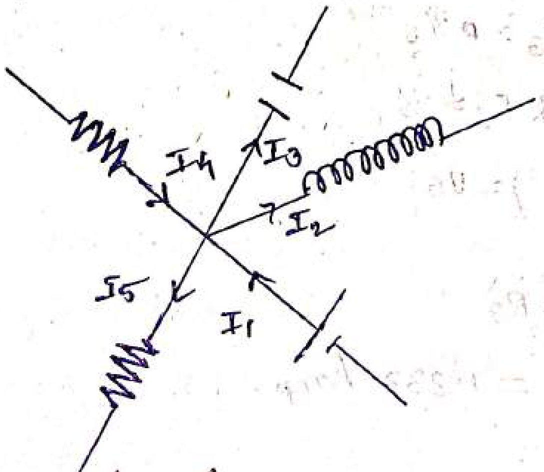
1. kirchoff's Current law (kcL)
2. kirchoff's Voltage law (kVL).

kirchoff's Current law

It states that algebric sum of Currents at a Node in a network at any instant of time is Zero.

(or)

The sum of the currents forming towards a Node is Equal to sum of the Currents leaving from that Node.

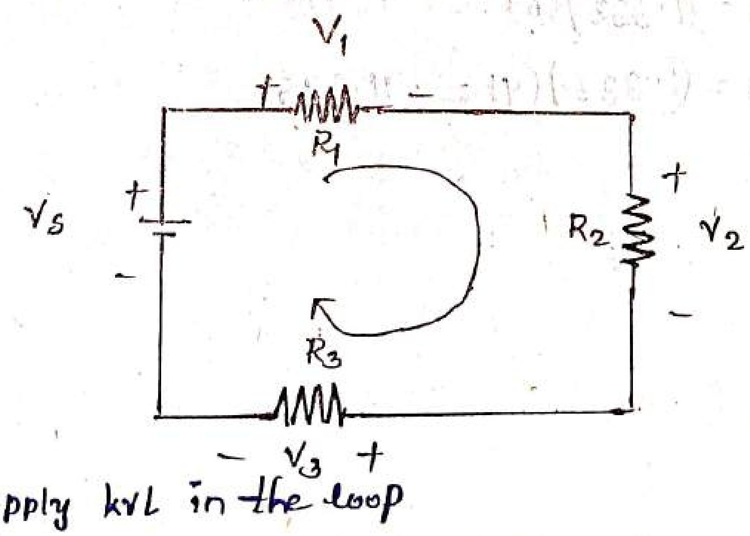


$$I_1 - I_2 - I_3 + I_4 - I_5 = 0$$

$$I_1 + I_4 = I_2 + I_3 + I_5.$$

kirchoff's Voltage law:

It states that at any instant of time the algebraic Sum of the voltages in a closed loop is Zero.

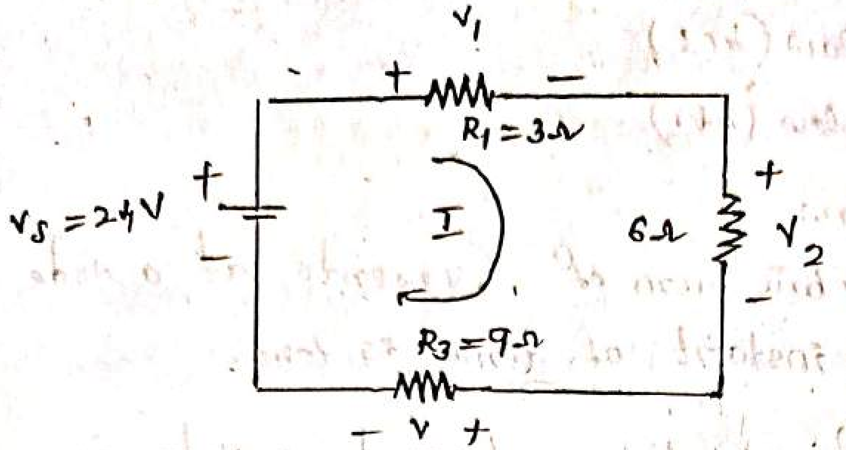


Apply kVL in the loop

$$V_s - V_1 - V_2 - V_3 = 0$$

$$V_s = V_1 + V_2 + V_3$$

Problem find V_1, V_2, V_3 in the below circuit diagram.



Apply KVL in the loop

$$-V_S + V_1 + V_2 + V_3 = 0$$

$$V_1 + V_2 + V_3 = V_S$$

$$IR_1 + IR_2 + IR_3 = V_S$$

$$I(R_1 + R_2 + R_3) = V_S$$

$$I = \frac{V_S}{R_1 + R_2 + R_3}$$

$$I = \frac{24}{3+6+9} = 1.333 \text{ Amp.}$$

Therefore

$$V_1 = IR_1 = (1.333)(3)$$

$$= 3.999$$

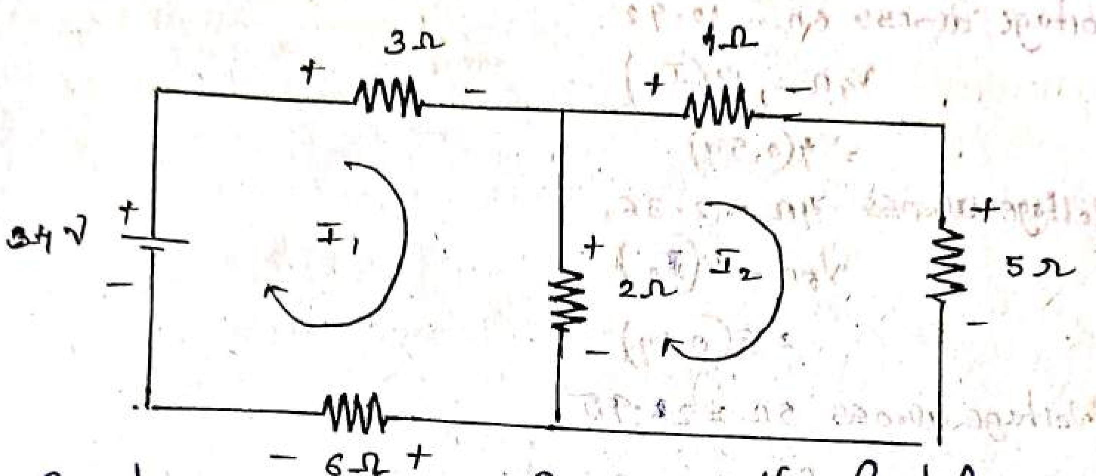
$$V_2 = IR_2 = (1.333)(6) = 7.998$$

$$V_3 = IR_3 = (1.333)(9) = 11.997$$

Mesh Analysis:

Finding Solution in a Circuit by applying and Solving KVL Equations is Called Mesh Analysis.

Find the voltage across Each element in the following Circuit.



Consider Current I_1 is flowing in the first loop and Current I_2 is flowing in the second loop and the directions are assumed in the circuit.

Apply KVL in loop ①

$= -24 + 3I_1 + 2(I_1 - I_2) + 6I_1$

$= -24 + 3I_1 + 2I_1 - 2I_2 + 6I_1$

$= -24 + 11I_1 - 2I_2 \Rightarrow 11I_1 - 2I_2 = 24 \rightarrow ①$.

Apply KVL in loop ②.

$4I_2 + 5I_2 - 2I_2 + 2I_1$

$\Rightarrow 4I_2 + 5I_2 + 2I_2 + 2I_1$

$7I_2 + 2I_1 = 0 \rightarrow ②$.

Solve ① & ② Equations

$I_1 = 2.07$ Amp.

$I_2 = 0.59$ Amp.

∴ Voltage across 3Ω resistance

$V_{3\Omega} = 3I_1$

$= 3(2.07)$

Voltage across $3\Omega = 6.21$

$V_{2\Omega} = 2(I_1 - I_2)$

$= 2(2.07 - 0.59)$

$= 2(1.48)$

Voltage across $2\Omega = 2.96$

$V_{6\Omega} = 6(I_1)$

$= 6(2.07)$

Voltage across $6\Omega = 12.42$

$V_{4\Omega} = 4(I_2)$

$= 4(0.59)$

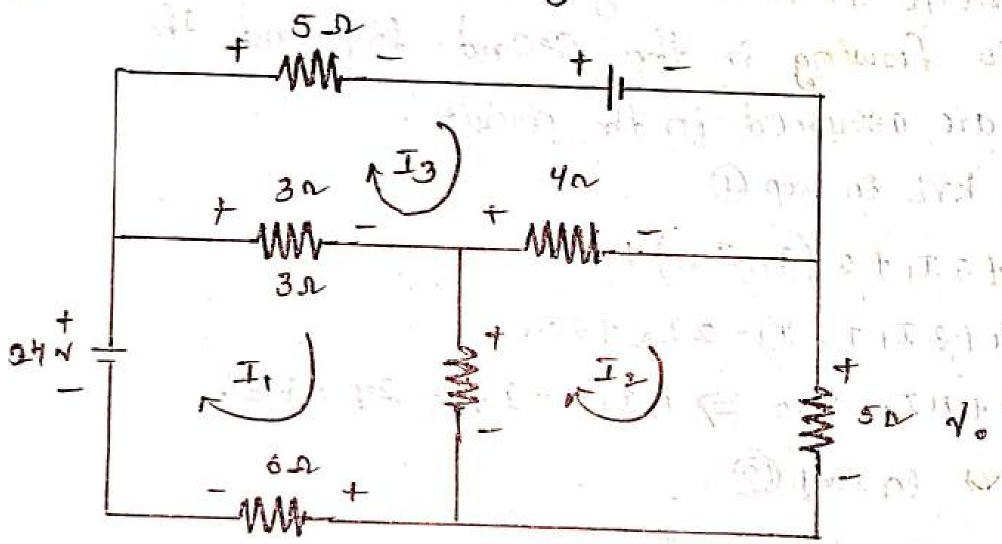
Voltage across $4\Omega = 2.36$

$V_{5\Omega} = 5(I_2)$

$= 5(0.59)$

Voltage across $5\Omega = 2.95$

(2) Find V_o in the following circuit.



Apply KVL in the loop (1)

$-24 + 3(I_1 - I_3) + 2(I_1 - I_2) + 6I_1 = 0.$

$-24 + 3I_1 - 3I_3 + 2I_1 - 2I_2 + 6I_1 = 0.$

$11I_1 - 2I_2 - 3I_3 = 24 \rightarrow (2).$

Apply KVL in the loop (2)

$4(I_2 - I_3) + 5(I_2 - 2(I_2 - I_1) = 0.$

$4I_2 - 4I_3 + 5I_2 - 2I_2 + 2I_1 = 0.$

$2I_1 + 7I_2 - 4I_3 = 0 \rightarrow (2).$

Apply KVL in the loop (3)

$6 + 4(I_3 - I_2)$

$6 - 4(I_3 - I_2) - 3(I_3 - I_1) + 6(I_3)$

$6 - 4I_3 + 4I_2 - 3I_3 + 3I_1 + 6I_3$

$3I_1 + 4I_2 - I_3 = -6 \rightarrow (3)$

Solve (1), (2) & (3) Equations

$\therefore I_1 = 0.26$ Amp

$I_2 = 2.96$ Amp

$I_3 = 5.05$ Amp

$V_{5\Omega} = 5(I_2)$

$V_{5\Omega} = 5(2.96)$

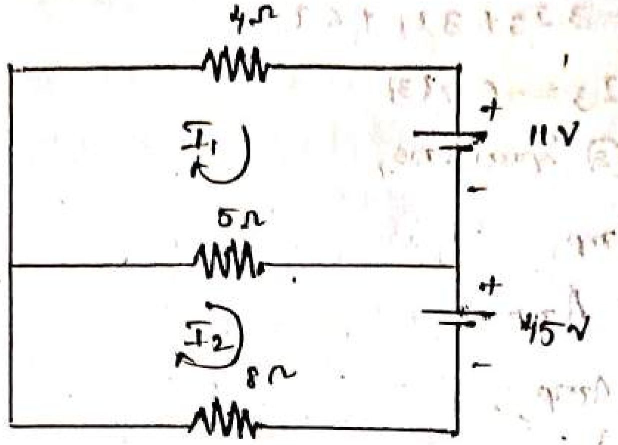
$= 14.8$

$\therefore$ Voltage across 5Ω = 14.8 volts.

Points to remember for loop analysis.

1. While assuming loop currents make sure that atleast one loop current links with every element
2. No two loops should be identical
3. Choose minimum number of loop currents.
4. Convert current sources if present into their equivalent voltage sources for loop analysis. Wherever possible.
5. If current in a particular branch is required then try to choose loop currents insuch a way that only one loop current links with that Branch.

(3).



find I_1 & I_2

Apply KVL in loop (1).

$$4I_1 + 11 + 5(I_1 - I_2) = 0.$$

$$4I_1 + 11 + 5I_1 - 5I_2 = 0$$

$$4I_1 + 5I_1 - 5I_2 = -11$$

$$9I_1 - 5I_2 = -11 \rightarrow (1).$$

Similarly Apply KVL in loop (2)

$$5(I_2 - I_1) + 45 + 8I_2 = 0$$

$$5I_2 - 5I_1 + 8I_2 = -45$$

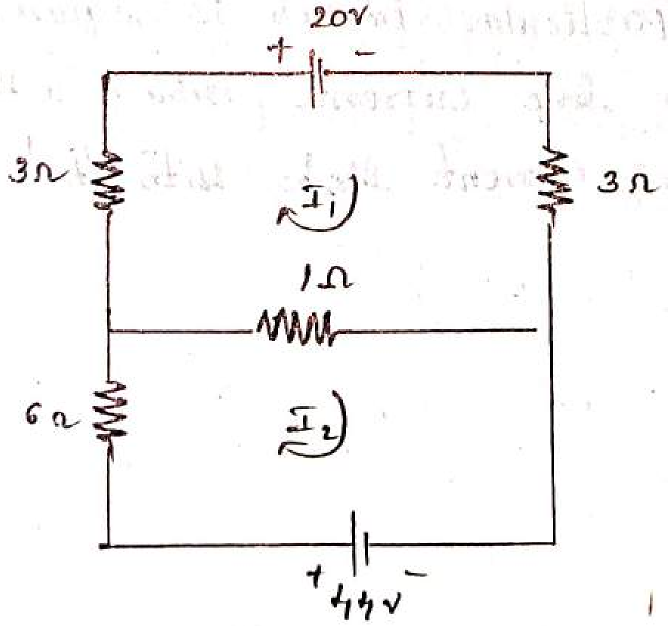
$$-5I_1 + 13I_2 = -45 \quad -(2).$$

Solve Equ (1) & (2)

$I_1 = -4A$

$I_2 = -5A$.

(4).



Apply KVL in loop (1)

$$20 + 3I_1 + 1(I_1 - I_2) + 3I_1 = 0$$

$$20 + 3I_1 + I_1 - I_2 + 3I_1 = 0$$

$$7I_1 - I_2 = -20 \rightarrow (1)$$

Apply kvl in loop (2)

$$6I_2 + 1(I_2 - I_1) - 44 = 0$$

$$6I_2 + I_2 - I_1 = 44$$

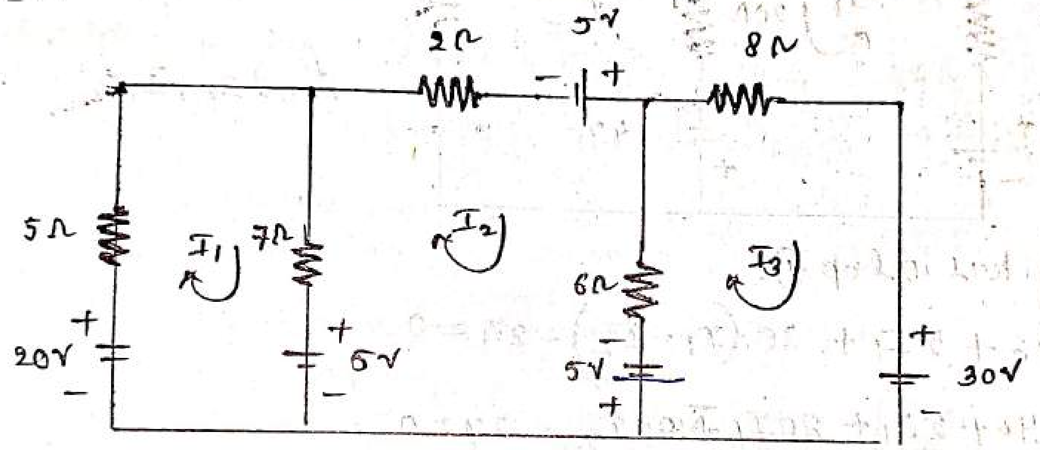
$$7I_2 - I_1 = 44 \quad - (2)$$

Solving (1) & (2).

$$I_1 = -2A$$

$$I_2 = 6A$$

(5).



find I_1, I_2 & I_3 in the circuit.

Apply kvl in loop (1).

$$-20 + 5I_1 + 7(I_1 - I_2) + 5 = 0$$

$$5I_1 + 7I_1 - 7I_2 - 15 = 0$$

$$12I_1 - 7I_2 = 15 \quad - (1)$$

Apply kvl in loop (2)

$$-5 + 7(I_2 - I_1) + 2I_2 - 5 + 6(I_2 - I_3) - 5 = 0$$

$$-5 + 7I_2 - 7I_1 + 2I_2 - 5 + 6I_2 - 6I_3 - 5 = 0.$$

$$-7I_1 + 15I_2 - 6I_3 - 15 = 0$$

$$-7I_1 + 15I_2 - 6I_3 = 15.$$

Apply kvl in loop 3.

$$5 + 6(I_3 - I_2) + 8I_3 + 30 = 0$$

$$5 + 6I_3 - 6I_2 + 8I_3 + 30 = 0$$

$$-6I_2 + 14I_3 + 35 = 0$$

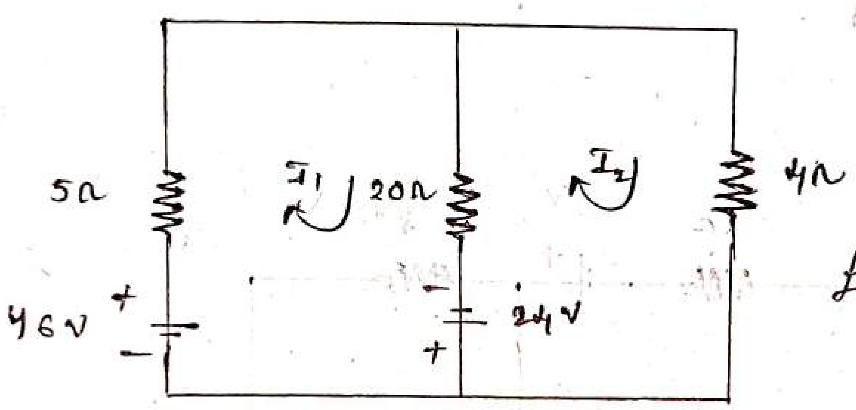
$$-6I_2 + 14I_3 = -35.$$

$I_1 = 1.86A$

$I_2 = 1.04A$

$I_3 = -2.05A$

6.



find I_1, I_2.

Apply kvl in loop ①.

$$-46 + 5I_1 + 20(I_1 - I_2) - 24 = 0$$

$$-46 + 5I_1 + 20I_1 - 20I_2 - 24 = 0.$$

$$25I_1 - 20I_2 - 70 = 0.$$

$$25I_1 - 20I_2 = 70$$

Apply kvl in loop ②.

$$24 + 20(I_2 - I_1) + 4I_2 = 0.$$

$$24 + 20I_2 - 20I_1 + 4I_2 = 0.$$

$$-20I_1 + 24I_2 = -24$$

$I_1 =$ ~~20.16A~~ 6A

$I_2 =$ ~~0.080A~~ 4A

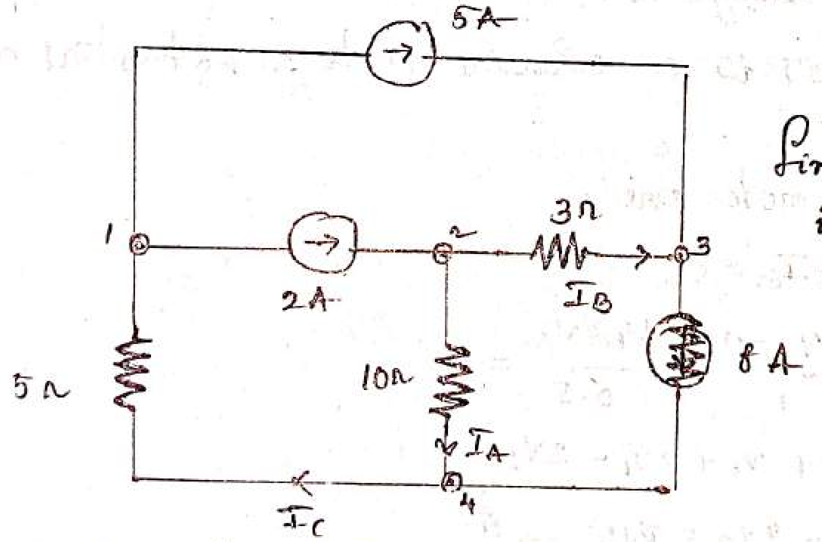
Nodal Analysis:

Finding solution in any type of circuit by writing and solving KCL Equations. is called nodal analysis.

Points to remember: in nodal Analysis:

1. By assuming branch currents make sure that each unknown Branch Current Considered atleast once.
2. Convert the voltage source present into their equivalent current source for nodal analysis. wherever possible.
3. Follow the same sign Convention Currents entering at a node are to be Considered as positive while Currents leaving the node are to Considered as negative.
4. As per as possible Select the directions of various branch Currents leaving the respective nodes.

1.



Find I_A, I_B, & I_C in the circuit

A) Apply KCL at node (3).

$$-5 - I_B + 8 = 0$$

$$5 + I_B = 8$$

$$\boxed{I_B = 3A}$$

$$\boxed{I_A = -1A}$$

Apply KCL at node (2)

$$-2 + I_A + I_B = 0$$

$$I_A + I_B = 2$$

$$I_A = 2 - I_B$$

Apply KCL at ①

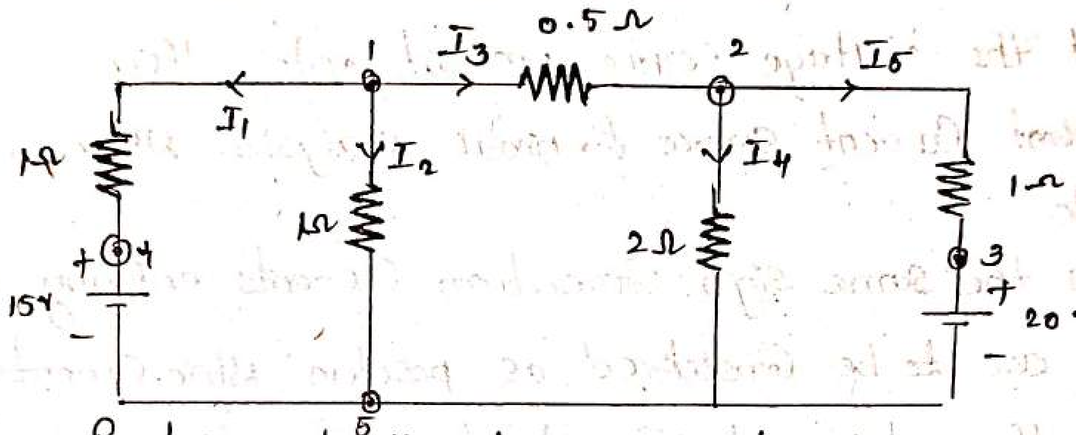
$-8 - I_A + I_C = 0$

$I_C = I_A + 8$

$I_C = -1 + 8$

$\boxed{I_C = 7A}$

②



Find current through each resistance by using nodal Analysis.

A. Solution :

Consider a voltage of V_1 at node 1, a voltage V_2 at node 2, node 5 is a reference node ∴ potential at node 5 is 0 volts.

Apply KCL at node one.

$I_1 + I_2 + I_3 = 0.$

$$\frac{V_1 - 15}{1} + \frac{V_1 - 0}{1} + \frac{V_1 - V_2}{0.5} = 0$$

$V_1 - 15 + V_1 + 2V_1 - 2V_2 = 0$

$4V_1 - 2V_2 = 15 \rightarrow ①.$

Apply KCL at node ②.

$-I_3 + I_4 + I_5 = 0$

$$-\frac{(V_2 - V_1)}{0.5} + \frac{V_2 - 0}{2} + \frac{V_2 - 20}{1} = 0$$

$$-2V_2 + 2V_1 + \frac{V_2}{2} + V_2 - 20 = 0.$$

$-4V_2 + 4V_1 + V_2 + 2V_2 - 40 = 0.$

$-4V_1 + 7V_2 = 40 \rightarrow ②.$

Solving Eqn ① & ②.

$V_1 = \cancel{1}9.25$ V

$V_2 = \cancel{25}\ 11$ V

i.e., Currents through each resistance $I_1 = \frac{V_1 - 15}{1} = \frac{9.25 - 15}{1}$

$I_1 = -5.75$ A

$I_2 = \frac{V_1 - 0}{1} = V_1$

$\boxed{I_2 = 9.25\text{ A}}$

$I_3 = \frac{V_1 - V_2}{0.5} = \frac{9.25 - 11}{0.5}$

$I_3 = -3.5$ A

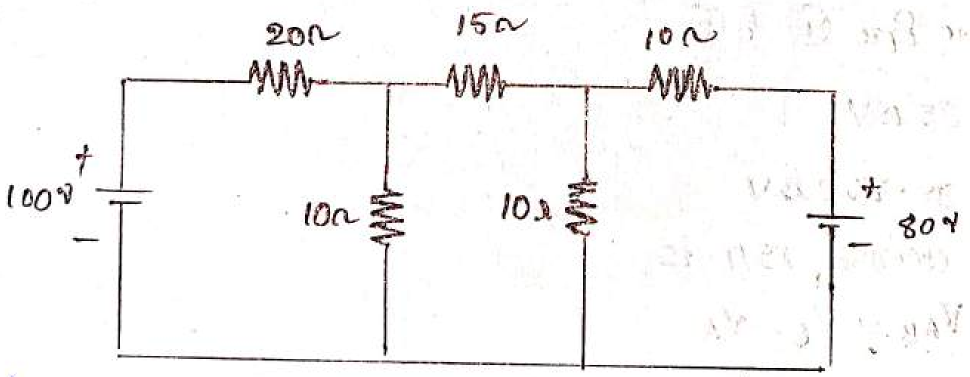
$I_4 = \frac{V_2 - 0}{2} = \frac{V_2}{2} = \frac{11}{2}$

$\boxed{I_4 = 5.5\text{ A}}$

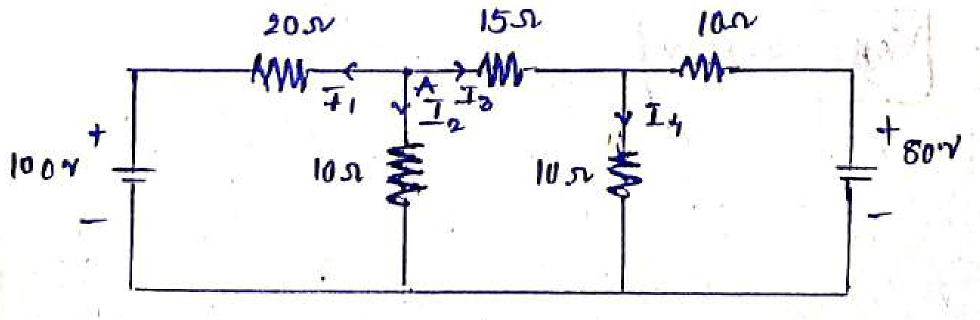
$I_5 = \frac{V_2 - 20}{1} = 11 - 20$

$\boxed{I_5 = -9\text{ A}}$

3. Calculate the voltage across 15Ω resistor in the network shown below using nodal analysis.



A



1. Identify nodes A, B in the circuit.

2. Assume V_A, V_B are the potentials at the nodes.

3. indicate the Current directions also in the circuit.

4. Apply KCL at Node A.

$$-I_1 - I_2 - I_3 = 0$$

$$I_1 + I_2 + I_3 = 0$$

$$\frac{V_A - 100}{20} + \frac{V_A - 0}{10} + \frac{V_A - V_B}{15} = 0.$$

$$\frac{3V_A - 300 + 6V_A + 4V_A - 4V_B}{60} = 0$$

$$13V_A - 4V_B = 300 \longrightarrow (1).$$

Apply KCL at Node B.

$$I_3 - I_4 - I_5 = 0.$$

$$\frac{V_A - V_B}{15} - \frac{V_B - 0}{10} - \frac{V_B - 80}{10} = 0.$$

$$\frac{2V_A - 2V_B - 3V_B - 3V_B + 240}{30} = 0.$$

$$2V_A - 8V_B = -240 \quad -(2).$$

Solve Equ (1) & (2)

$$V_A = 35 V$$

$$V_B = 38.750 V$$

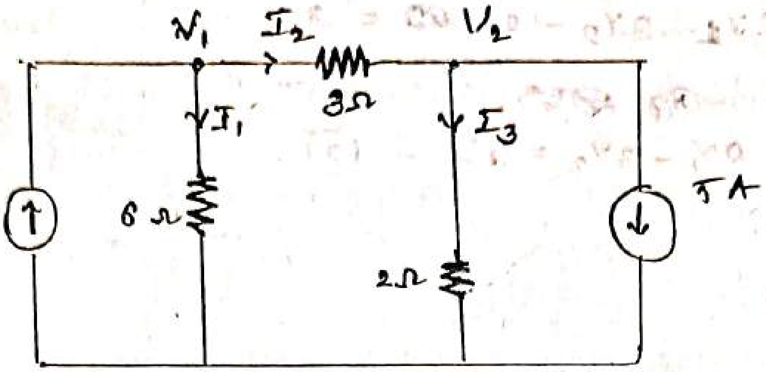
Voltage across 15Ω is

$$V_{AB} = V_B - V_A$$

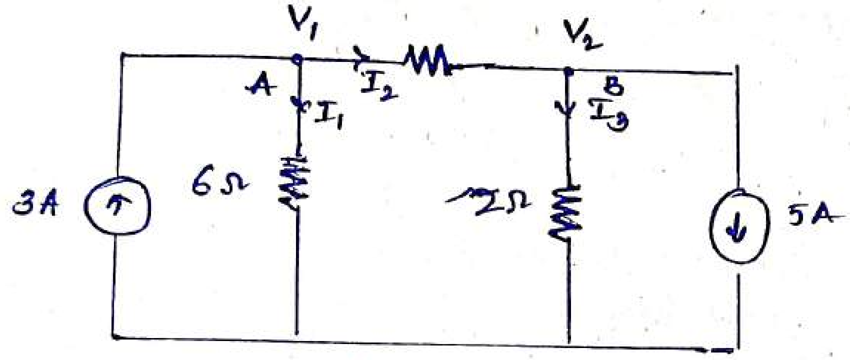
$$= 38.75 - 35$$

$$\boxed{V_{AB} = 3.75 V}$$

2. For the circuit given in below figure find the branch currents I_1, I_2, I_3 & node voltages V_1, V_2.



Sol:-



Apply kcl at A.

$-I_1 - I_2 + 3 = 0.$

$I_1 + I_2 = +3$

$$\frac{V_1 - 3}{6} + \frac{V_2 - 3}{3} = 3.$$

$$\frac{V_1 - 3 + 2V_2 - 6}{6} = 3$$

$$V_1 - 3 + 2V_2 - 6 = 18.$$

$$V_1 + 2V_2 - 9 - 18 = 0$$

$$V_1 - 2V_2 - 27 = 0. \rightarrow (1).$$

Apply kcl at B

$$I_2 - I_3 - 5 = 0$$

$$I_2 - I_3 = 5.$$

$$\frac{V_2 - 5}{3} - \frac{V_2 - 5}{2} = 5.$$

$$\frac{2v_1 - 10 - 3v_2 - 15}{6} = 5$$

$$2v_1 - 3v_2 - 25 = 30.$$

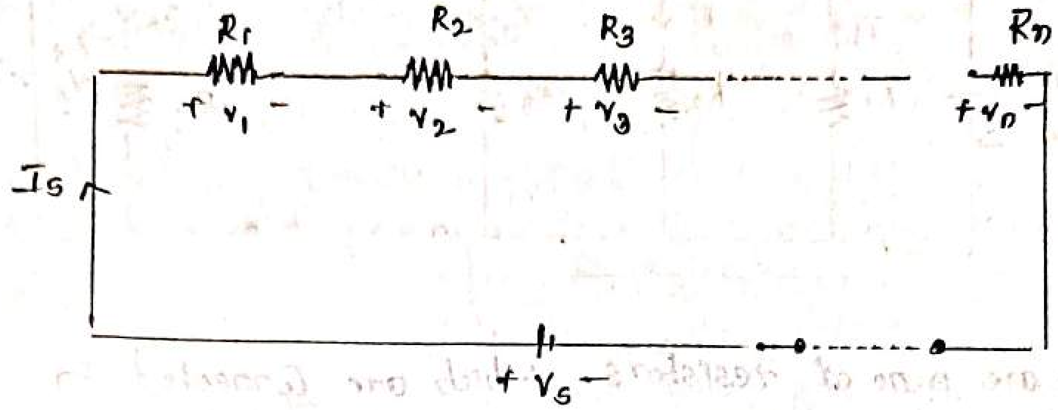
$$2v_1 - 3v_2 = 55 \quad - \quad 2.$$

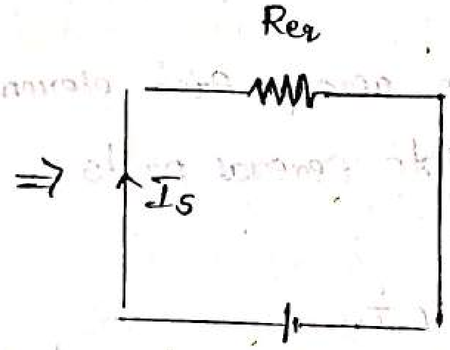
$v_1 = 4.$

$v_2 =$

Resistive Circuits

i) Series Connection :





There are n number of resistors which are connected in Series. in Series Connection the current flowing through each element is same.

Apply KVL in the above circuit.

According to ohm's law. $\boxed{V_S = I_S R_{eq}}$

$$-V_S + V_1 + V_2 + V_3 + \cdots + V_n = 0.$$

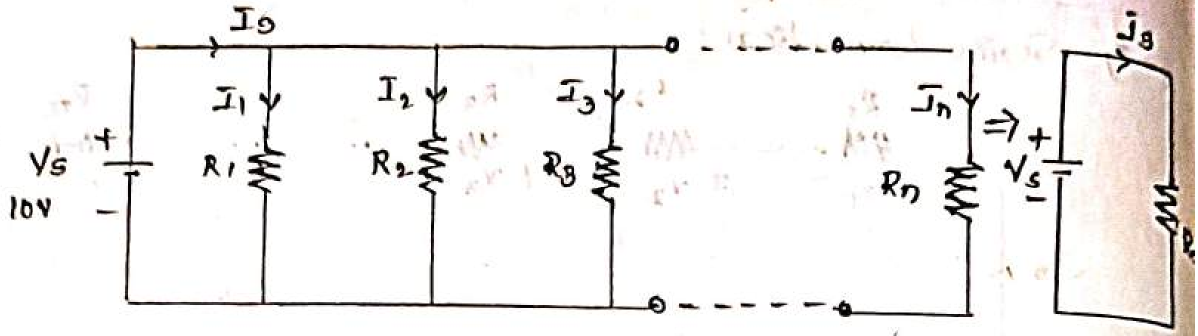
$$V_S = V_1 + V_2 + V_3 + \cdots + V_n$$

$$I_S R_{eq} = I_S R_1 + I_S R_2 + I_S R_3 + \cdots I_S R_n$$

$$\cancel{I_S} R_{eq} = \cancel{I_S} \left[R_1 + R_2 + R_3 + \cdots + R_n\right]$$

$$\boxed{R_{eq} = R_1 + R_2 + R_3 + \cdots + R_n}$$

2. Parallel Connection:



There are n no of resistors which are Connected in Parallel.

In parallel Connection voltage across each element is same but Current will divide into several parts.

Apply KCL at node one (1).

$$I_S = I_1 + I_2 + I_3 + \cdots + I_n$$

$$\frac{V_S}{R_{equ}} = \frac{V_S}{R_1} + \frac{V_S}{R_2} + \frac{V_S}{R_3} + \cdots + \frac{V_S}{R_n}$$

$$\frac{V_S}{R_{eq}} = V_S\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}\right)$$

$$\frac{1}{R_{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots \frac{1}{R_n}\right)$$

if $n = 2$

then

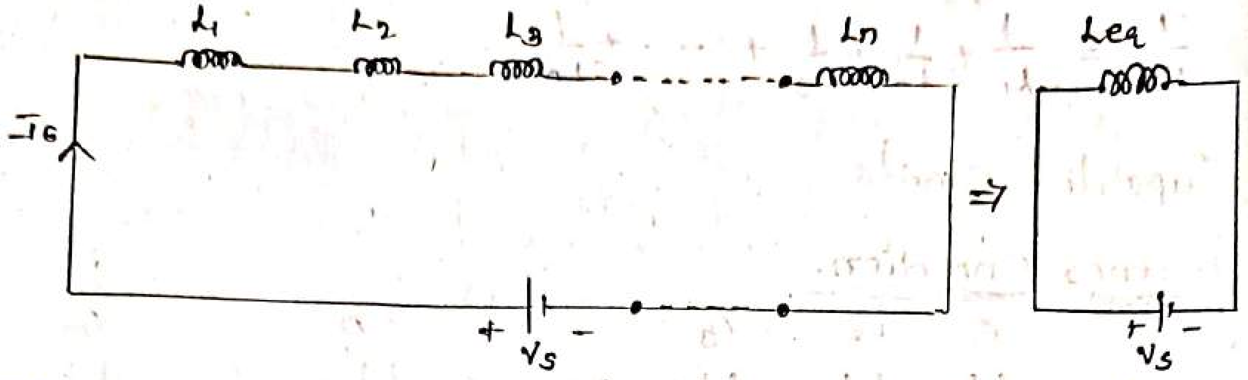
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$= \frac{R_1 + R_2}{R_1 R_2}$$

$$\boxed{R_{eq} = \frac{R_1 R_2}{R_1 + R_2}}$$

Inductive Circuits

1. Series Connection



There are n number of inductors which are connected in series.

Apply KVL

$$-V_s + V_1 + V_2 + V_3 + \cdots + V_n = 0$$

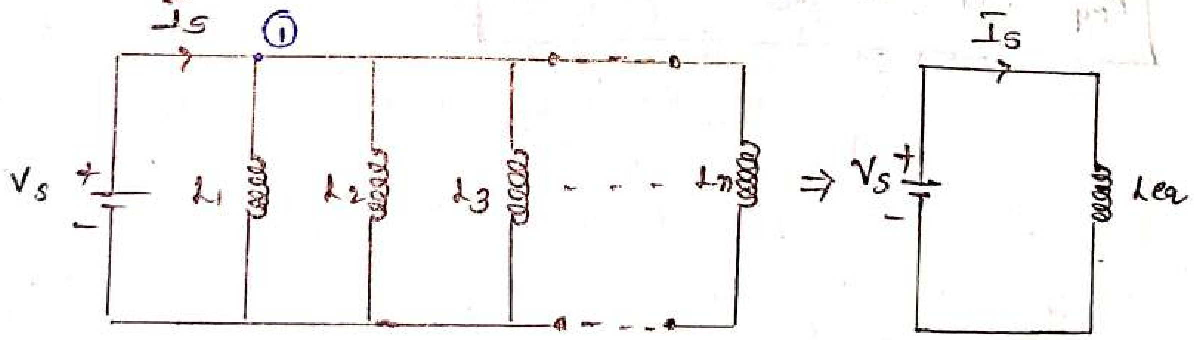
$$V_s = V_1 + V_2 + V_3 + \cdots + V_n$$

$$L_{eq}\frac{dI_s}{dt} = L_1\frac{dI_s}{dt} + L_2\frac{dI_s}{dt} + L_3\frac{dI_s}{dt} + \cdots L_n\frac{dI_s}{dt}$$

$$L_{eq}\cancel{\frac{dI_s}{dt}} = \cancel{\frac{dI_s}{dt}}\left(L_1 + L_2 + L_3 + \cdots + L_n\right)$$

$$L_{eq} = \left(L_1 + L_2 + L_3 + L_4 + \cdots + L_n\right).$$

2. Parallel Connection



There are n number of inductors which are connected in Parallel.

Apply KCL at node 1

$$I_s = I_1 + I_2 + I_3 + \cdots + I_n$$

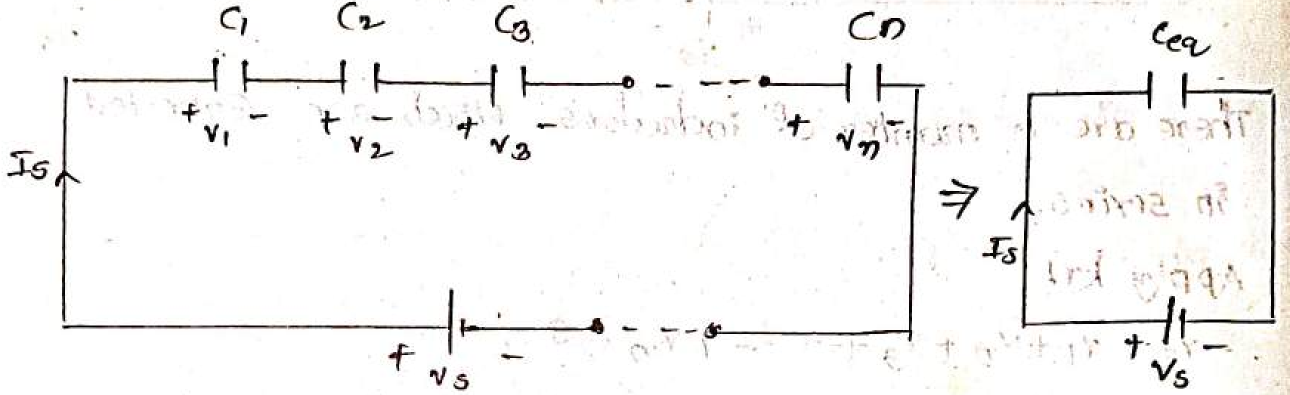
$$\frac{1}{L_{eq}}\int v_s\,dt = \frac{1}{L_1}\int v_s\,dt + \frac{1}{L_2}\int v_s\,dt + \cdots + \frac{1}{L_n}\int v_s\,dt.$$

$$\frac{1}{L_{eq}}\int v_s\,dt = \int v_s\,dt\left(\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{L_3}+\cdots+\frac{1}{L_n}\right)$$

$$\frac{1}{L_{eq}} = \frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{L_3}+\cdots+\frac{1}{L_n}.$$

Capacitive Circuits:

1. Series Connection.



Apply kvl

$$-v_s + v_1 + v_2 + v_3 + \cdots + v_n = 0$$

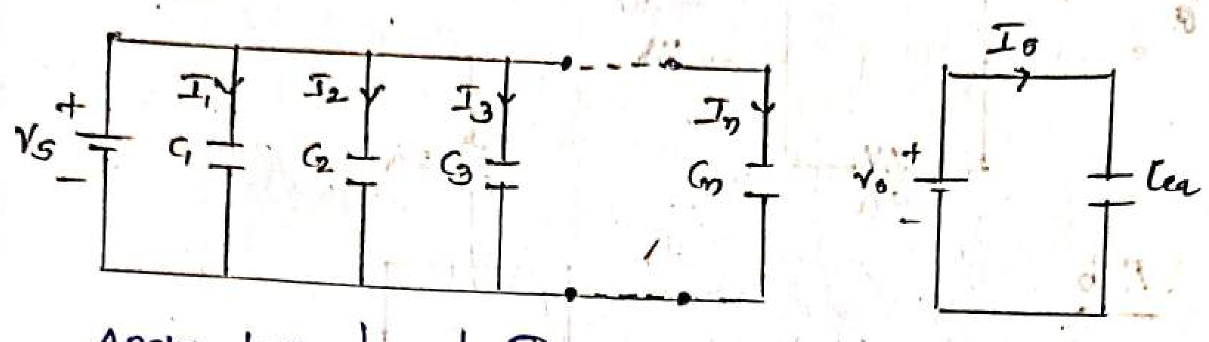
$$v_s = v_1 + v_2 + v_3 + \cdots + v_n$$

$$\frac{1}{C_{eq}}\int I_c\,dt = \frac{1}{C_1}\int I_c\,dt + \frac{1}{C_2}\int I_c\,dt + \frac{1}{C_3}\int I_c\,dt + \cdots + \frac{1}{C_n}\int I_c\,dt$$

$$\frac{1}{C_{eq}}\int I_c\,dt = \int I_c\,dt\left(\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}+\cdots+\frac{1}{C_n}\right)$$

$$\boxed{\frac{1}{C_{eq}} = \frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}+\cdots+\frac{1}{C_n}}$$

2. Parallel Connection.



Apply KCL at node ①.

$$I_S = I_1 + I_2 + I_3 + \cdots + I_n$$

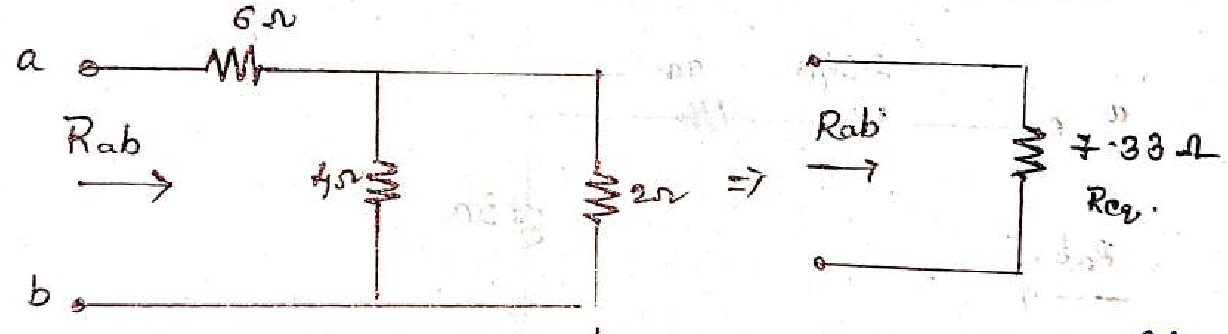
$$C_{eq}\frac{dv_c}{dt} = C_1\frac{dv_c}{dt} + C_2\frac{dv_c}{dt} + C_3\frac{dv_c}{dt} + \cdots + C_n\frac{dv_c}{dt}$$

$$C_{eq}\cancel{\frac{dv_c}{dt}} = \cancel{\frac{dv_c}{dt}}\left(C_1 + C_2 + C_3 + \cdots + C_n\right)$$

$$\boxed{C_{eq} = C_1 + C_2 + C_3 + \cdots + C_n}$$

1. Find Equivalent resistance between A, B terminals in the below circuit.

A.



Now the resistors 4Ω and 2Ω are in parallel so that

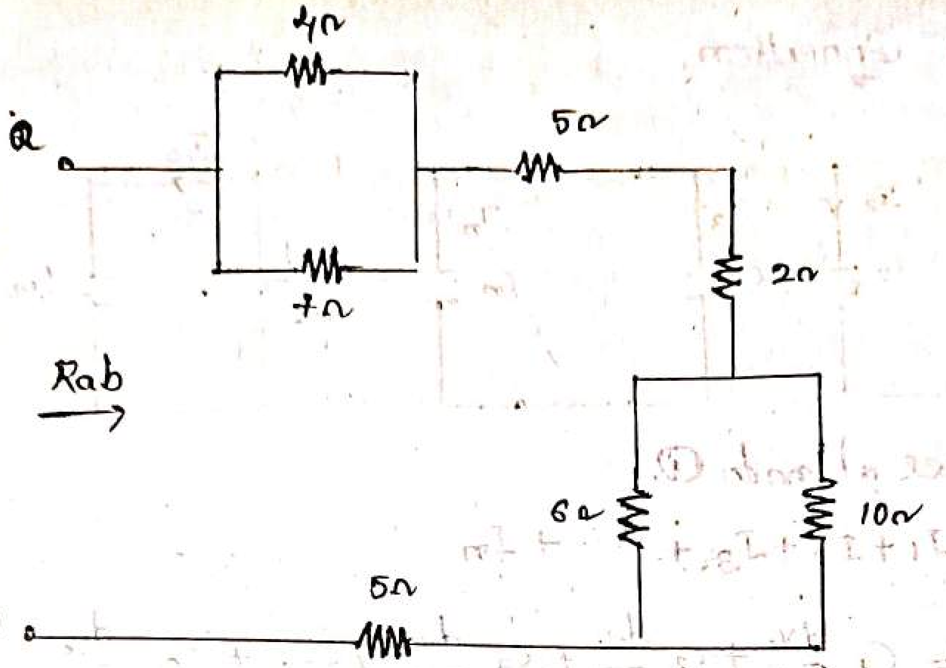
$$R_{equ_1} = \frac{(4)(2)}{4+2} = \frac{\cancel{8}\,4}{\cancel{6}\,3} = \frac{4}{3} = 1.333$$

Now the resistors 6Ω and 1.33 are in series so that

$$R_{equ} = 6\Omega + 1.33\Omega = 7.33\Omega$$

$$\boxed{R_{ab} = 7.33}$$

2.

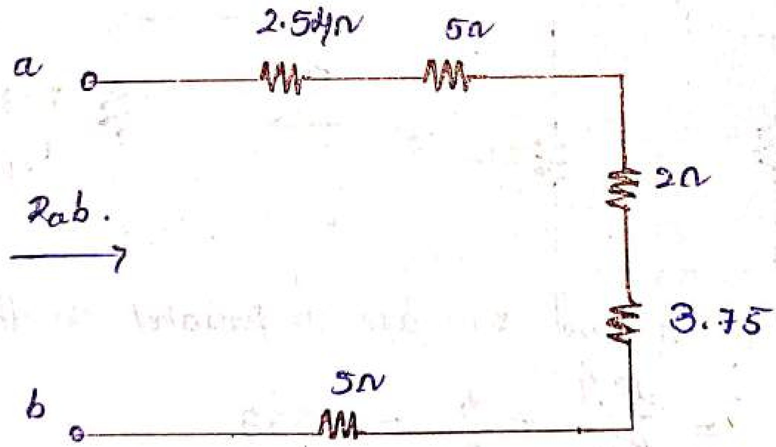


Both 4Ω and 7Ω are connected in parallel as well as 6Ω and 10Ω are connected in parallel i.e.,

$$R_{equ\,1} = \frac{(7)(4)}{7+4} = \frac{28}{11} = 2.54\,\Omega$$

$$R_{equ\,2} = \frac{(6)(10)}{16} = \frac{60}{16} = 3.75\,\Omega$$

Therefore the circuit becomes as,

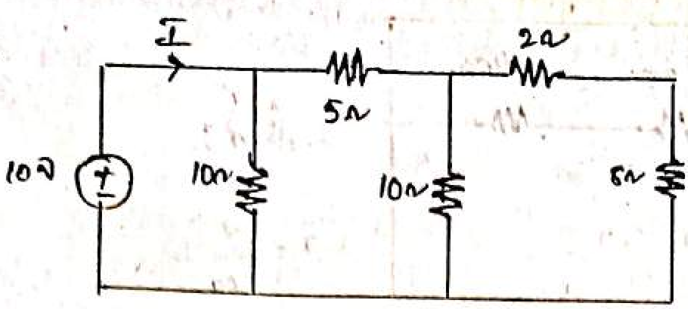


In the above circuit all the resistors are connected in series i.e total resistance betwe A - B is

$$R_{ab} = 2.54 + 5 + 2 + 3.75 + 5$$

$$\boxed{R_{ab} = 18.29\,\Omega}$$

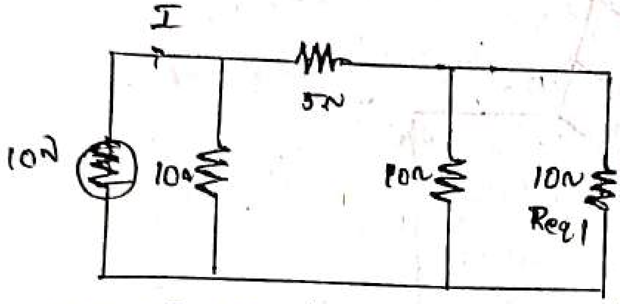
③



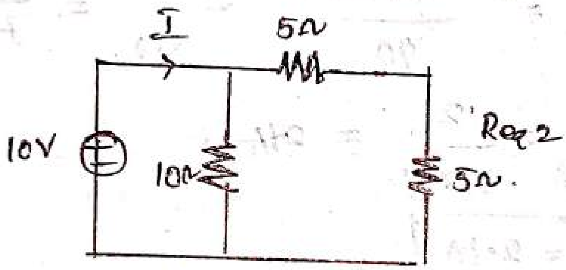
find I

Now 2Ω and 8Ω and are in series so that Requ 1 is

$2\Omega + 8\Omega = 10\Omega$.

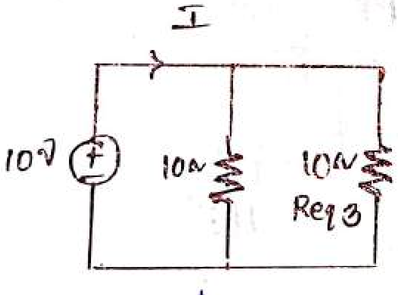


Now the resistances 10Ω and Requ 1 are in Parallel. so that Requ 2 $= \dfrac{10 \times 10}{10+10} = \dfrac{10\cancel{0}}{2\cancel{0}} = 5\Omega$



Now the resistance 5Ω and 5Ω are in series so that

Req 3 $= 5+5 = 10\Omega$



Now 10Ω and Req3 are in parallel. So that

$$Req = \frac{10 \times 10}{10+10} = \frac{10\cancel{0}}{2\cancel{0}} = 5\Omega.$$

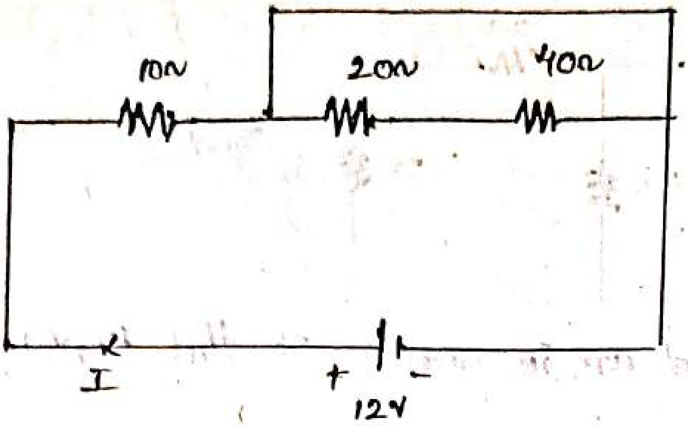
$\boxed{Req = 5}$

We know that $V = 10$, $I = ?$, $R = 5$

$$I = \frac{V}{R} \Rightarrow \frac{10}{5} = 2A$$

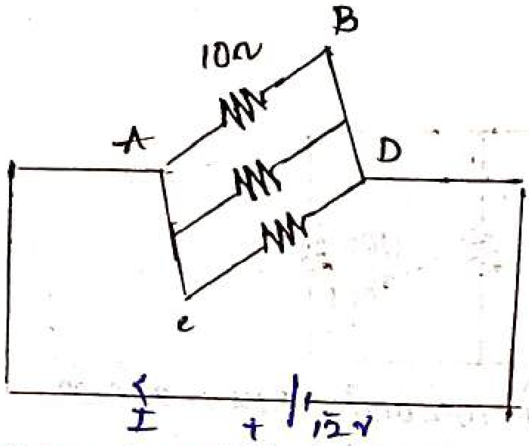
$\boxed{I = 2A}$

4.



find I

$$I = \frac{V}{R_{eq}}$$



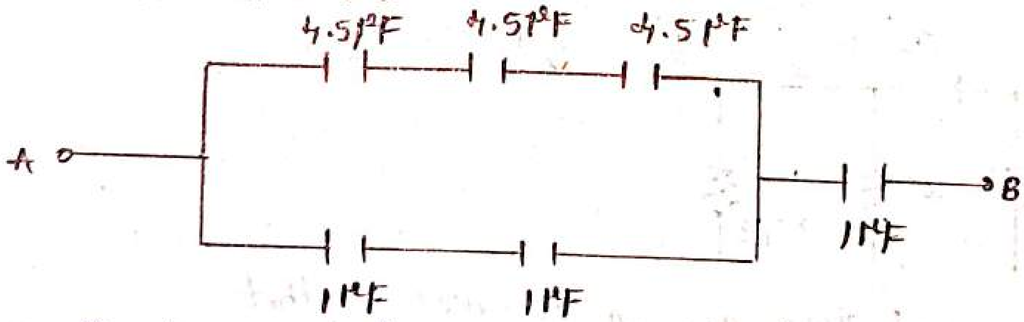
$$\frac{1}{R_{eq}} = \frac{1}{10} + \frac{1}{20} + \frac{1}{40}$$

$$= \frac{4+2+1}{40} = \frac{7}{40} = \frac{40}{7} = 5.714Ω$$

$$I = \frac{V}{R_{eq}} = \frac{12}{5.714} = 2.1A$$

$$\boxed{I = 2.1A}$$

5. find out the Equivalent Capacitance between the terminals a and b



In the given Circuit the Capacitors are in series

$$\therefore \frac{1}{C_{eq}} = \frac{1}{4.5μF} + \frac{1}{4.5μF} + \frac{1}{4.5μF}$$

$$= \frac{3}{4.5μF}$$

$$C_{eq1} = \frac{4.5}{3} μF$$

$\therefore C_{eq1} = 1.5\,\mu F$

In the given circuit two capacitors are in series.

$$\frac{1}{C_{eq2}} = \frac{1}{1\,\mu F} + \frac{1}{1\,\mu F}$$

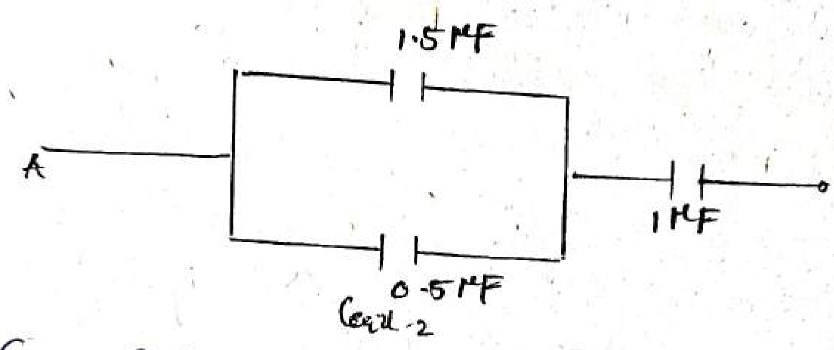
$$= \frac{1+1}{1\,\mu F}$$

$$\frac{1}{C_{eq2}} = 2\,\mu F$$

$$C_{eq2} = \frac{1}{2}\,\mu F$$

$\therefore C_{eq2} = 0.5\,\mu F$

So, the given circuit becomes.



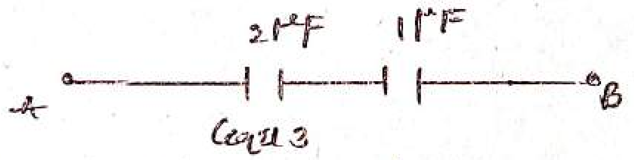
C_{eq1} & C_{eq2} are in parallel.

$$C_{eq3} = C_{eq1} + C_{eq2}$$

$$= 1.5 + 0.5$$

$\therefore C_{eq3} = 2\,\mu F$

Again the circuit becomes



C_{eq3} and $1\,\mu C$ capacitors are in series

$$\therefore \frac{1}{C_{eq}} = \frac{1}{2} + \frac{1}{1}$$

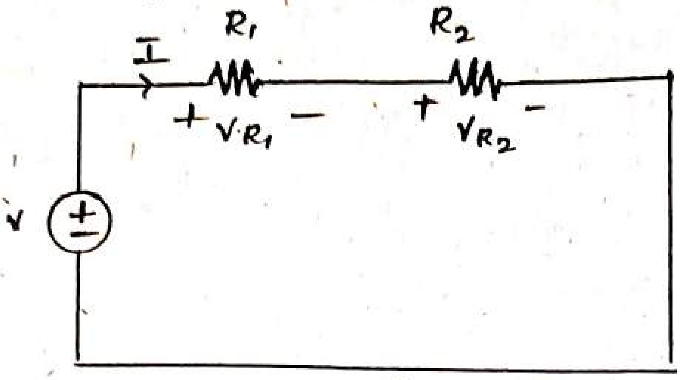
$$= \frac{1+2}{2}$$

$$\frac{1}{C_{eq}} = 3/2$$

$$C_{eq} = 2/3$$

$$C_{eq} = 0.67\,\mu F.$$

Voltage Division in series circuit of resistors.



* Consider a series circuit of R and R_2 of two resistors connected to source of v volts.
* As two resistors are connected in series, the current flowing through both the resistors is same i.e., I

Apply KVL in the circuit.

$$V = V_{R_1} + V_{R_2}$$

$$V = IR_1 + IR_2$$

$$I(R_1 + R_2) = V$$

$$I = \frac{V}{R_1 + R_2}$$

From the ckt,

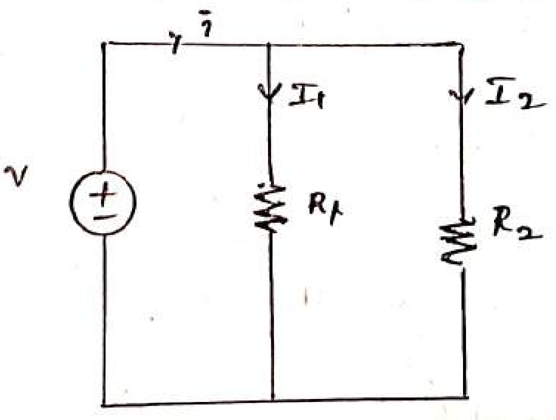
$$V_{R_1} = IR_1 = \left(\frac{V}{R_1 + R_2}\right) R_1.$$

$$V_{R_1} = V\left(\frac{R_1}{R_1 + R_2}\right)$$

$$V_{R_1} = IR_2 = \left(\frac{V}{R_1 + R_2}\right) R_2$$

$$V_{R_2} = V\left(\frac{R_2}{R_1 + R_2}\right).$$

Current Division in parallel circuit of resistor.



Consider a parallel circuit of resistors R_1 & R_2 connected across a source of v volts.

The total current drawn from the source is I_T.

Apply KVL in the circuit.

$$I_T = I_1 + I_2$$

from the circuit (ckt)

$$V = I_1 R_1 = I_2 R_2$$

$$I_1 = \frac{R_1}{R_2} I_2$$

$\therefore$ Total current $I_T = I_1 + I_2$

$$= \frac{R_2}{R_1} I_2 + I_2$$

$$I_T = I_2\left(\frac{R_2}{R_1}+1\right)$$

$$= \left(\frac{R_1+R_2}{R_1}\right) I_2$$

Total current, $I_2 = \frac{I_T R_1}{R_1+R_2}$

$$I_2 = \left(\frac{R_1}{R_1+R_2}\right) I_T.$$

from total current $I_T = I_T - I_2$

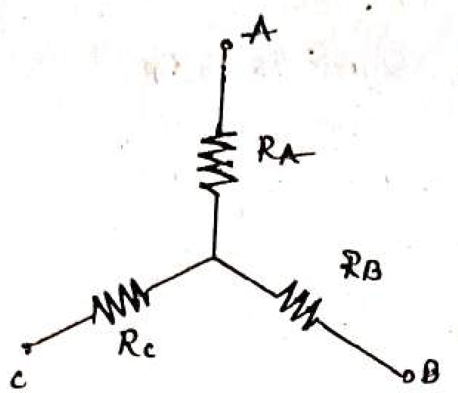
$$I_1 = I_T - \left(\frac{R_1}{R_1+R_2}\right) I_T.$$

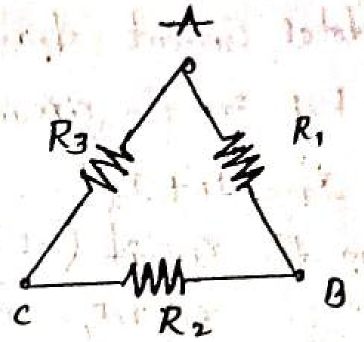
$$= I_T\left(1 - \frac{R}{R_1+R_2}\right)$$

$$= \frac{R_2 I_T}{R_1+R_2}$$

$$\therefore I_1 = \left(\frac{R_2}{R_1+R_2}\right) I_T.$$

Delta - star Conversion:





The Effective resistances between each 2 two nodes of above star Connection is

$$R_{AB} = R_A + R_B \rightarrow (1)$$

$$R_{BC} = R_B + R_C \rightarrow (2)$$

$$R_{CA} = R_C + R_A \rightarrow (3)$$

The effective resistances between each 2 terminals above Delta Connection are

$$R_{AB} = \frac{R_1(R_2+R_3)}{R_1+R_2+R_3} \quad - (4)$$

$$R_{BC} = \frac{R_2(R_1+R_3)}{R_2+R_1+R_3} \quad - (5)$$

$$R_{CA} = \frac{R_3(R_1+R_2)}{R_1+R_2+R_3} \quad - (6)$$

From the above all Equations we Can write

$$R_A + R_B = \frac{R_1(R_2+R_3)}{R_1+R_2+R_3} \rightarrow (7)$$

$$R_B + R_C = \frac{R_2(R_1+R_3)}{R_1+R_2+R_3} \rightarrow (8)$$

$$R_C + R_A = \frac{R_3(R_1+R_2)}{R_1+R_2+R_3} \rightarrow (9)$$

Add Equ (1), (2) & (3).

$$R_{AB} + R_{BC} + R_{CA} = R_A + R_B + R_B + R_C + R_C + R_A$$

$$R_{AB}+R_{BC}+R_{CA} = 2(R_A+R_B+R_C) \longrightarrow (10).$$

Add Equ (4) (5) & (6) we get.

$$R_{AB}+R_{BC}+R_{CA} = \frac{R_1(R_2+R_3)}{R_1+R_2+R_3} + \frac{R_2(R_1+R_3)}{R_1+R_2+R_3} + \frac{R_3(R_1+R_2)}{R_1+R_2+R_3}$$

$$R_{AB}+R_{BC}+R_{CA} = \frac{R_1R_2+R_1R_3+R_1R_2+R_2R_3+R_1R_3+R_2R_3}{R_1+R_2+R_3}.$$

$$R_{AB}+R_{BC}+R_{CA} = \frac{2(R_1R_2+R_2R_3+R_3R_1)}{R_1+R_2+R_3} \longrightarrow (11).$$

From Equ (10) & (11) we can write.

$$2(R_A+R_B+R_C) = \frac{2(R_1R_2+R_2R_3+R_3R_1)}{R_1+R_2+R_3}$$

$$R_A+R_B+R_C = \frac{R_1R_2+R_2R_3+R_3R_1}{R_1+R_2+R_3} \quad -(12)$$

Subtract Equ (8) from Equ (12) we get.

$$(R_A+R_B+R_C)-(R_B+R_C) = \frac{R_1R_2+R_2R_3+R_3R_1}{R_1+R_2+R_3} - \frac{R_1R_2+R_2R_3}{R_1+R_2+R_3}$$

$$R_A = \frac{\cancel{R_1R_2}+\cancel{R_2R_3}+R_3R_1-\cancel{R_1R_2}-\cancel{R_2R_3}}{R_1+R_2+R_3}$$

$$\boxed{R_A = \frac{R_1R_3}{R_1+R_2+R_3}} \quad -(13)$$

Similarly

$$\boxed{R_B = \frac{R_1R_2}{R_1+R_2+R_3}} \quad -(14).$$

Similarly

$$\boxed{R_C = \frac{R_2R_3}{R_1+R_2+R_3}} \quad -(15).$$

Subtract Eqn 9 from 12 we get.

$$(R_A+R_B+R_C)-(R_C+R_A) = \frac{R_1R_2+R_2R_3+R_3R_1}{R_1+R_2+R_3} - \frac{R_3(R_1+R_2)}{R_1+R_2+R_3}$$

Star-Delta Conversion

From Eqn (13), (14) & (15) multiply and add each two Equations we get

$$R_AR_B+R_BR_C+R_CR_A =$$

$$\frac{R_1^2R_2R_3}{(R_1+R_2+R_3)^2}+\frac{R_1R_2^2R_3}{(R_1+R_2+R_3)^2}+\frac{R_1R_2R_3^2}{(R_1+R_2+R_3)^2}.$$

$$= \frac{R_1R_2R_3\cancel{(R_1+R_2+R_3)}}{(R_1+R_2+R_3)^{\cancel{2}}} = \frac{R_1R_2R_3}{R_1+R_2+R_3} \rightarrow (16).$$

Divide Eqn (16) with Eqn (15) we get.

$$\frac{\left(\frac{R_1R_2R_3}{R_1+R_2+R_3}\right)}{\left(\frac{R_2R_3}{R_1+R_2+R_3}\right)} = \frac{R_AR_B+R_BR_C+R_CR_A}{R_C}$$

$$\boxed{R_1 = \frac{R_AR_B+R_BR_C+R_CR_A}{R_C}} \rightarrow (17)$$

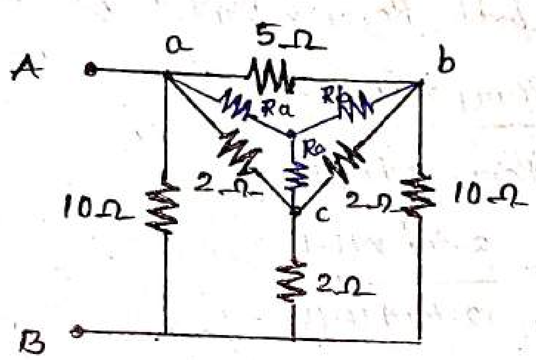
Divide Eqn (16) with (14) we get

$$\frac{\left(\frac{R_1R_2R_3}{R_1+R_2+R_3}\right)}{\frac{R_1R_2}{R_1+R_2+R_3}} = \frac{R_AR_B+R_BR_C+R_CR_A}{R_B}$$

$$R_2 = \frac{R_AR_B+R_BR_C+R_CR_A}{R_A}$$

$$R_3 = \frac{R_AR_B+R_BR_C+R_CR_A}{R_B}.$$

1. Find the Equal resistance between



Identify Delta Connection in the given ckt.

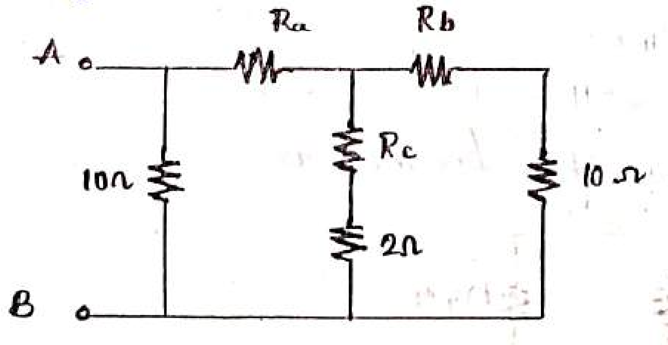
i.e, delta(abc) Convert into star.

$$\therefore R_a = \frac{5\times 2}{5+2+2} = 1.11\,\Omega$$

$$R_b = \frac{5\times 2}{5+2+2} = 1.11\,\Omega$$

$$R_c = \frac{2\times 2}{5+2+2} =$$

The given Circuit becomes as.



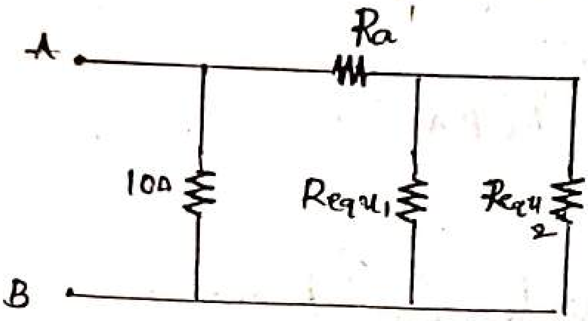
In the above ckt both (R_c, 2Ω) & (R_b, 10Ω) are in Series

$\therefore R_{equ1} = R_c + 2$ $\qquad R_{equ2} = R_b + 10$

$= 0.44 + 2$ $\qquad = 1.11 + 10$

$R_{equ1} = 2.44\,\Omega$ $\qquad = 11.11\,\Omega$

The the ckt becomes as.



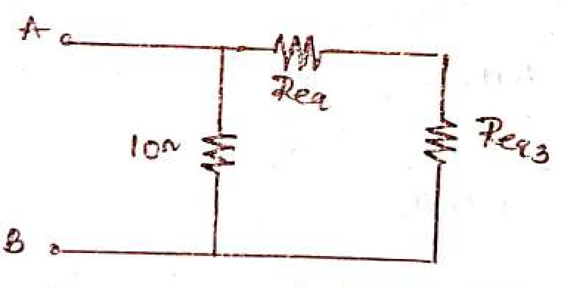
In the above ckt both R_{equ1} & R_{equ2} are in parallel.

$$R_{equ3} = \frac{R_{equ1} \times R_{equ2}}{R_{equ1} + R_{equ2}}$$

$$R_{equ3} = \frac{2.44 \times 11.11}{2.44 + 11.11}$$

$$R_{equ3} = 2\,\Omega$$

Therefore the ckt becomes as.



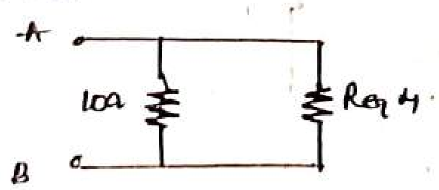
In the above cirk, both R_{eq} & R_{eq3} are in series

$R_{equ4} = R_{eq} + R_{eq3}$

$= 1.11 + 2$

$R_{equ4} = 3.11\,\Omega$

Therefore the above ckt becomes as.

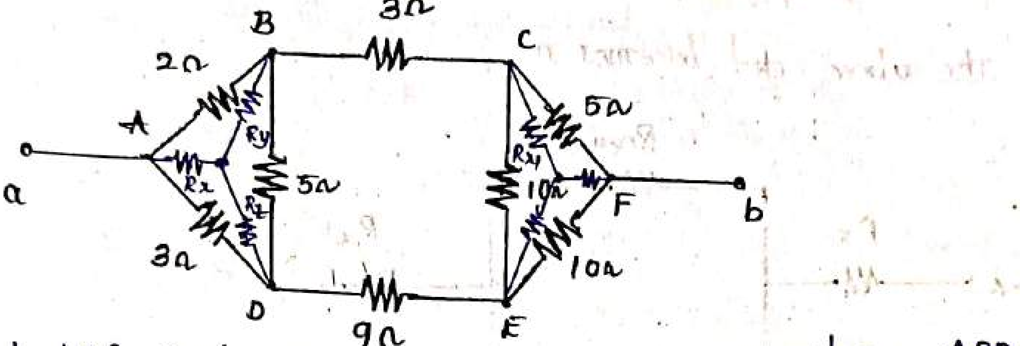


In the above ckt both 10Ω & R_{eq} are in parallel.

$$R_{AB} = \frac{10 \times R_{eq4}}{10 + R_{eq4}} = \frac{10 \times 3.11}{10 + 3.11}$$

$$\boxed{R_{AB} = 2.372\,\Omega}$$

(2) -



Identitfy Delta Connection in the given circuit is Δ^{ABD}

Convert into Star.

$$R_x = \frac{2 \times 3}{2+3+5} = 0.6\,\Omega$$

$$R_y = \frac{2 \times 5}{2+3+5} = 1\,\Omega$$

$$R_z = \frac{3 \times 5}{2+3+5} = 1.5\,\Omega$$

Identify another Delta Connection given in circuit Δ^{CFE}

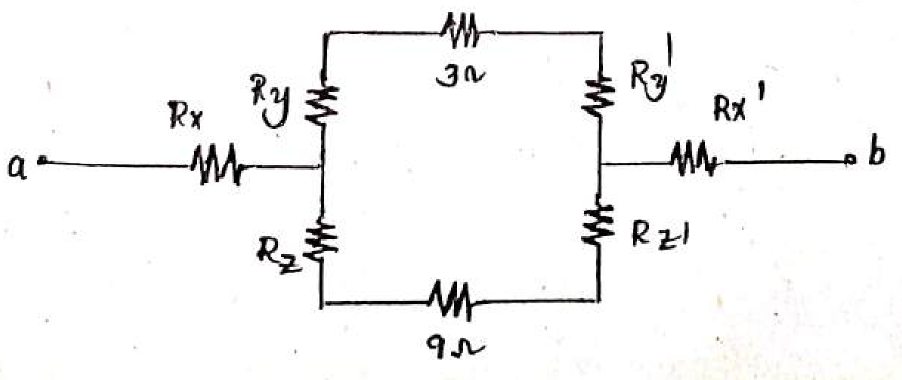
Converts into Star.

$$R_x' = \frac{5 \times 10}{5+10+10} = 2\,\Omega$$

$$R_y' = \frac{5 \times 10}{5+10+10} = 2\,\Omega$$

$$R_z' = \frac{10 \times 10}{5+10+10} = 4\,\Omega$$

Therefore the given ckt becomes as.



In the above ckt both (R_y , 3Ω, R_y') & (R_3 , 9Ω , R_z') are in series therefore.

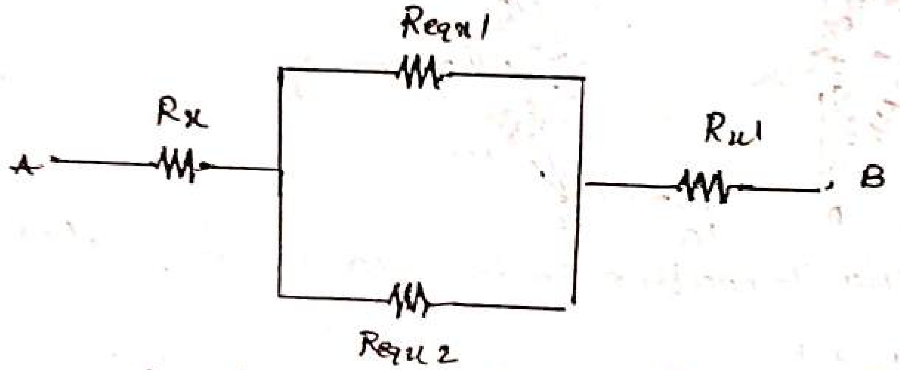
$$R_{eqx1} = 1 + 3 + 2$$

$$= 6$$

$$R_{eq2} = 1.5 + 9 + 4$$

$$= 14.5 .$$

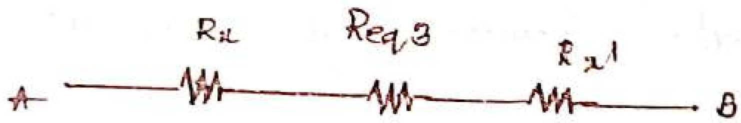
The above ckt becomes as.



The above ckt Reqx1 & Req2 are in parallel.

$$R_{eqx3} = \frac{R_{eqx1} \times R_{eq2}}{R_{eqx1} + R_{eqx2}}$$

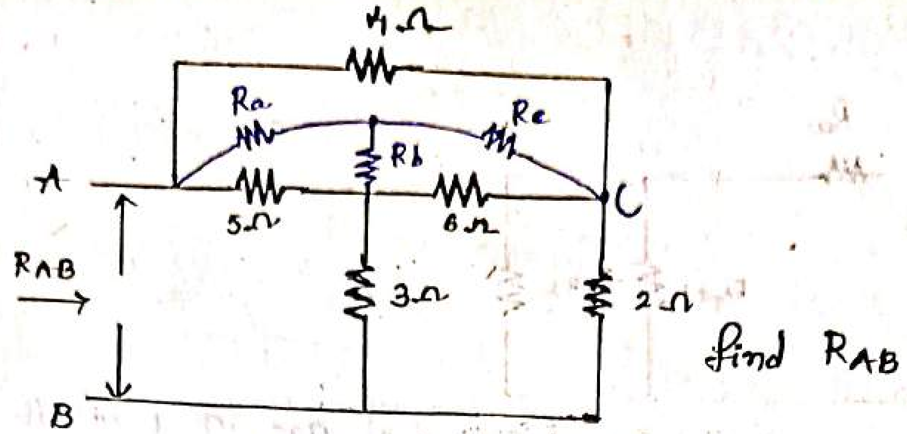
$$= \frac{6 \times 14.5}{6 + 14.5} = 4.243.$$



$$R_{eqx} = 0.6 + 4.24 + 2$$

$$\boxed{R_{eqx} = 6.84}$$

(3).



find R_{AB}

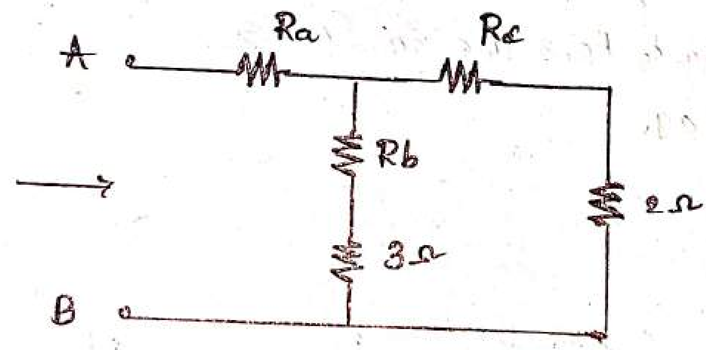
In the given ckt Identify Delta Connection Δ^{abc} Convert into star.

$$R_a = \frac{4 \times 5}{4+5+6} = 1.33\Omega$$

$$R_b = \frac{5 \times 6}{4+5+6} = 2\Omega$$

$$R_c = \frac{4 \times 6}{4+5+6} = 1.6\Omega$$

therefore the circuit becomes as.

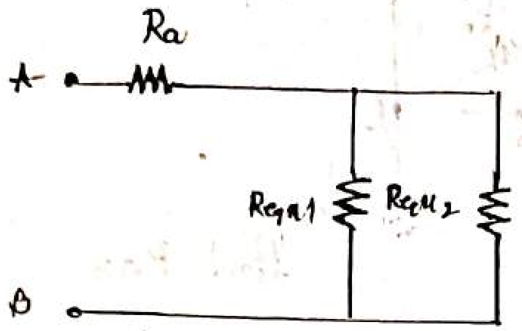


Both $(R_b, 3\Omega)$ & $(R_c$ & $2\Omega)$ are in Series.

$$R_{equ\,1} = R_b + 3\Omega$$
$$= 2+3 = 5\Omega$$

$$R_{equ\,2} = R_c + 2\Omega$$
$$= 1.6 + 2\Omega$$
$$= 3.6\Omega.$$

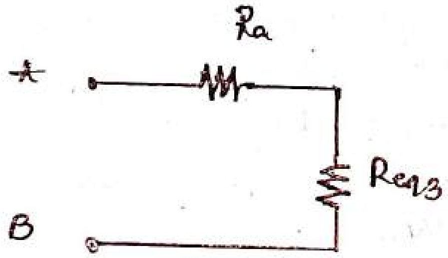
therefore the circuit becomes as.



Now Both Requ1 & Requ 2 are in parallel.

$$Requ\,3 = \frac{5 \times 3.6}{5+3.6}$$

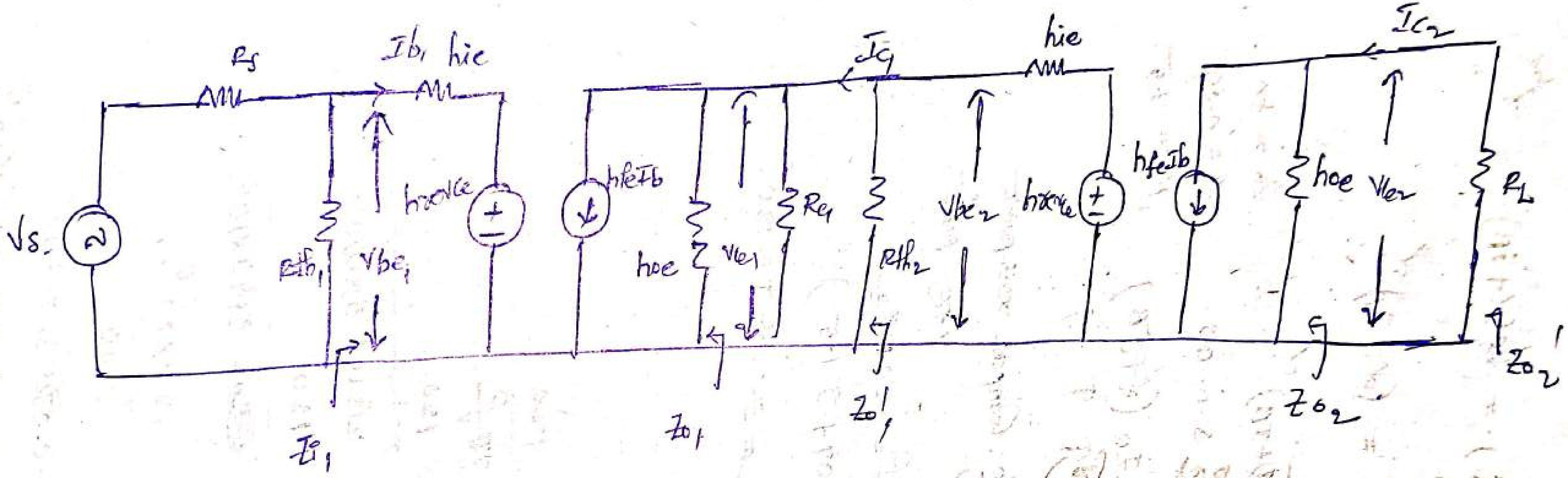
$$= \frac{18}{8.6} = 2.09$$



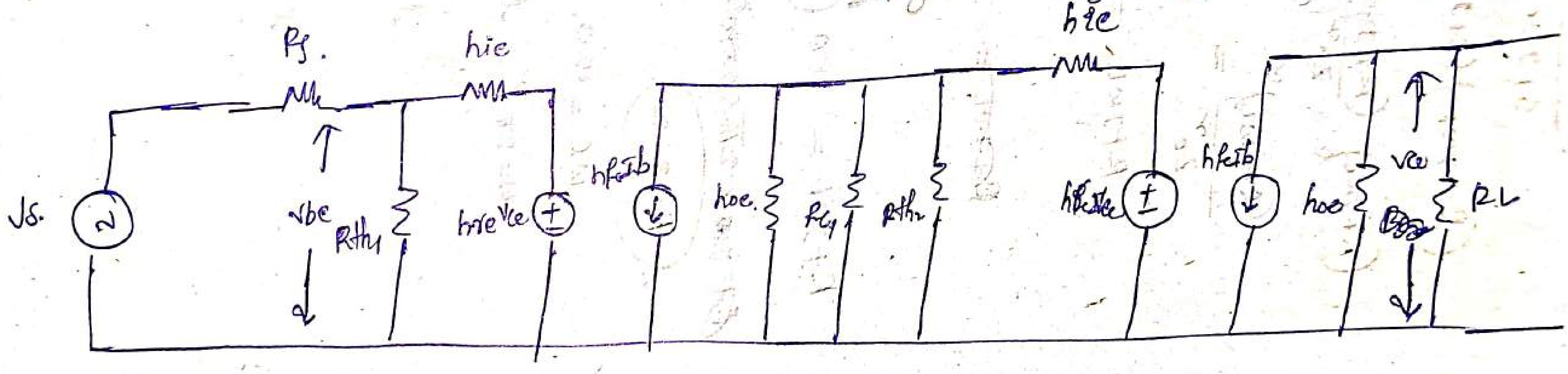
Now Both Ra & Req3 are in Series

$$Req = 1.33 + 2.09$$

$$= 3.42\ \Omega$$



Rs
Ib1 hie
Vs.
Rth1
Vbe1
hre Vce
hfeIb
hoe
Vce1
Re1
Ic1
Rth2
Vbe2
hie
hre Vce
hfeIb
hoe
Vce2
Ic2
RL
Zi1
Zo1
Zo'1
Zo2
Zo'2



Rs.
hie
Vs.
vbe
Rth1
hre Vce
hfeIb
hoe.
Re1
Rth2
hie
hfeIb
hoe
vce
RL

1. We know that $Z_0 = \sqrt{\frac{R+j\omega l}{G+j\omega C}}$

$$\frac{R+j\omega l}{G+j\omega C} = Z_0^2 \Rightarrow G+j\omega C = \frac{R+j\omega L}{Z_0^2}$$

$$(R+j\omega L)(G+j\omega C) = (\alpha+i\beta)^2 = \gamma^2$$

$$(R+j\omega L)\frac{(R+j\omega L)}{Z_0^2} = (\alpha+i\beta)^2$$

$$\frac{(R+j\omega L)^2}{Z_0^2} = (\alpha+j\beta)^2$$

S.O.B.S.

$Z_0 = 80\,\Omega$

$\alpha = 0.04$

$\beta = 1.5$

$$R+j\omega L = Z_0(\alpha+j\beta)$$

~~R+jωL =~~ Z_0. Compare real part

imag part

$f = 500\,MHz$

$$R+j\omega l = 80(0.04+j\,1.5)$$

$\cancel{R+j\omega \times 2\pi \times 5\times10^8 L} = 3.2 + j\times 120$

$R = 3.2$

$\omega l = 120$

$\omega = 2\pi f \quad l = \frac{120}{2\pi f}$

$$l = \frac{120}{2(3.14) \cdot 500\times10^6}$$

$$l = \frac{120}{3140\times10^6}$$

$$= \frac{120\times10^{-6}}{3140}$$

$= \cancel{0.03822\times10^{-6}}$

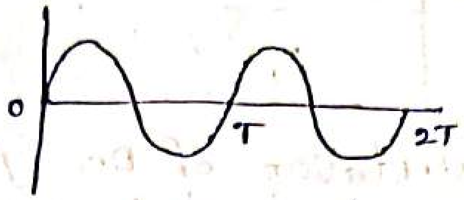
$= 0.038216\times10^{-6}$

Unit - II

Single phase AC circuits

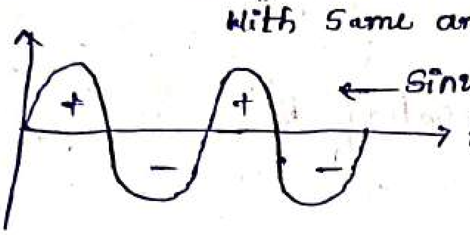
Periodic signals: A same signal occurs continuously at regular intervals of time.

$$x(t) = x(t+T)$$

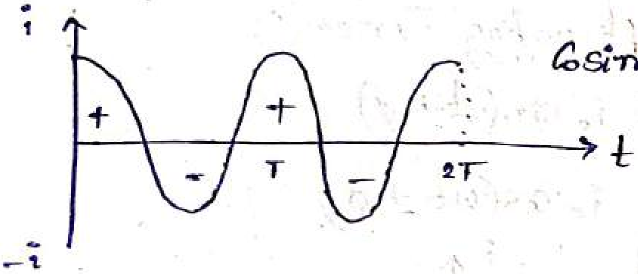


Alternating Current: Current which periodically reversing its direction with same amplitude is called AC current.

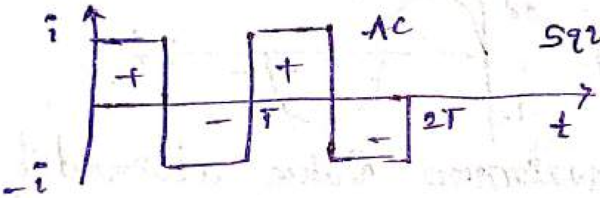
Ex:



← sinusoidal wave
sine wave

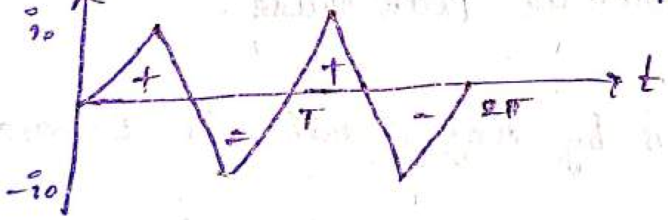


Cosine wave

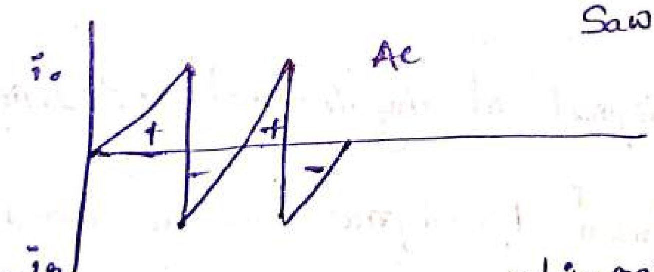


AC

Square wave.



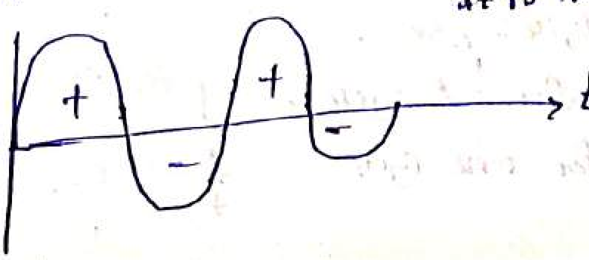
Triangular wave.

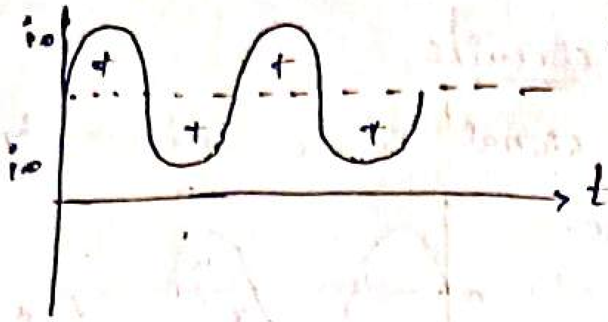


AC

Sawtooth wave

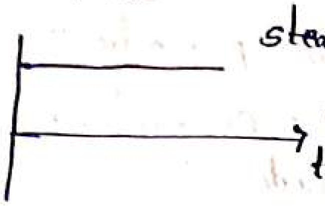
It is not an AC



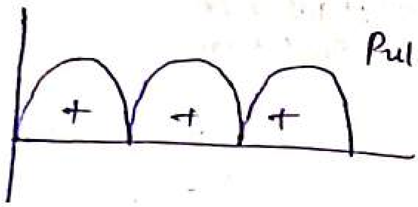


It is not AC

Representation of Dc signals.



steady signal
steady Dc signal.



Pulsating Dc

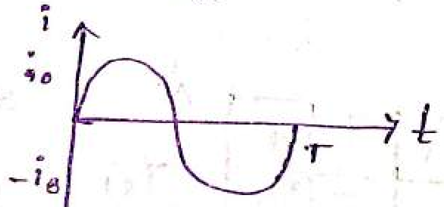
Representation of Alternating Currents:

for sine wave, $i = i_0 \sin(\omega t \pm \phi)$

for cos wave, $i = i_0 \cos(\omega t \pm \phi)$

~~present~~

$i = i_0 \sin \omega t$



Ampiitude: It is the maximum value atained by any signal. It is also called as peak value.

Time period: (T)

It is the time taken by any signal in seconds to complete one cycle.

Instataneous value:

the value of any signal at any instant of time "t"

frequency:

the no of cycles produced by signal in one second.

Units: Hertz (HZ). (or) cycles/sec.

The time taken for f cycles = 1 sec)

time taken for one cycle = $\frac{1}{f}$ sec.

$$\boxed{T = \frac{1}{f}}$$

Angular frequency: (ω) : The no of cycles that can be converted into radians in one second.

$$\boxed{\omega = \frac{2\pi}{T} = 2\pi f} \text{ rad/sec}$$

Average value of alternating current: The amount of charge transferred in a given interval of time is called average value.

→ Let sinusoidal wave of alternating current.

$i = i_0 \sin \omega t$ $\qquad\qquad i = \frac{dq}{dt}$

$i_{avg} = \frac{q_{avg}}{t}$ (Average charge transferred in given interval of time). $\qquad q = \int i\,dt$.

$$i_{avg} = \frac{\int_0^t i\,dt}{\int_0^t dt}.$$

for half cycle $(0 - T/2)$

$$i_{avg} = \frac{\int_0^{T/2} i_0 \sin\omega t\,dt}{\int_0^{T/2} dt}$$

$$i_{avg} = \frac{\int_0^{T/2} i_0 \sin\omega t\,dt}{\left(\frac{T}{2}\right)}$$

$$i_{avg} = \frac{2i_0}{T}\int_0^{T/2} \sin\omega t\,dt$$

$$i_{avg} = \frac{2i_0}{T}\left[\frac{-\cos\omega t}{\omega}\right]_0^{T/2}$$

$$i_{avg} = \frac{2i_0}{T\omega}\left[-\cos\omega\left(\frac{T}{2}\right) + \cos(0)\right]$$

$$i_{avg} = \frac{2i_0}{\omega T}\left[-\cos\left(\frac{2\pi}{T}\right)\left(\frac{T}{2}\right) + 1\right]$$

$$= \frac{2i_0}{\omega T}[1-\cos\pi]$$

$$= \frac{2i_0}{\left(\frac{2\pi}{T}\right)T}[1-(-1)]$$

$$i_{avg} = \frac{i}{\pi}[1+1]$$

$$\boxed{i_{avg} = \frac{2i_0}{\pi}}$$

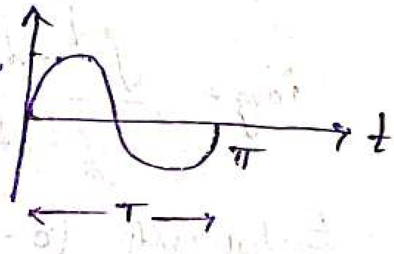
$$\boxed{i_{avg} = \frac{2I_m}{\pi}}$$

If the signal is voltage signal $\boxed{V_{avg} = \frac{2V_m}{\pi}}$

Rms value of sinusoidal Alternating Current.

$i = i_0 \sin\omega t$

$$\therefore I_{rms} = \sqrt{\frac{1}{T}\int_0^T i^2\,dt}$$



where, $T = \pi$

$$I_{rms} = \sqrt{\frac{1}{\pi}\int_0^{\pi} i_0^2 \sin^2\omega t\,dt}$$

$$= \sqrt{\frac{i_0^2}{\pi}\int_0^{\pi}\left(\frac{1-\cos 2\omega t}{2}\right)dt}.$$

$$= \sqrt{\frac{i_0^2}{2\pi}\int_0^{\pi}(1-\cos 2\omega t)\,dt}$$

$$= \sqrt{\frac{i_0^2}{2\pi}\left[t - \frac{\sin 2\omega t}{2\omega}\right]_0^{\pi}}$$

$$I_{rms} = \sqrt{\frac{i_0^2}{2\pi}\left(\pi - \frac{\sin 2\omega\pi}{2\omega} - 0 + \frac{\sin 0}{2\omega}\right)}$$

$$I_{rms} = \sqrt{\frac{i_0^2}{2\pi}\left(\pi - \frac{\sin 2\left(\frac{2\pi}{T}\right)T}{2\left(\frac{2\pi}{T}\right)} - 0 + 0\right.}$$

$(\sin 4\pi = 0)$

$\sin n\pi = 0$

$$i_{rms} = \sqrt{\frac{i_0^2}{2T}(T)}$$

$$\boxed{i_{rms} = \frac{i_0}{\sqrt{2}}}$$

$$\boxed{\therefore I_{rms} = \frac{I_m}{\sqrt{2}}}$$

$$\boxed{V_{rms} = \frac{V_m}{\sqrt{2}}}$$

Form factor:

It is defined as the ratio of rms value to the average value.

$$\text{Form factor} = \frac{\text{Rms Value}}{\text{Average Values}}$$

→ For a sinusoidal Alternating Current form factor

$$= \frac{I_{Rms}}{I_{avg}} = \frac{(i_0/\sqrt{2})}{\left(\frac{2i_0}{\pi}\right)} = \frac{1}{\sqrt{2}} \times \frac{\pi}{2}$$

$$\boxed{\text{Form factor} = 1.11}$$

Peak factor: It is defined as the ratio of peak value (maximum value) to the rms value is called peak factor.

$$\text{Peak factor} = \frac{\text{Peak (Max) value}}{\text{Rms Value}}$$

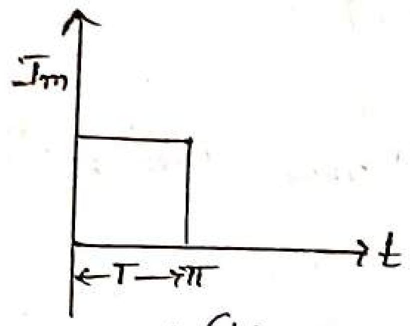
→ For a sinusodial A-C

$$\text{Peak factor} = \frac{i_0}{\left(\frac{i_0}{\sqrt{2}}\right)} = \sqrt{2}.$$

$$\boxed{\text{Peak factor} = 1.414}$$

Problems

find Form factor and peak factor for a given Square wave shown in the below figure.



$i(t) = I_m$

Average value:

$$I_{avg} = \frac{1}{T}\int_0^T i(t)\,dt$$

from the given wave form $T = \pi$.

$$\therefore I_{avg} = \frac{1}{\pi}\int_0^\pi I_m\,dt.$$

$$= \frac{I_m}{\pi}(t)_0^\pi$$

$$= \frac{I_m}{\cancel{\pi}}(\pi - 0)$$

$$\boxed{I_{avg} = I_m}$$

Rms value:

$$I_{rms} = \sqrt{\frac{1}{T}\int_0^t i(t)^2\,dt}$$

$$= \sqrt{\frac{1}{\pi}\int_0^\pi I_m^2\,dt}$$

$$= \sqrt{\frac{I_m^2}{\pi}(t)_0^\pi}$$

$$= \sqrt{\frac{I_m^2}{\cancel{\pi}}(\cancel{\pi} - 0)}$$

$$\boxed{I_{rms} = I_m}$$

$$\text{Form factor} = \frac{\text{Rms Value}}{\text{Avg Value}}$$

$$= \frac{I_{rms}}{I_m}$$

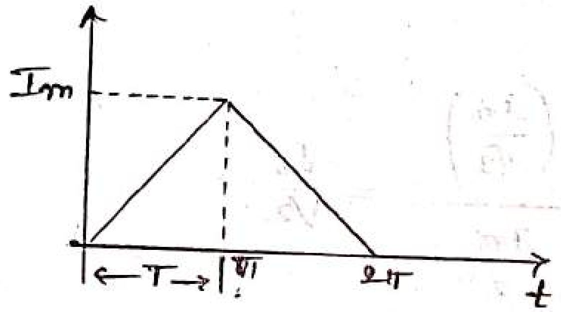
$$\text{form factor} = 1$$

$$\text{Peak factor} = \frac{\text{Max Value}}{\text{Rms Value}}$$

$$= \frac{I_m}{I_m}$$

$$\boxed{\text{Peak factor} = 1}$$

2. Find form factor and peak factor for given triangular wave form.



$(0,0)$ (π, I_m)

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1)$$

$$y - 0 = \frac{I_m - 0}{\pi - 0}(x - 0)$$

$$\boxed{i(t) = \frac{I_m}{\pi} t}$$

Average Value:-

$$I_{avg} = \frac{1}{T}\int_0^T i(t)\,dt$$

From the given form $T = \pi$

$$I_{avg} = \frac{1}{\pi}\int_0^{\pi} \frac{I_m}{\pi} t\,dt$$

$$= \frac{1}{\pi^2}\int_0^{\pi} I_m (t)\,dt.$$

$$\frac{I_m}{\pi^2}\int_0^{\pi}(t)\,dt.$$

$$= \frac{I_m}{\pi^2}\left(\frac{t^2}{2}\right)_0^{\pi}$$

$$= \frac{I_m}{\cancel{\pi^2}}\frac{\cancel{\pi^2}}{2} = \boxed{\frac{I_m}{2} = I_{avg}}$$

Peak value

$$I_{rms} = \frac{1}{T}\int_0^{\pi} i(t)^2\, dt$$

$$= \sqrt{\frac{1}{\pi}\int_0^{\pi}\left(\frac{I_m}{\pi}\right)^2 t^2\, dt}$$

$$= \sqrt{\frac{I_m^2}{\pi^3}\int_0^{\pi} t^2\, dt}$$

$$= \sqrt{\frac{I_m^2}{\pi^3}\left[\frac{t^3}{3}\right]_0^{\pi}}$$

$$= \sqrt{\frac{I_m^2}{\cancel{\pi^3}}\left(\frac{\cancel{\pi^3}}{3} - 0\right)}$$

$$= \sqrt{\frac{I_m^2}{3}}$$

$$\therefore I_{rms} = \frac{I_m}{\sqrt{3}}$$

3. Form factor: $\dfrac{\text{Rms Value}}{\text{Avg value}} = \dfrac{\left(\frac{I_m}{\sqrt{3}}\right)}{\frac{I_m}{2}} = \dfrac{\sqrt{3}}{1/2}$

$$= 2/\sqrt{3} = 1.154.$$

$$\boxed{\text{Form factor} = 1.154}$$

$$\text{Peak Factor} = \frac{\text{Peak value}}{\text{Rms value}} = \frac{\cancel{I_m}}{\frac{\cancel{I_m}}{\sqrt{3}}}$$

$$= \sqrt{3}$$

$$\boxed{\therefore \text{Peak factor} = 1.73}$$

1. If an alternating quantity is represented by $i(t) = 20 \sin(314t + 45°)$. find frequency, timeperiod, avg Value, Rms Value, form factor, Peak factor, instateneous Value at $t = 2m$ sec.

A Compare the given value with standard value, $v(t) = V_m \sin(\omega t + \phi)$

1. Angular frequency $\omega = 314$

$$2\pi f = 314$$

$$f = \frac{314}{2\pi}$$

$$\boxed{f = 50}$$

2. Time period $= \frac{1}{f} = \frac{1}{50} = 0.02 sec.$

3. Avg Value $= \frac{2V_m}{\pi}$

$$= \frac{2(20)}{3.14} = \frac{40}{3.14} = 12.73.$$

4. Rms Value =

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{20}{\sqrt{2}} = 14.142.$$

5. form factor

$$\frac{\text{Rms Value}}{\text{avg value}} = \frac{14.142}{12.73} = 1.110.$$

6. peak factor

$$\frac{\text{Peak Value}}{\text{avg Value}} = \frac{20}{12.73} = 1.4104.$$

7. instateneous value at $t = 2ms$.

3. If an alternating quantity is represented by a Sinusoid $v(t) = 200 \sin(314t + 30°)$ volts. Find its frequency, Time period, average, Rms Value, form factor, peak factor, and finally instataneous value at $t = 2$ milli sec

A. Given

$$v(t) = 200 \sin(314t + 30°)$$

Compare the given value with standard value

$$v(t) = V_m \sin(\omega t + \phi) \qquad V_m = 200$$
$$\omega = 314$$

i) Angular frequency.

$$\omega = 2\pi f$$
$$314 = 2\pi f$$
$$\frac{314}{2\pi} = f$$
$$f = \frac{314}{2(3.14)} = \frac{314}{6.28}$$
$$\boxed{\therefore f = 50 \text{ HZ}}$$

ii) Time period:

$$T = \frac{1}{f}$$
$$= \frac{1}{50}$$
$$= 0.02 \text{ Sec}$$

$\therefore T = 20$ milli Sec / 0.02 sec

iii) Average Value:

$$V_{avg} = \frac{2V_m}{\pi}$$
$$= \frac{2(200)}{3.14}$$
$$= \frac{400}{3.14}$$
$$\therefore V_{avg} = 127.38$$

iv) V_{rms} :

$$\frac{V_m}{\sqrt{2}} = \frac{200}{\sqrt{2}} = 141.421$$

v) form factor :

$$\frac{\text{Rms value}}{\text{Average value}} = \frac{141.421}{127.38} = 1.11$$

$\therefore$ form factor $= 1.11$.

vi) Peak factor :

$$\frac{\text{Peak Value}}{\text{Rms Value}} = \frac{200}{141.42} = 1.41$$

$\therefore$ Peak Value $= 1.41$

vii) Instataneous Value at $t = 2$ milli sec

$$v(t) = 200 \sin\left[(314 \times 2 \times 10^{-3}) + 30^\circ\right]$$

$$= 200 \sin\left[(314 \times 0.002) + 30^\circ\right]$$

$$= 200 \sin(0.628 + 30^\circ)$$

$$= 200 \sin\left[\left(0.628 \times \frac{180}{\pi}\right) + 30^\circ\right]$$

$$= 200 \sin\left[113.04/3.14 + 30^\circ\right]$$

$$= 200 \sin\left[36^\circ + 30^\circ\right]$$

$$= 200 \sin 66^\circ$$

$$= 200(0.913)$$

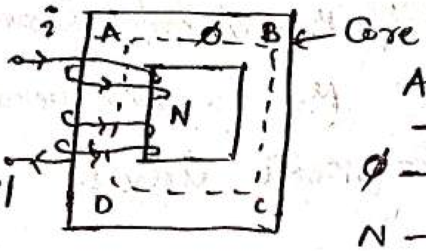
$$= 182.7$$

$\sqrt{(A)^2}$

Unit -3

Magnetic circuits

1. Magnetic circuit is defined as a closed path followed by the magnetic flux.
2. Magnetic circuit consists of a core of magnetic materials having high permeability like Iron, soft steel etc.. It is because these materials offer very small opposition to the flow of magnetic flux.



ABCDA $\rightarrow$ closed path for flux

$\phi \rightarrow$ magnetic ~~field~~ flux

N $\rightarrow$ Numbers of turns of coil.

i $\rightarrow$ Current through the coil.

Magnetic flux:-

This is produced due to the flow of current in a wire or coil and it is denoted by ϕ.

units: Webers (Wb)

magnetic flux density:

magnetic flux density is defined as magnetic flux per unit area. It is denoted by B

$$\text{magnetic flux density} = \frac{\text{magnetic flux}}{\text{Area}} \quad \frac{Wb}{m^2}$$

units: $\frac{Wb}{m^2}$

Magneto motive force (MMF)

This is the ability of the coil to produce magnetic flux.

$$\boxed{MMF = N \times i}$$

Where N = Number of turns of coil.

i = Current through the coil.

units = Ampere turns (At).

Reluctance ! It is defined as the oposition offered by magnetic material to the flow of magnetic flux.

$$MMF = \phi \times S$$

$$S = \frac{MMF}{\phi} \quad At/wb$$

units = At/wt. $\boxed{S = \frac{l}{A\mu}}$ where l = length of the Coil.

μ = permeability.

$\mu = \mu_0 \mu_r$.

μ_0 = absolute permeability.

μ_r = relative permeability.

Analogy between Electric & magnetic circuits :

Electric circuit	Magnetic circuit.
$E \rightarrow$ Electromotive force (EMF) $\rightarrow$ volts	$F \rightarrow$ Magnetomotive force (MMF) $\rightarrow$ (A-T).
$I \rightarrow$ Electric Current $\rightarrow$ (Amp)	$\phi \rightarrow$ Magnetic flux $\rightarrow$ (wb).
$R \rightarrow$ Resistance $\rightarrow$ (Ω)	$R \rightarrow$ Reluctance (m/A-T) $\left(\frac{A-T}{wb}\right)$
$\sigma \rightarrow$ Conductivity $(\Omega - m)^{-1}$	$\mu \rightarrow$ permeability $\left[\frac{(A-T)}{m^2}\right]$
ohms law: $\boxed{E = IR}$	ohms law = $\boxed{F = \phi R}$
$R = \frac{\rho l}{A} = \frac{l}{\sigma A}$	$\boxed{R = \frac{l}{\mu A}}$
$R_{se} = R_1 + R_2 + R_3 + \dots + R_n$	$R_{se} = R_1 + R_2 + \dots + R_n$
$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$.

Electric circuit

Current: the rate of flow of electrons in a conductor is called as electric current (I)
units : ampere (A)

Resistance: oppose the flow of electrons in a conductor
units: ohm (Ω)

EMF : Electromotive force
It is a driving force for current
Measured in volts.

Conductance $= \frac{1}{\text{Resistance}}$
$= \frac{1}{R}$ (Ω-m).

ohms law = $\boxed{E = IR}$

$$R = \frac{\rho l}{A} = \frac{l}{\sigma A}$$

Magnetic circuit

Magnetic flux: the total numbers of lines of forces in a magnetic field (ϕ).
units : Weber (wb).

Reluctance: oppose the flow of magnetic flux in magnetic materials. (AT/wb).

MMF : magnetomotive force
It is a driving force for flux.
measured in ampere-turn (AT).

Permeance: $\frac{1}{\text{Reluctance}}$
$= \frac{1}{S}$.

ohms law:

$$F = \phi R$$

$$R = \frac{l}{\mu A}$$

Faraday's law of Electro-magnetic induction.

Faraday's First law.

Whenever a Conductor is placed in a varying magnetic field an electromotive force (emf) is induced across the Conductor.

Faraday's second law.

The induced Emf in a Conductor is directly proportional to rate of change of flux linkages.

$$e \propto \frac{d\varphi}{dt}$$

$\varphi \rightarrow$ flux linkages.

$$e = N\frac{d\varphi}{dt}.$$

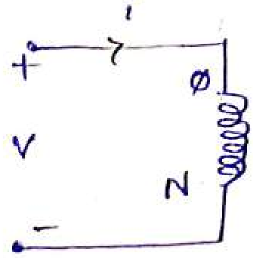
-ve sign indicates lentz's law.

$N \rightarrow$ No of turns of coil.

$\frac{d\phi}{dt} \rightarrow$ rate of change of flux.

Self inductance

The inductance offered by a coil whenever the flux produced by the coil links with the same Coil itself.



$$\phi \propto i$$

$$\phi = ki \Rightarrow k = \frac{\phi}{i} \rightarrow ①$$

Differentiate w.r.t on both sides w.r.t "t"

$$\frac{d\varphi}{dt} = k\frac{di}{dt} \rightarrow ②$$

We know that

$$V = -N\frac{d\phi}{dt}$$

$$V = -Nk\frac{di}{dt} \quad (\text{from Eqn } ②)$$

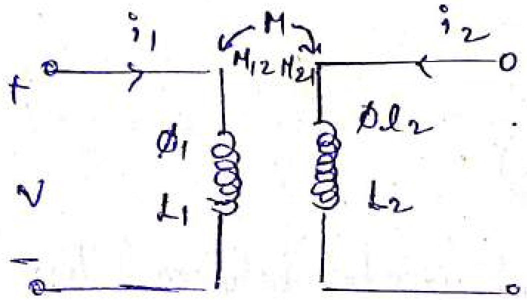
$V = -N \frac{\phi}{i} \frac{di}{dt}$ from Equ ①.

$$\boxed{V = L \frac{di}{dt}}$$

Where $\boxed{L = \frac{N\phi}{i}}$ Henry.

Mutual inductance

In inductance offered by the coil whenever the flux produced by one coil links with another coil.



$\phi_2 \propto i_1$

$\phi_2 = k i_1 \Rightarrow k = \frac{\phi_2}{i_1} \rightarrow ①$

Differentiate on both sides w.r.t "t".

$\frac{d\phi_2}{dt} = k \frac{di_1}{dt} \rightarrow ②$.

We know that,

$V = -N \frac{d\phi}{dt}$

$V = -Nk \frac{di_1}{dt}$ (from Equ ②)

$V = -N \frac{\phi_2}{i_1} \frac{di_1}{dt}$ (from Equ ①)

$$\boxed{V = M_{12} \frac{di_1}{dt}}$$

where, $\boxed{M_{12} = \frac{N\phi_2}{i_1}}$

Similarly,

$\phi_1 \propto i_2$

$\phi_1 = k i_2 \Rightarrow k = \frac{\phi_1}{i_2} \rightarrow ③$

D.w.t "t" we get.

$$\frac{d\phi_1}{dt} = k\frac{di_2}{dt} \rightarrow (9).$$

w.r.t

$$V = -N\frac{d\phi}{dt}$$

$$V = -Nk\frac{di_2}{dt}$$

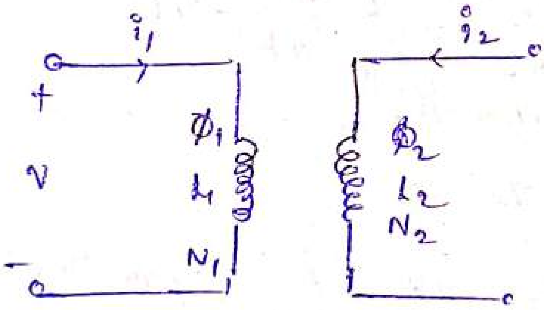
$$V = -N\frac{\phi_1}{i_2}\frac{di_2}{dt}$$

$$\boxed{V = M_{21}\frac{di_2}{dt}}$$

imp . where, $M_{21}' =$

Coefficients of Coupling (k)

The degree of Coupling that exists between two Coupled coils.



From the circuit,

$$L_1 = \frac{N_1\phi_1}{i_1} \; ; \; L_2 = \frac{N_2\phi_2}{i_2}.$$

$$M_{12} = k\frac{N_2\phi_1}{i_1} \; ; \; M_{21} = k\frac{N_1\phi_2}{i_2}$$

Assume, $M_{12} = M_{21} = m$

Multiplying both

$$M_{12} \times M_{21} = k\frac{N_2\phi_1}{i_1} \times k\frac{N_1\phi_2}{i_2}$$

$$M \times M = k^2\frac{N_1\phi_1}{i_1} \times \frac{N_2\phi_2}{i_2}.$$

$$M^2 = k^2 L_1 L_2$$

$$\boxed{M = k\sqrt{L_1 L_2}}$$

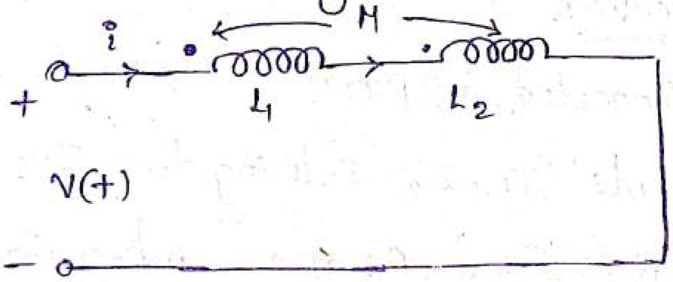
$$\boxed{k = \frac{M}{\sqrt{L_1 L_2}}}$$

Coupled circuits.

- Series Connected
  - Series aiding
  - Series opposing
- Parallel Connected.
  - Parallel aiding
  - Parallel opposing.

Series Aiding Connected:

If 2 coils Connected in series Such that the Current ~~flowing through 2 coils~~ at entering Both the coils at doted terminals is. is Called Series aiding Connection.



Apply KVL in the circuit,

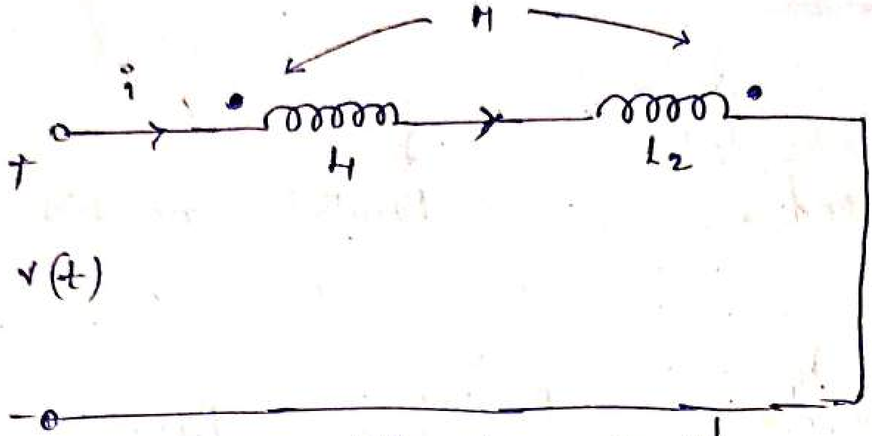
$$v(t) = L_1\frac{di}{dt} + M\frac{di}{dt} + L_2\frac{di}{dt} + M\frac{di}{dt}$$

$$L_{eq}\cancel{\frac{di}{dt}} = \cancel{\frac{di}{dt}}\left[L_1 + L_2 + 2M\right]$$

$$\boxed{L_{eq} = L_1 + L_2 + 2M}$$ Henry.

Series opposing Connect

If two Coils Connected in series in Such a way that the current entering ~~Both the coils~~ one coil at doted terminal another at undoted terminal is Called Series opposing Connect.



Apply KVL in the above circuit.

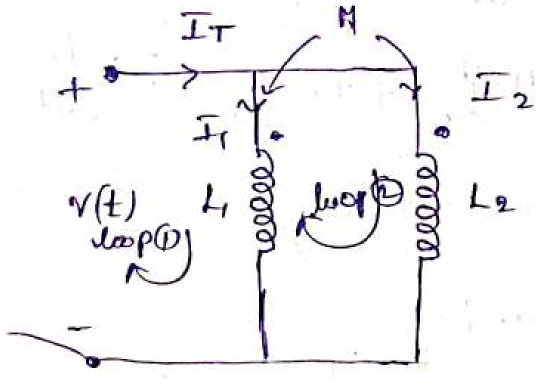
$$v(t) = L_1 \frac{di}{dt} - M \frac{di}{dt} + L_2 \frac{di}{dt} - M \frac{di}{dt}$$

$$L_{eq} \left(\frac{di}{dt}\right) = \left(\frac{\cancel{di}}{dt}\right) \left[L_1 + L_2 - 2m\right]$$

$$\boxed{L_{eq} = L_1 + L_2 - 2m} \text{ Henry.}$$

Parallel aiding Connection

If two Coils are Connected in parallel in such a way that the two Currents I_1, I_2 entering the Coils at doted terminals it is Called parallel aiding Connection



$V = IZ$

$Z \rightarrow$ impedence

$Z = R + jX$

L

Inductive Reactance, X_L

Capacitive Reactance, X_C

$X_L = \omega L$

$X_C = \frac{1}{\omega C}$

Apply KCL in the ckt,

$I_T = I_1 + I_2 \rightarrow (1)$

Apply KVL in loop (1).

$$-v(t) + J\omega L_1 I_1 + J\omega M I_2 = 0$$

$$v(t) = J\omega L_1 I_1 + J\omega M I_2 \rightarrow ①$$

Apply KVL in loop ②

$$-v(t) + J\omega L_2 I_2 + J\omega M I_1 = 0.$$

$$v(t) = J\omega M I_1 + J\omega L_2 I_2 \rightarrow ②$$

$$\begin{bmatrix} v(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} J\omega L_1 & J\omega M \\ J\omega M & J\omega L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

where,

$$\Delta = \begin{bmatrix} J\omega L_1 & J\omega M \\ J\omega M & J\omega L_2 \end{bmatrix}$$

$$\Delta_1 = \begin{bmatrix} v(t) & J\omega M \\ v(t) & J\omega L_2 \end{bmatrix}$$

$$\Delta_2 = \begin{bmatrix} J\omega L_1 & v(t) \\ J\omega M & v(t) \end{bmatrix}$$

$$|\Delta| = (J\omega L_1)(J\omega L_2) - (J\omega M)(J\omega M)$$

$$= J^2\omega^2 L_1 L_2 - J^2\omega^2 M^2.$$

$$= (-1)\omega^2 L_1 L_2 + \omega^2 M^2$$

$$|\Delta| = \omega^2 (M^2 - L_1 L_2)$$

$$|\Delta_1| = (v(t))(J\omega L_2) - (v(t))(J\omega m)$$

$$= v(t) J\omega (L_2 - M)$$

$$|\Delta_2| = v(t)(J\omega L_1) - v(t)(J\omega m)$$

$$= v(t) J\omega [L_1 - M].$$

$$I_1 = \frac{|\Delta_1|}{|\Delta|} = \frac{v(t) J\omega (L_2 - m)}{\omega^2 [M^2 - L_1 L_2]}$$

$$= \frac{v(t) J\omega [L_2 - M]}{-\omega^2 [L_1 L_2 - M^2]}$$

$$I_1 = \frac{V(t)\, J\omega\,(L_2 - M)}{J^2\omega^2 (L_1L_2 - M^2)}$$

$$I_1 = \frac{V(t)\,(L_2 - M)}{J\omega\,(L_1L_2 - M^2)}$$

$$I_2 = \frac{|\Delta_2|}{|\Delta|} = \frac{V(t)\,J\omega\,(L_1 - M)}{\omega^2\,[M^2 - L_1L_2]}$$

$$= \frac{V(t)\,J\omega\,(L_1 - M)}{J^2\omega^2\,[L_1L_2 - M^2]}$$

$$I_2 = \frac{V(t)\,[L_1 - M]}{J\omega\,(L_1L_2 - M^2)}$$

Substitute I_1 & I_2 in Equ ①

$$I_{total} = I_1 + I_2$$

$$I_{total} = \frac{V(t)\,(L_2 - M)}{J\omega\,(L_1L_2 - M^2)} + \frac{V(t)\,(L_1 - M)}{J\omega\,(L_1L_2 - M^2)}$$

$$I_T = \frac{V(t)}{J\omega}\left[\frac{L_2 - M + L_1 - M}{L_1L_2 - M^2}\right]$$

$$I(t) = \frac{V(t)}{J\omega}\left[\frac{L_1 + L_2 - 2m}{L_1L_2 - M^2}\right]$$

$$\frac{I(t)}{V(t)} = \frac{(L_1 + L_2 - 2m)}{J\omega\,(L_1L_2 - m^2)}$$

$$\frac{V(t)}{I(t)} = \frac{J\omega\,(L_1L_2 - m^2)}{L_1 + L_2 - 2m.}$$

$$Z(t) = \frac{J\omega\,[L_1L_2 - M^2]}{(L_1 + L_2 - 2m)}$$

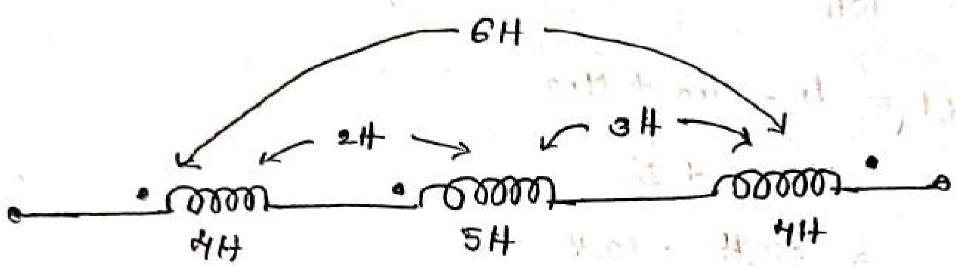
$$Z = \overset{0}{R} + JX$$

$$Z = JX$$

$$j\omega L_{eq} = \frac{j\omega\left(L_1L_2 - M^2\right)}{L_1+L_2-2M}$$

$$\boxed{L_{eq} = \frac{L_1L_2 - M^2}{L_1+L_2-2M}}$$

(1).



L_{equ1} = Equivalent inductance of first Coil i.e.

$$L_{equ1} = L_1 + M_{12} - M_{13} = 4+2-6$$

$$= 0 \text{ Henry.}$$

Equivalent inductance of 2nd Coil i.e.,

$$L_{equ2} = L_2 + M_{21} - M_{31}$$

$$= 5+2-3$$

$$= 4 \text{ Henry}$$

Equivalent inductance of 3rd coil ie...

$$L_{equ3} = L_3 - M_{31} - M_{32}$$

$$= 4-3-6$$

$$= -5H$$

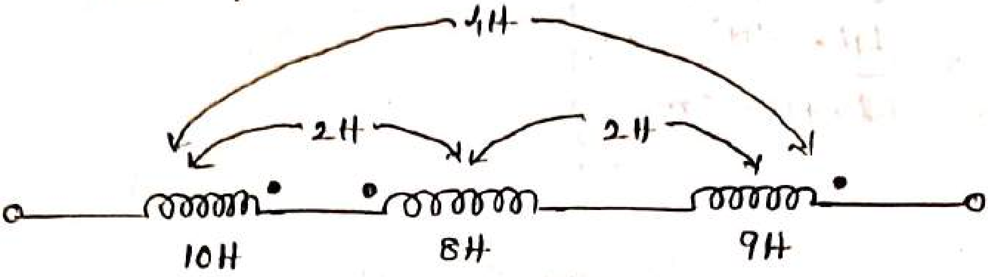
total Equivalent inductance

$$L_{eq} = L_{equ1} + L_{equ2} + L_{equ3}.$$

$$= 0+4-5$$

$$= -1 \text{ Henry.}$$

(5) Find out the Equivalent inductance of the given circuit ?



A. $L_{eq \cdot 1} = L_1 - M_{12} + M_{13}$

$= 10 - 2 + 4.$

$= 12 H$

$L_{equ\,2} = 8 - M_{21} + M_{23}.$

$= 8 - 2 - 2$

$= 4 H$

$L_{equ\,3} = L_3 + M_{31} - M_{32}$

$= 9 + 4 - 2$

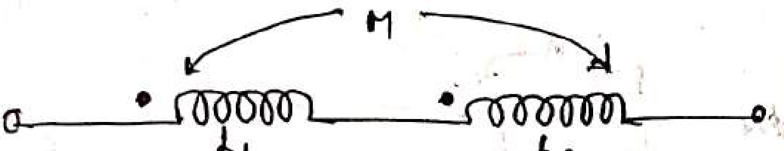
$= 11$ Henry

Total Equivalent $= 12 + 4 + 11$

$= 27$

(3). 2 coils connected in series aiding passion have total inductance of 250 milli henry. If the coil is connected in series oposing passion have total inductance of 150 milli henry. The first coil have an inductance of L_1 is equal to 3 times of the inductance of 2nd coil L_2. Find the value of L_1, L_2, M, and also find coefficient of coupling.

(A).

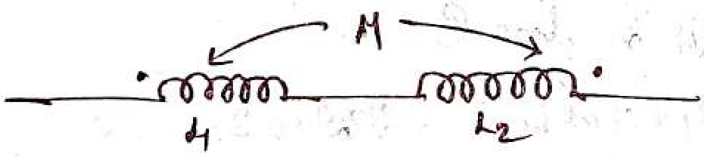


Series Aiding

$L_1 + L_2 + 2m = 250mH$

$L_1 + L_2 + 2m = 250 \times 10^{-3}$

$L_1 + L_2 + 2m = 0.25 \rightarrow (1)$



Series opposing

$L_1 + L_2 - 2m = 150 mH$

$L_1 + L_2 - 2m = 150 \times 10^{-3}$

$L_1 + L_2 - 2m = 0.15 \rightarrow (2)$.

$L_1 = 3L_2$

$L_1 - 3L_2 = 0$

Save Equ (1), (2) & (3).

$L_1 = 0.15 H$

$L_2 = 0.05 H$

$M = 0.025 H$

Coefficient of Coupling $k = \frac{M}{\sqrt{L_1 L_2}} = \frac{0.025}{\sqrt{(0.15)(0.05)}}$

KVL Equations in Couple Circuits:

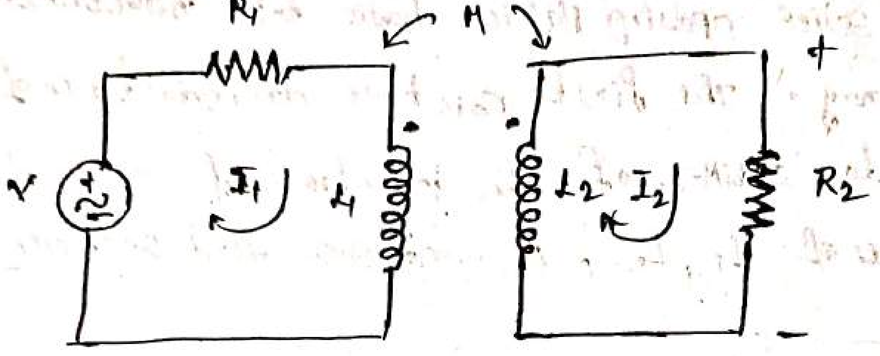


fig (1)

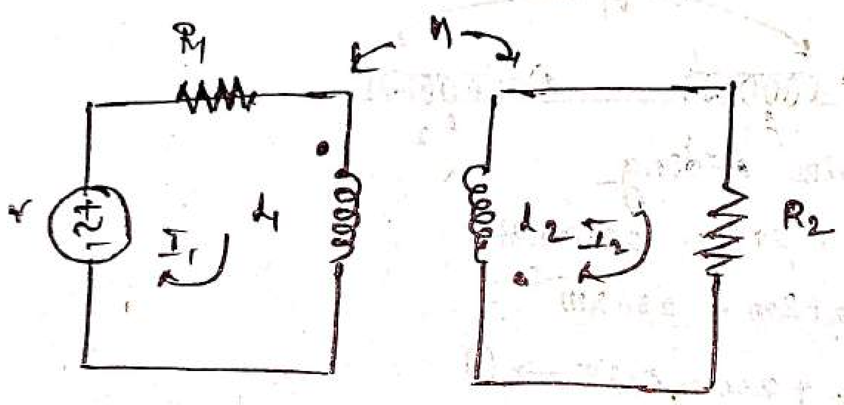


fig (2)

Apply KVL in fig (1) in loop ①,

$$-V + I_1R_1 + j\omega L_1 I_1 - j\omega M I_2 = 0$$

Apply KVL in loop ②

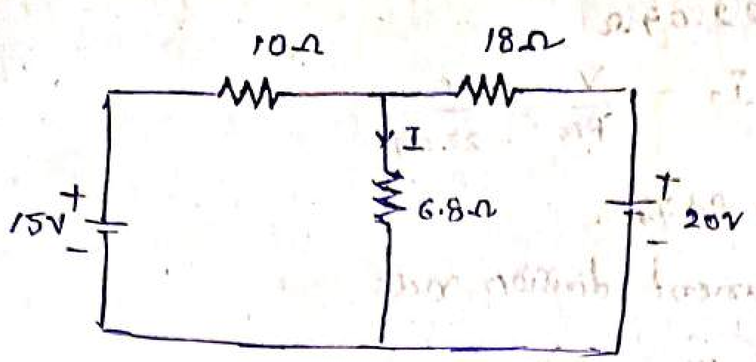
$$I_2R_2 + j\omega L_2 I_2 - j\omega M I_1 = 0.$$

Apply KVL in fig (2) in loop ①

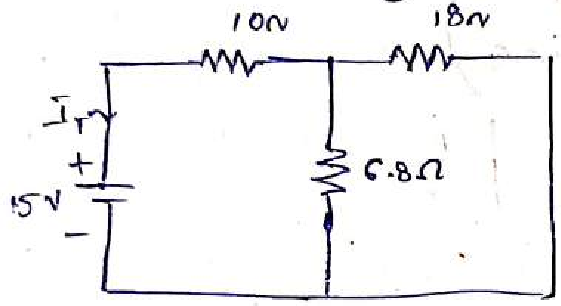
$$-V + I_1R_1 + j\omega L_1 I_1 + j\omega M I_2 = 0$$

Apply KVL in loop ②,

$$I_2R_2 + j\omega L_2 I_2 + j\omega M I_1 = 0.$$



A) 15 V Source acting alone.



$$R_{eq} = \frac{18 \times 6.8}{18 + 6.8} + 10$$

$$R_{eq} = 14.93\,\Omega$$

Total Current

$$I_T = \frac{V}{R_{eq}} = \frac{15}{17.93}$$

$$I_T = 1.004\,A$$

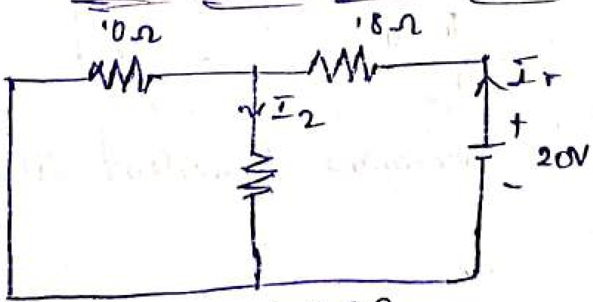
By using Current division rule,

$$I_1 = I_T \left(\frac{18}{18 + 6.8}\right)$$

$$I_1 = 1.004 \left(\frac{18}{18 + 6.8}\right)$$

$$\boxed{I_1 = 0.728\,A}$$

20V Source acting alone



$$R_{eq} = \frac{10 \times 6.8}{10 + 6.8} + 18$$

$R_{ee} = 22.04\,\Omega$

Total Current, $I_T = \frac{V}{R_{eq}} = \frac{20}{22.04}$

$I_T = 0.907A.$

By using Current division rule

$I_2 = I_T \left(\frac{10}{10+6.8}\right)$

$= 0.907 \left(\frac{10}{10+6.8}\right)$

$\boxed{I_2 = 0.539\,A}$

Principle of Superposition

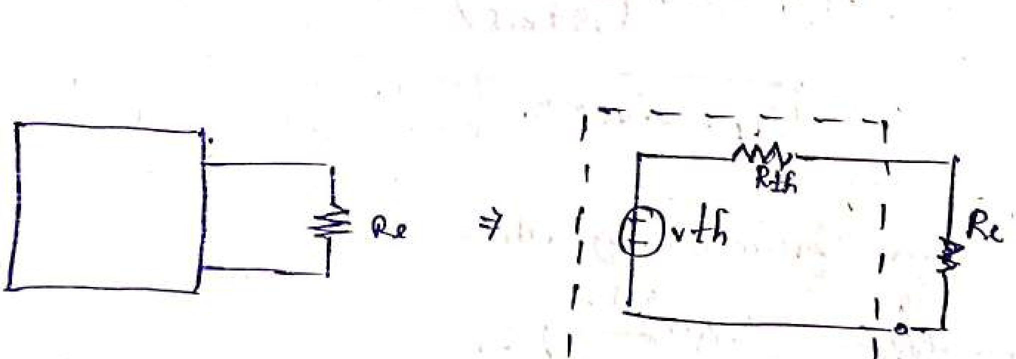
$I = I_1 + I_2$

$I = 0.728 + 0.539$

$I = 1.267\,A.$

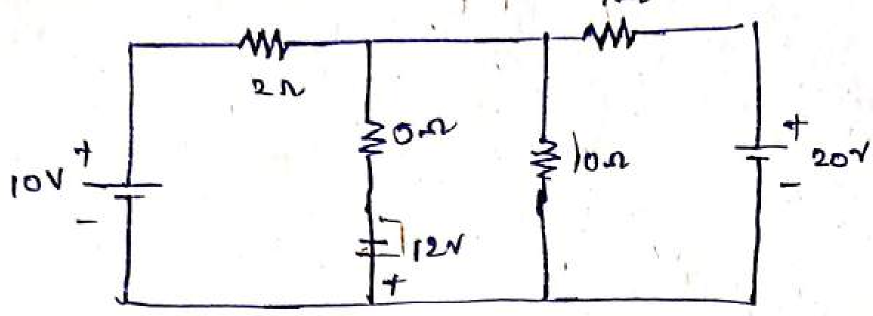
Therinen's theorem

Statement: Any linear 2 terminal ~~bg~~ bi lateral network Consisting of Several Voltage Sources, Several Current sources, and several resistors.. It can be replaced by and Equivalent circuit Consisting of a Voltage Source in series with a resistor.

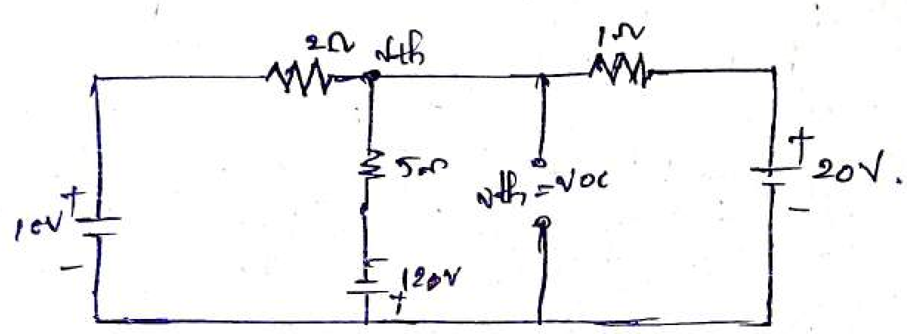


Therinins Equivalent circuit

In the below network find the current through 6Ω resistor using thevinins theroem



$R_L = 10\Omega$



Finding Vth

By using nodal analysis.

$$\frac{V_{th}-10}{2} + \frac{V_{th}+12}{5} + \frac{V_{th}-20}{1} = 0$$

$$\frac{5V_{th} - 50 + 2V_{th} + 24 + 10V_{th} - 200}{10} = 0$$

$$17V_{th} - 226 = 0$$

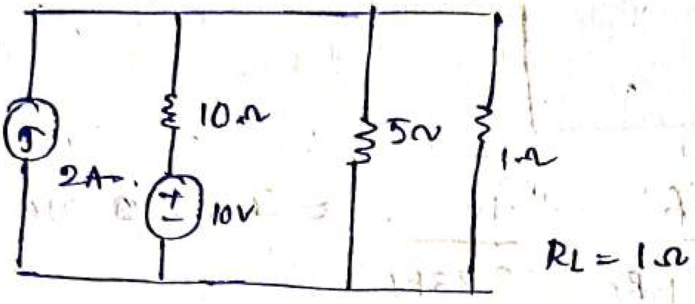
$$V_{th} = \frac{226}{17}$$

$$\boxed{V_{th} = 13.29\,V}$$

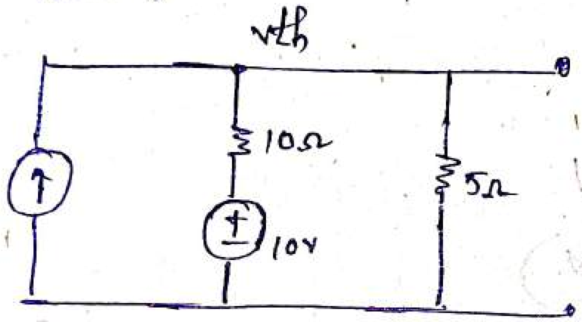
Finding Rth

1. In the ckt shown below find the power loss in 1 Ω resistor by using thevinin's theorem.

A.



finding Vth



using nodal Analysis

$$\frac{Vth-10}{10} + \frac{Vth-0}{5} = 2$$

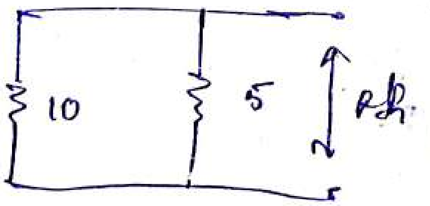
$$\frac{Vth-10+2Vth}{10} = 2$$

$$3Vth-10 = 20$$

$$3Vth = 30$$

$$Vth = 10$$

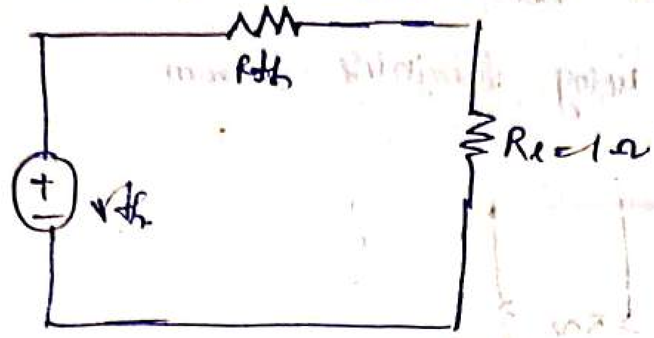
finding Rth



$$Rth = \frac{10\times5}{10+5}$$

$$\boxed{Rth = 3.33\Omega}$$

Thevinin's Equivalent ckt



$$I_{1\Omega} = \frac{V_{th}}{R_{th} + R_L} = \frac{10}{3.33+1} = I_{1\Omega} = 2.31A$$

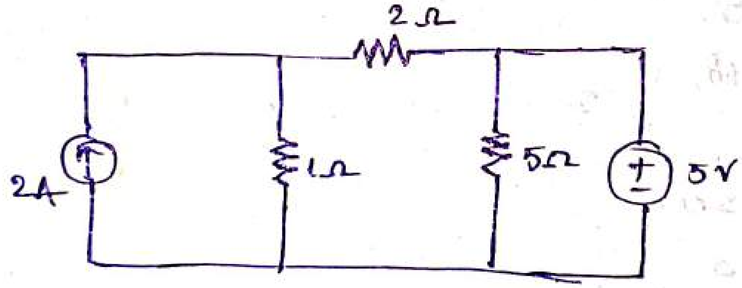
∴ Power loss in 1Ω resistor in

$$P_{1\Omega} = I^2 R_L$$

$$= (2.31)^2 (1)$$

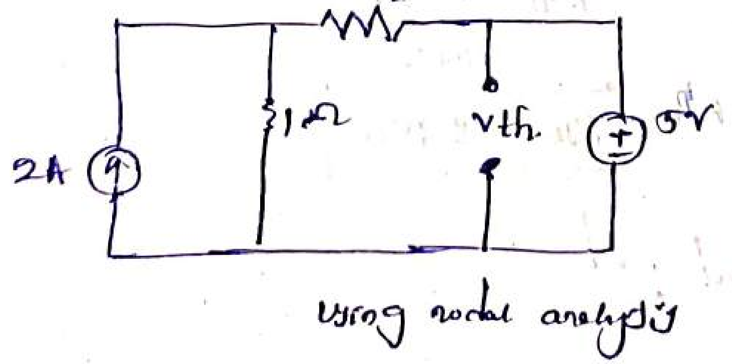
$$P = 5.33\,W$$

3. Determine the current through 5Ω resistor in the circuit shown below.



$R_L = 5\Omega$

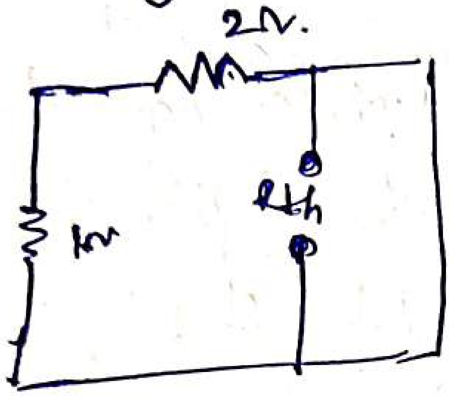
finding V_{th}

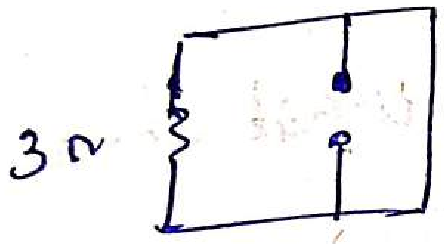


using nodal analysis

$V_{th} = 5V$.

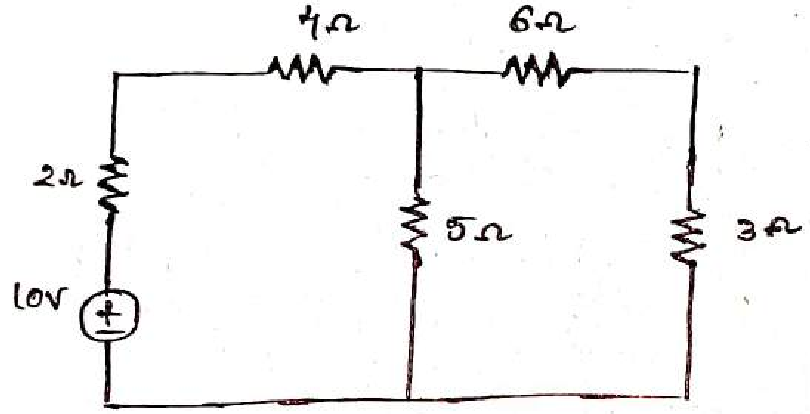
Finding Rth





The value of Current due to a Single source in any Particular branch of a circuit is Equal to the Value of Current in the original branch where the source was replaced when the source is shifted to that Particular branch of circuit

Apply the principle of reciprocity to the circuit shown below



Finding Current through 3Ω resistor for the actual Circuit.

$R_{equ_1} = 6+3$

$R_{equ_1} = 9\Omega$

$$R_{equ} = \frac{R_{equ_1} \times 5}{R_{equ_1} + 5} = \frac{9 \times 5}{9+5}$$

$R_{equ_2} = 3.214\Omega$

$\therefore R_{equ} = R_{eq_2} + 4 + 2$

$R_{eq_2} = 3.214 + 4 + 2$

$R_{equ} = 9.214\Omega$

$\therefore$ Total Current Coming from the Source

$$I_T = \frac{V}{R_{eq}} = \frac{10}{9.214} = 1.085A$$

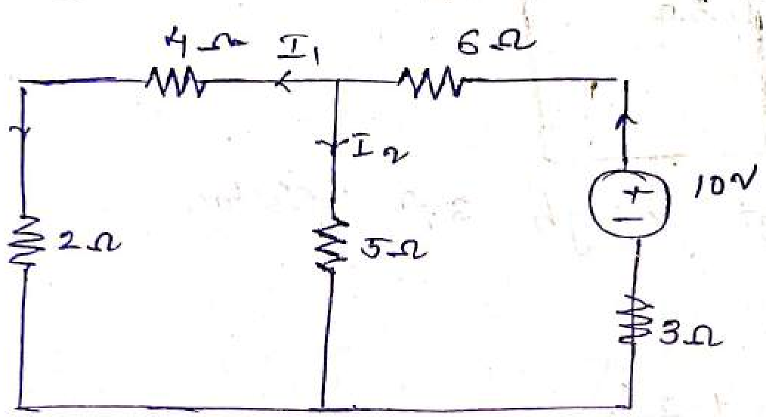
By using Current division Rule

$$I_1 = I_T\left(\frac{5}{5+6+3}\right)$$

$$I_1 = 1.085\left(\frac{5}{14}\right)$$

$$\boxed{I_1 = 0.387\ A} \quad - (1)$$

Finding Current when the Source is replaced.



$Requ_1 = 4+2 = 6$

$Requ_1 = 6.$

$$Requ_2 = \frac{6 * 5}{6+5} = \frac{30}{11} = 2.72\ \Omega$$

$Req_3 = Requ_2 + 6 + 3$

$= 2.72 + 6 + 3$

$= 11.72\ \Omega$

$$I_T = \frac{V}{R} = \frac{10}{11.72} = 0.85324.$$

By using Current division rule.

$$I_1 = I_T\left(\frac{5}{4+5+2}\right)$$

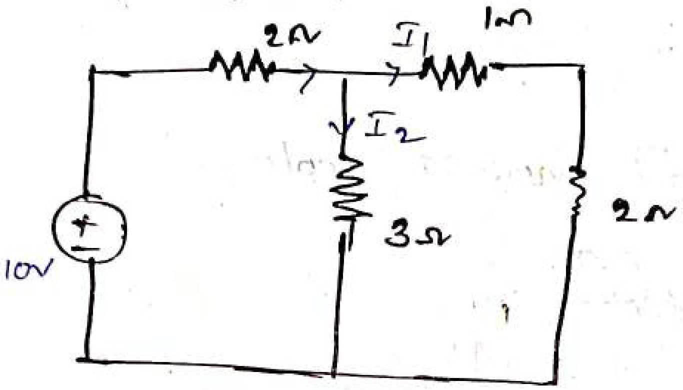
$$= 0.85324\left(\frac{5}{11}\right)$$

$= 0.85324 \times 0.4540$ $\boxed{I_1 = 0.38779\ \Omega}$ $- (2)$

from ① & ②.

$I = I_1'$.

② Show the application of reciprocity theorem in the circuit shown below



finding current through 3Ω resistor.

$R_{equ_1} = 3$.

$R_{equ_2} = \frac{3 \times 3}{3+3} + 2$.

$= \frac{9}{6} + 2$

$= 1.5 + 2$

$= 3.5\,\Omega$

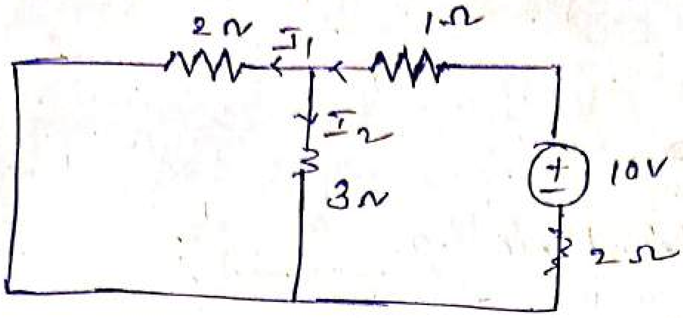
$I_T = \frac{V}{R_e} = \frac{10}{3.5} = 2.8571$

$I_\varphi =$

$I_1 = 2.857\left(\frac{3}{3+1+2}\right)$

$I_1 = 2.857 \times 0.5$

$= 1.42$



$$R_{eq\,x_1} = \frac{3\times 2}{3+2} = \frac{6}{5}$$

$$R_{eq\,x_2} = 1.2 + 1$$

$$R_{eq\,x_2} = 2.2 + 2 = 4.2$$

$$I_1 = \frac{V}{R} = \frac{10}{4.2} =$$

$$I_1 = \left(\frac{3}{2+3}\right)$$

$$= \left(\frac{3}{5}\right)$$

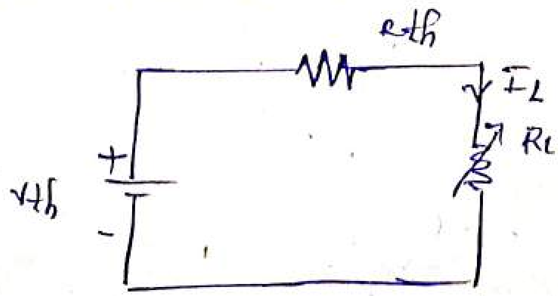
$$= \times 0.6.$$

$$=$$

Maximum Power transfer theorem

According maximum power transfer theorem the condition for maximum power flow through load resistor R_L can be achieve when the load resistance is equal to thevinins equivalent resistance.

Take thevinins equivalent circuit



From the circuit

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

Power in load resistance is.

$$P_L = I_L^2 R_L$$

$$P_L = \left(\frac{V_{th}}{R_{th} + R_L}\right)^2 R_L \Rightarrow \frac{V_{th}^2 R_L}{(R_{th} + R_L)^2}$$

To get maximum current differentiate w.r.t to R_L and equal it to zero.

ie $\frac{dP_L}{dR_L} = 0$.

$$\frac{d}{dR_L}\left[\frac{V_{th}^2 R_L}{(R_{th} + (R_L))^2}\right] = 0.$$

$$\frac{d}{dR_L}\left[\frac{V_{th}^2 R_L}{R_{th}^2 + R_L^2 + 2R_{th}R_L}\right] = 0.$$

$$(V_{th}^2 R_L)(0 + 2R_L + 2R_{th}) - (R_{th}^2 + R_L^2 + 2R_{th}R_L)V_{th}^2 = 0$$

$$2V_{th}^2 R_L^2 + \cancel{2V_{th}^2 R_L R_{th}} - R_{th}^2 V_{th}^2 - R_L^2 V_{th}^2 - \cancel{2V_{th}^2 R_{th} R_L} = 0$$

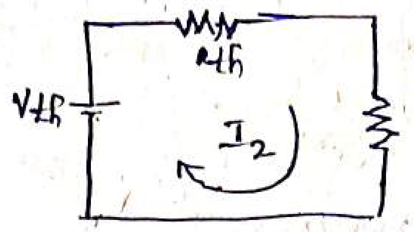
$$V_{th}^2 R_L^2 - R_{th}^2 V_{th}^2 = 0$$

$$V_{th}^2 (R_L^2 - R_{th}^2) = 0.$$

$$R_L^2 = R_{th}^2$$

$$\boxed{R_L = R_{th}}$$

Consider



$\Rightarrow I_L = \dfrac{V_{th}}{R_{th}+R_L}$

Power at load in

$P_L = I_L^2 R$

$P_L = \left(\dfrac{V_{th}}{R_{th}+R_L}\right)^2 R_L$

$\therefore P_{max} = \left(\dfrac{V_{th}}{R_{th}+R_{th}}\right)^2 R_{th} \quad (\because R_L = R_{th})$

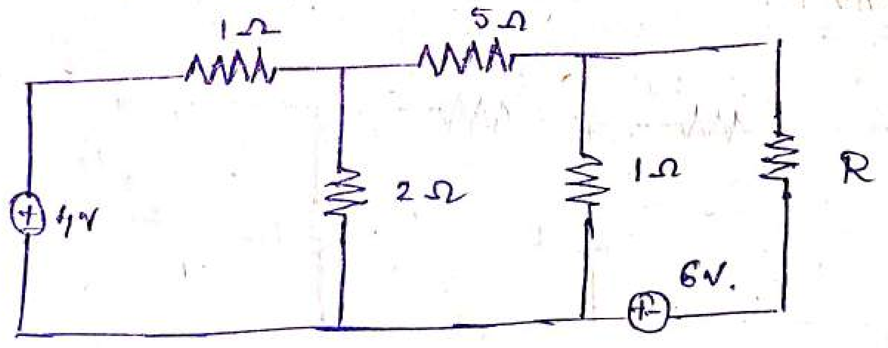
$P_{max} = \dfrac{V_{th}^2 R_{th}}{(2R_{th})^2}$

$P_{max} = \dfrac{V_{th}^2 \cancel{R_{th}}}{4 R_{th}^{\cancel{2}}}$

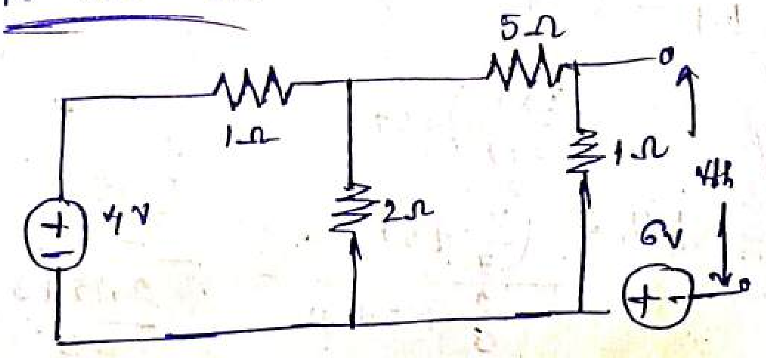
$$\boxed{P_{max} = \dfrac{V_{th}^2}{4R_{th}}}$$

(2) find value of R in the circuit shown below such that maximum power transfer takes place what is the amount of this power?

(A).



Sol To find Vth



Equivalent resistance of above ckt:

$$R_{equ} = \frac{(5+1)\,2}{5+1+2} + 1$$

$$R_{equ} = \frac{5}{2} = 2.5\,\Omega$$

∴ Total Current, $I_T = \frac{V}{R_{eq}} = \frac{4}{(5/2)} = 8/5$

$$I_T = 1.6\,A.$$

using Current division rule.

$$I_{1\Omega} = I_T\left(\frac{2}{2+5+1}\right)$$

$$= 1.6\left(\frac{2}{2+5+1}\right)$$

$$I_{1\Omega} = 1.6\left(\frac{2}{8}\right)$$

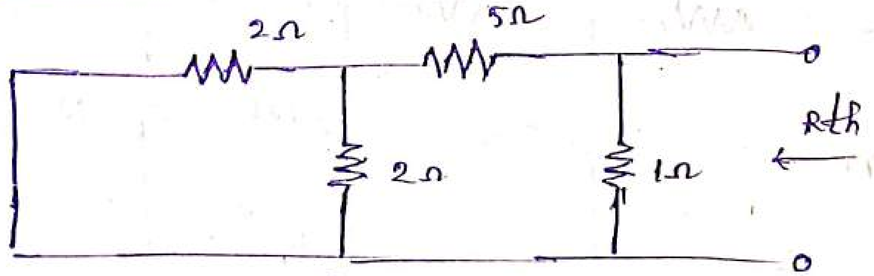
$$\boxed{I_{1\Omega} = 0.4\,A}$$

Voltage across resistor,

$$V_{1\Omega} = V_{th} = 1 \times 0.4$$

$$\boxed{V_{th} = 0.4 \text{ Volts}}$$

To find R_{th}



$$R_{th} = \frac{\left(\frac{2\times 1}{2+1} + 5\right) \times 1}{\left(\frac{2\times 1}{2+1}\right) + 5 + 1}$$

$$R_{th} = \frac{\left(\frac{2}{3} + 5\right)}{\frac{2}{3} + 5 + 1} = \frac{\frac{2+15}{\cancel{3}}}{\frac{2+15+3}{\cancel{3}}} = \frac{17}{20} = 0.85$$

∴ At max power flow

$$\boxed{R = R_{th} = 0.85\,\Omega}$$

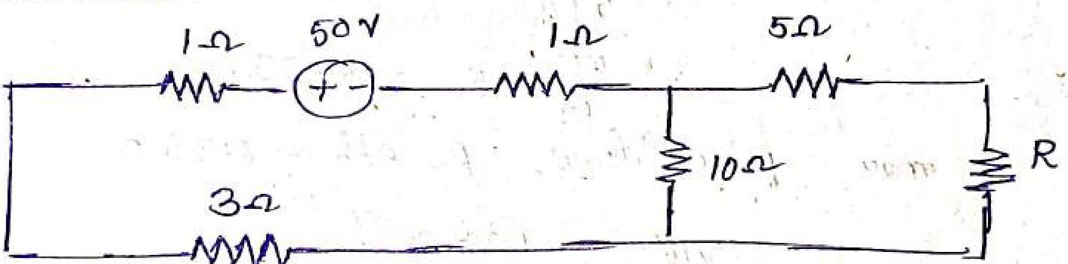
∴ maximum power flow is

$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

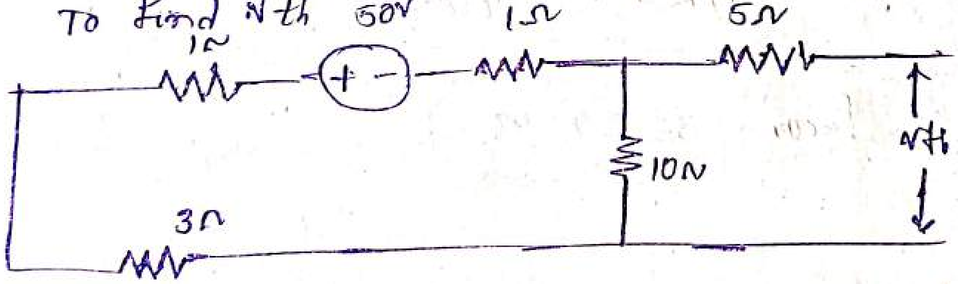
$$P_{max} = \frac{(6.4)^2}{4(0.85)}$$

$$\boxed{P_{max} = 12\,W}$$

Assuming maximum power transfer from the source to R find the value of this amount of power in the below circuit.



To find Vth



$$R_{eq} = \frac{5\times10}{5+10} + 5$$

$$= \frac{\cancel{50}\,10}{\cancel{15}\,3} + 5$$

$$= \frac{10+15}{3}$$

$$= \frac{\cancel{25}\,8}{\cancel{3}}$$

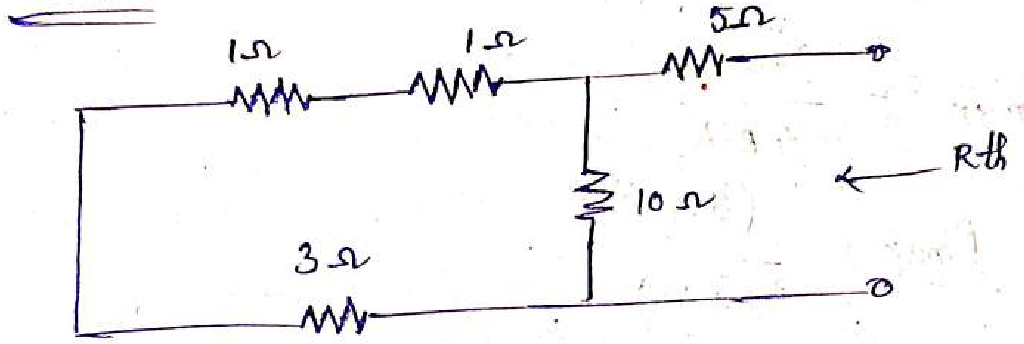
$$= 8.33$$

∴ Total current, $I_T = \frac{V}{R_{eq}} = \frac{50}{\cancel{8.33}\;15} = 3.33\,\Omega$

$V_{10\Omega} = 10 \times 3.33$

$V_{10\Omega} = V_{th} = 33.33$

R_{th}



$$R_{th} = \frac{(3+1+1) \times 10}{3+1+1+10} + 5$$

$$= \frac{5 \times 10}{5+10} + 5$$

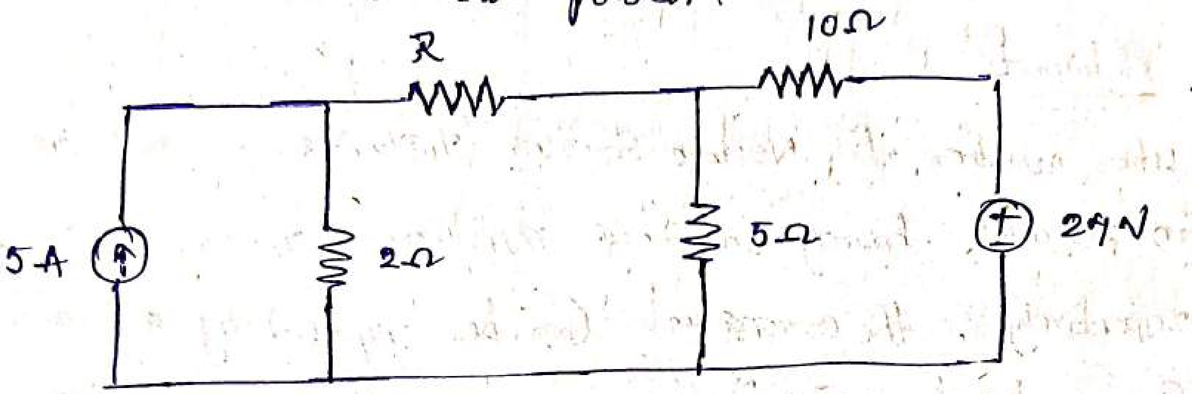
$$= \frac{50}{15} + 5 = 8.33$$

At max power flow, $R = R_{th} = 8.33\,\Omega$

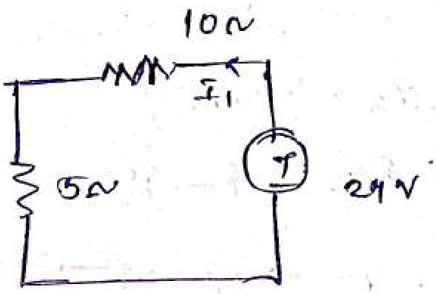
$$\therefore P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{(33.3)^2}{4(8.3)}$$

$P_{max} = 33.34$ w

What should be the value of R such that maximum power transfer can takes place from the rest of the network to R in the circuit shown below. obtain the amount of this power.



To find Vth.



$R_{eq} = 5+10 = 15\Omega$.

Total current $= I = \frac{V}{R_{eq}} = \frac{\cancel{24}^{8}}{\cancel{15}_{5}} = \frac{8}{5} = \cancel{2.66}\ 1.66$

$\boxed{I_T = 1.66}$

$I_1 \Omega = I_T$

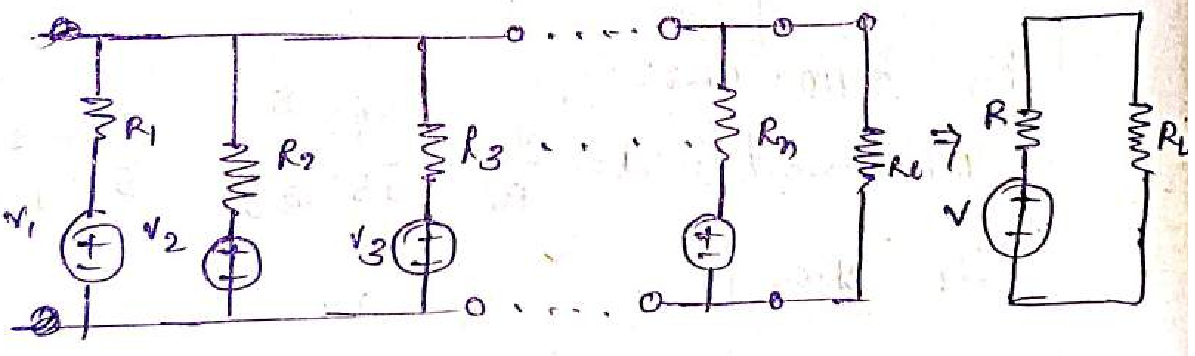
Millman's Theorem

The utility of this theorem is that any number of Parallel Voltage Sources can be reduced to one Equivalent Source.

statement

When number of Voltage Sources ($V_1, V_2, V_3, \ldots V_n$) are in parallel having internal resistances ($r_1, r_2, r_3, \ldots r_n$) respectively, the arrangement can be replaced by a single Equivalent voltage Source "V" in series with an Equivalent series resistance "R".

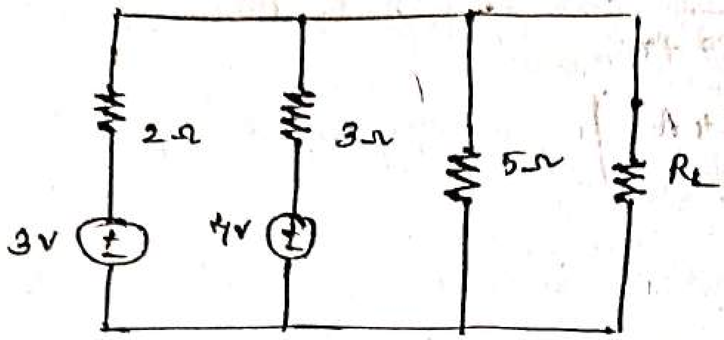
$$(V_1, V_2, V_3 \ldots V_n)$$
$$(R_1, R_2, R_3 \ldots R_n)$$



$$V = \frac{\pm V_1 G_1 \pm V_2 G_2 \pm V_3 G_3 + \cdots \pm V_n G_n}{G_1 + G_2 + G_3 + \cdots + G_n};$$

$$R = \frac{1}{G} = \frac{1}{G_1 + G_2 + G_3 + \cdots + G_n}$$

Find Current through load resistance $R_L = 2\Omega$ using millman's theorem for the circuit shown below.



A. We get Equavalent voltage source

$$V = \frac{\pm V_1 G_1 \pm V_2 G_2 \pm \cdots \pm V_n G_n}{G_1 + G_2 + G_3 + \cdots + G_n}$$

$$V = \frac{\pm \frac{V_1}{R_1} \pm \frac{V_2}{R_2} \pm \frac{V_3}{R_3} \cdots \pm \frac{V_n}{R_n}}{\frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_n}}$$

$$V = \frac{\pm \frac{3}{2} + \frac{4}{3} + \frac{0}{5}}{\frac{1}{2} + \frac{1}{3} + \frac{1}{5}}$$

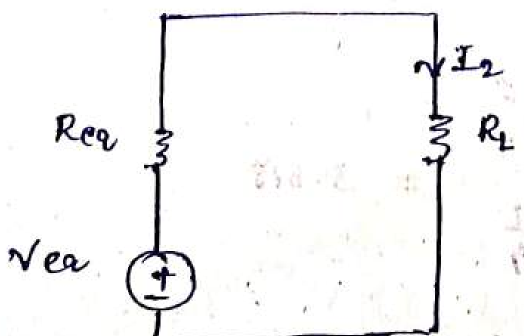
$$\boxed{V_{eq} = 2.74V}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$$

$$\frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5}$$

$$\boxed{R_{eq} = 0.967\Omega}$$

∴ Equivalent circuit is

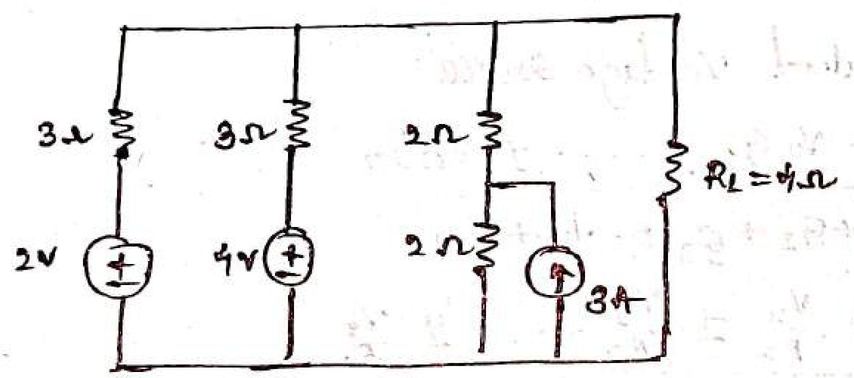


∴ Current through load resistance

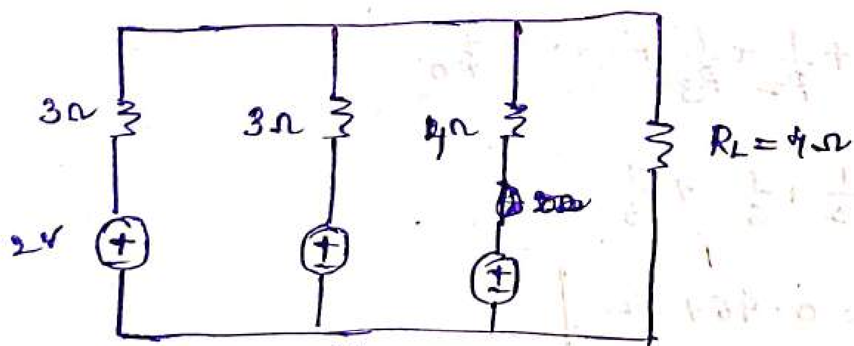
$$I_L = \frac{V_{eq}}{R_{eq} + R_L}$$

$$I_L = \frac{2.79}{0.967 + 2}$$

$$\boxed{I_L = 0.924\ A}$$



A. In the given circuit 3A and 2Ω resistor are both are connected in parallel. i.e., By using source transformation technique the above circuit is converted into.



$$V = \frac{+\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}}{\frac{1}{3} + \frac{1}{3} + \frac{1}{4}}$$

$$= \frac{\frac{2}{3} + \frac{4}{3} + \frac{6}{4}}{\frac{1}{3} + \frac{1}{3} + \frac{1}{4}} = 3.818$$

$$\frac{1}{R_{eq}} = \frac{1}{3} + \frac{1}{8} + \frac{1}{4}$$

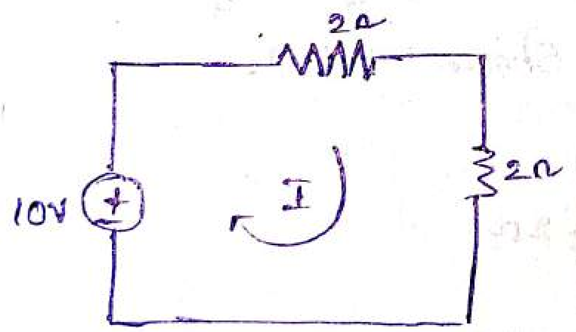
$$\boxed{R_{eq} = 1.09\,\Omega}$$

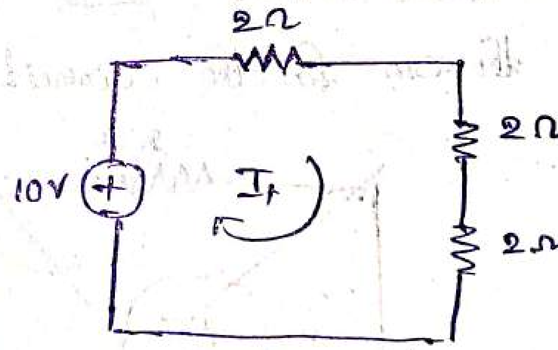
∴ the Equivalent circuit diagram is

Compensation theorem.

Any network Computing of linear of bi lateral impindence and independent. Source of a branch having Current (I) having impedance (Z) that increases by ΔZ then the change of voltage is same as the voltage ~~Source~~ of are Concurrent produced by an opposing voltage Source of Value I . ΔZ placed in that branch after replacing original Sources by their internal resistances.

1. Find Current ΔI from the given electrical circuit using Compensation theorem?





Current, $I = \frac{10}{2+2}$ $\qquad$ $I' = \frac{10}{2+2+2}$

$I = \frac{10}{4}$ $\qquad$ $I' = \frac{10}{6}$

$I = 2.5A$ $\qquad$ $I' = 1.66A$

∴ change in Current

$\Delta I = I - I'$

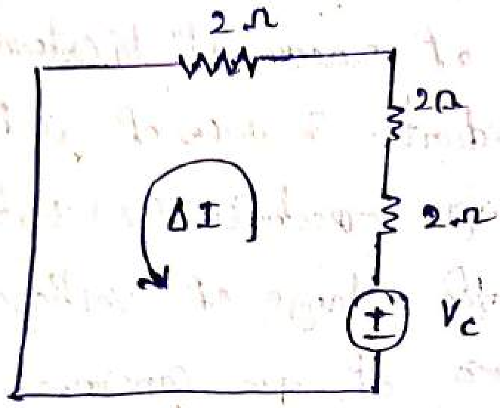
$= 2.5 - 1.66$

$$\boxed{\Delta I = 0.84A}$$

Opposing Voltage

$$V_c = I \Delta R$$
$$= (2.5)(2)$$
$$V_c = 5 \text{ volts.}$$

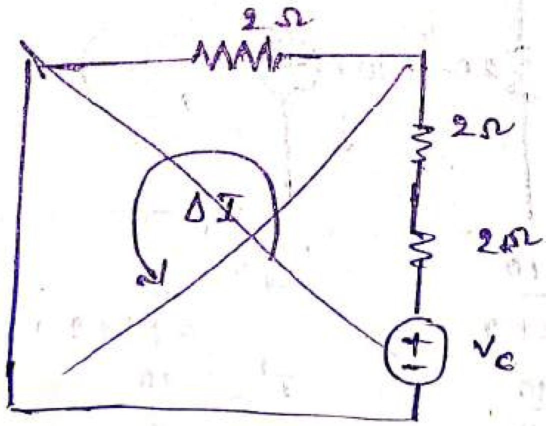
The circuit is redrawn as

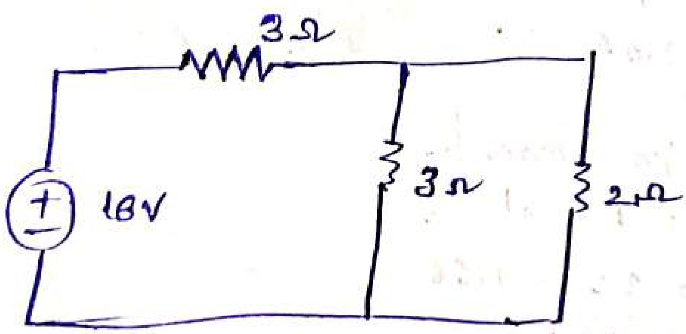


$$\Delta I = \frac{V_c}{2+2+2} = \frac{5}{2+2+2} = \frac{5}{6}.$$

$$\boxed{\Delta I = 0.833 A}$$

2. Find the current ΔI and ΔR equal to 1Ω added across 2Ω resistor by using compensation theorem for the circuit shown below.





$$R_{eq} = \frac{3 \times 2}{3+2} + 3$$

$$R_{equ} = 4.2\,\Omega$$

$$I_T = \frac{V}{R_{eq}} = \frac{16}{4.2}$$

$$I_T = 3.8\,A$$

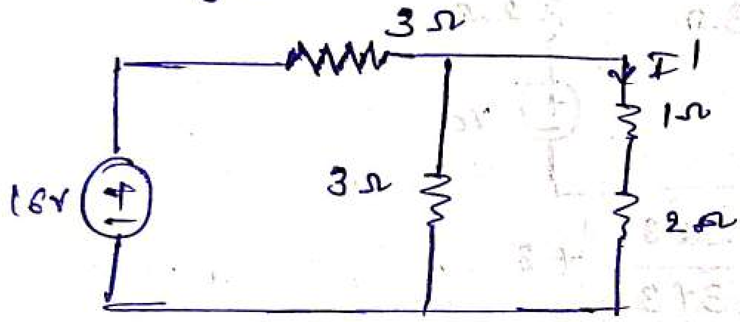
By using Current division rule.

$$I = I_T \left(\frac{3}{3+2} \right)$$

$$I = 3.8 \left(\frac{3}{5} \right)$$

$$\boxed{I = 2.28\,A}$$

changed circuit



$$R_{equ} = \frac{3 \times 3}{3+3} + 3$$

$$R_{equ} = 4.5\,\Omega$$

$$\therefore I_T = \frac{V}{R_{eq}} = \frac{16}{4.5}$$

$$\boxed{I_T = 3.55\,A}$$

By using Current division rule

$$I' = I_T \left(\frac{3}{3+3} \right)$$

$$I' = 3.55 \left(\frac{3}{6} \right)$$

$$\boxed{I' = 1.77\,A}$$

change in Current

$$\Delta I = I - I'$$

$$= 2.28 - 1.77$$

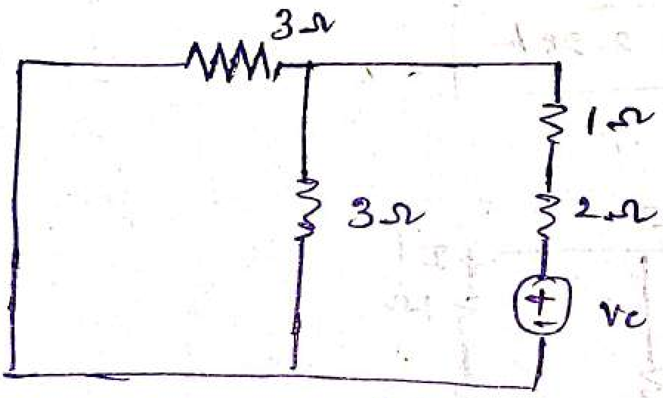
$$\boxed{\Delta I = 0.508A}$$

Therefore $V_c = I \Delta R$

$$V_c = (2.28)(1)$$

$$\boxed{V_c = 2.28V}$$

$\therefore$ the circuit is redrawn as.



$$R_{equ} = \frac{3 \times 3}{3+3} + 3$$

$$R_{eq} = 4.5\Omega$$

$\therefore$ Current

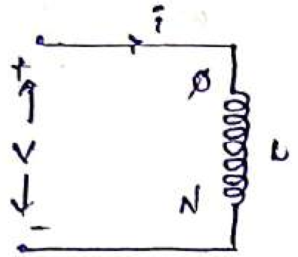
$$\Delta I = \frac{V_c}{R_{eq}}$$

$$\Delta I = \frac{2.28}{4.5}$$

$$\boxed{\Delta I = 0.506A}$$

1. What is self inductance ? and derive the Expression for Equivalent inductance of parallel aiding circuit. ?

A. The inductance offered by a coil whenever the flux produced by the coil links with the same coil itself.



$\phi \propto i$

$\phi = ki \Rightarrow k = \frac{\phi}{i} \rightarrow (1)$

Defferentiate w.r.t "t" on Both sides.

$$\frac{d\phi}{dt} = k\frac{di}{dt} \rightarrow (2).$$

We know that

$$V = -N\frac{d\phi}{dt}$$

$$V = -Nk\frac{di}{dt} \quad (\text{from } (2))$$

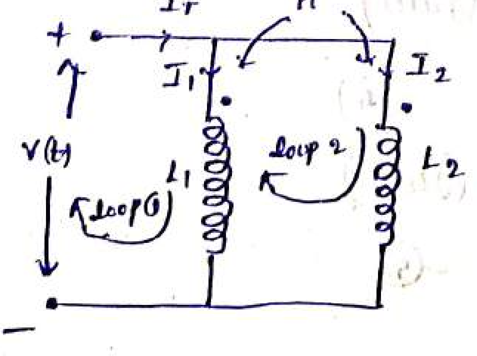
$$V = -N\left(\frac{\phi}{i}\right)\frac{di}{dt} \quad (\text{from } (1))$$

$$\boxed{V = -L\frac{di}{dt}}$$

where $\boxed{L = \frac{N\phi}{i}}$ Henry.

Parallel aiding Connection :

If two Coils are Connected in parallel in such a Way that the two Currents I_1, I_2 Entering the Coils at dotted terminals it is Called parallel aiding Connection.

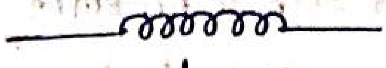


$V = IZ$

$Z \rightarrow$ impedance

$Z = R + JX$

Applying KCL in circuit.



L

Inductive reactance, X_L

Capacitive reactance, X_C

$X_L = \omega L$

$X_C = \frac{1}{\omega C}$

$I_T = I_1 + I_2 \longrightarrow (1)$

Apply KVL in loop (1).

$- v(t) + j\omega L_1 I_1 + j\omega M I_2 = 0$

$v(t) = j\omega L_1 I_1 + j\omega M I_2 \longrightarrow (1)$

Apply KVL in loop (2).

$- v(t) + j\omega L_2 I_2 + j\omega M I_1 = 0$

$v(t) = j\omega L_2 I_2 + j\omega M I_1 \rightarrow (2).$

$$\begin{bmatrix} v(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Where $\Delta = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix}$

$\Delta_1 = \begin{bmatrix} v(t) & j\omega M \\ v(t) & j\omega L_2 \end{bmatrix}$

$\Delta_2 = \begin{bmatrix} j\omega L_1 & v(t) \\ j\omega M & v(t) \end{bmatrix}$

$|\Delta| = (j\omega L_1)(j\omega L_2) - (j\omega M)(j\omega M)$

$= j^2 \omega^2 L_1 L_2 - j^2 \omega^2 m^2.$

$(-1)\, \omega^2 L_1 L_2 + \omega^2 m^2.$

$|\Delta| = \omega^2 (m^2 - L_1 L_2) \quad - (3)$

$|\Delta_1| = v(t)(j\omega L_2) - v(t)(j\omega M).$

$= v(t)\, j\omega (L_2 - m) \quad - (4)$

$|\Delta_2| = v(t)(j\omega L_1) - v(t)(j\omega M)$

$= v(t)\, j\omega (L_1 - m). \quad - (5)$

$$Z(t) = \frac{J\omega\left[L_1 L_2 - M^2\right]}{(L_1 + L_2 - 2M)}$$

$$Z = R + JX$$

$$R = 0$$

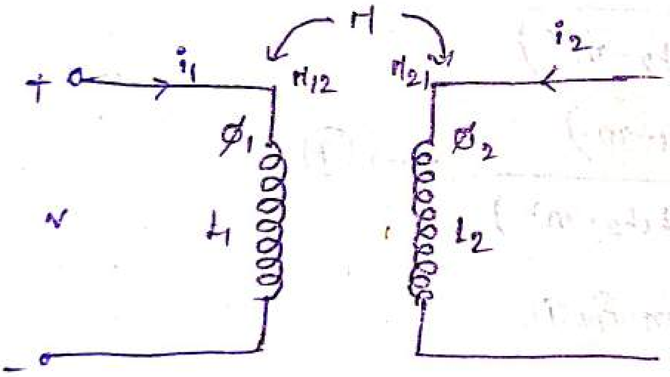
$$Z = JX.$$

$$\cancel{J\omega} L_{eq} = \frac{\cancel{J\omega}(L_1 L_2 - M^2)}{L_1 + L_2 - 2M}$$

$$\boxed{L_{equ} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}}$$

2. What is mutual inductance? And derive the Expression for Co-efficient of Coupling.

A. The inductance offered by the coil whenever the flux produced by one coil links with another coil.



$$\phi_2 \propto i_1$$

$$\phi_2 = K i_1 \Rightarrow K = \frac{\phi_2}{i_1} \rightarrow (1)$$

Diff. w.r.t to "t"

$$\frac{d\phi_2}{dt} = K \frac{di_1}{dt} \longrightarrow (2)$$

We know that

$$V = -N \frac{d\phi}{dt}$$

$$V = -N K \frac{di_1}{dt}$$

$$V = -N\left(\frac{\phi_2}{i_1}\right) \frac{di_1}{dt}$$

$$\boxed{V = -M_{12}\frac{di}{dt}}$$

Where $\boxed{M_{12} = N\left(\frac{\phi_2}{i_1}\right)}$

Similarly

$\phi_1 \propto i_2$

$\phi_1 = k\, i_2$

$K = \frac{\phi_1}{i_2}$

D.w.r.t "t"

$$\frac{d\phi_1}{dt} = k\,\frac{di_2}{dt}$$

We know that

$$V = -N\,\frac{d\phi}{dt}$$

$$V = -N(k)\,\frac{di_2}{dt}$$

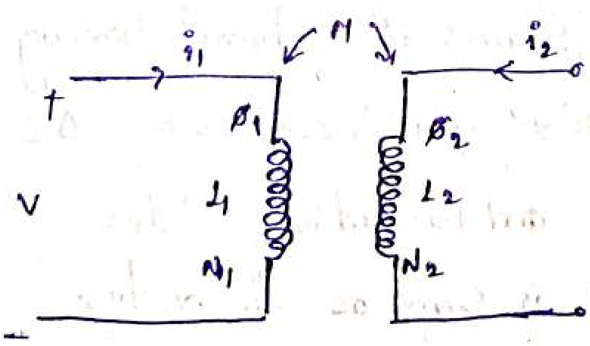
$$V = -N\left(\frac{\phi_1}{i_2}\right)\frac{di_2}{dt}$$

$$\boxed{V = m_{21}\,\frac{di_2}{dt}}$$

Where $\boxed{m_{21} = \frac{N\phi_1}{i_2}}$

<u>Coefficient of Coupling</u>

The degree of Coupling that Exists between two coupled Coils for the circuit.



$$L_1 = \frac{N_1 \phi_1}{i_1} \; ; \; L_2 = \frac{N_2 \phi_2}{i_2}$$

$$M_{12} = \frac{k N_2 \phi_1}{i_1} \; ; \; M_{21} = k \frac{N_1 \phi_2}{i_2}$$

Assume, $M_{12} = M_{21} = m$

$$M_{12} \times M_{21} = k \frac{N_2 \phi_1}{i_1} \times k \frac{N_1 \phi_2}{i_2}$$

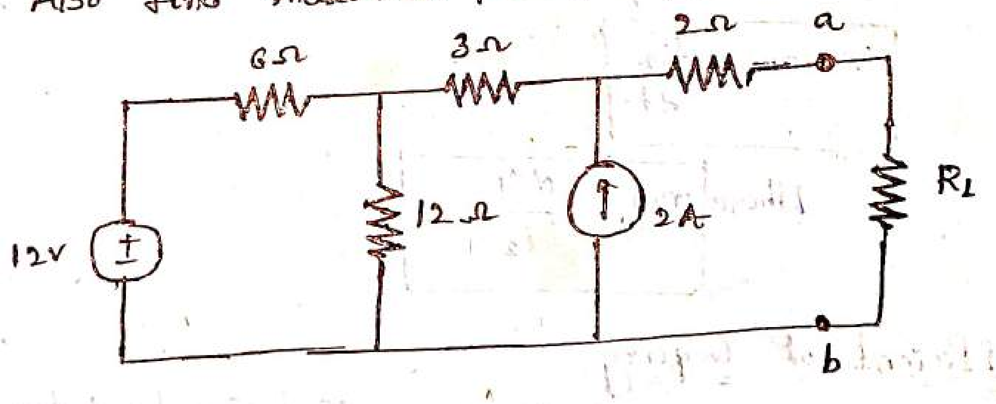
$$M \times M = k^2 \frac{N_1 \phi_1}{i_1} \times \frac{N_2 \phi_2}{i_2}$$

$$M^2 = k^2 L_1 L_2 .$$

$$\boxed{M = k \sqrt{L_1 L_2}}$$

$$\boxed{k = \frac{M}{\sqrt{L_1 L_2}}}$$

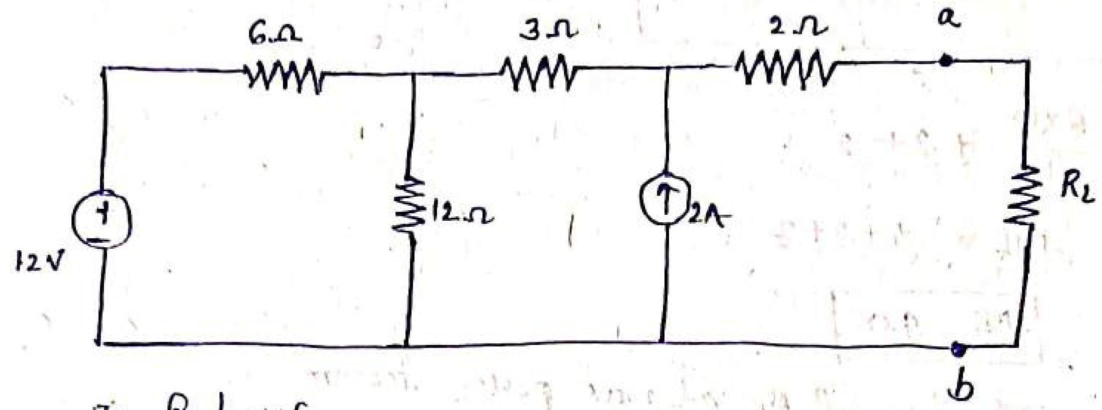
3. state Compensation theorem. find the value of R_L for maximum power transfer in the circuit shown below. Also find maximum power?



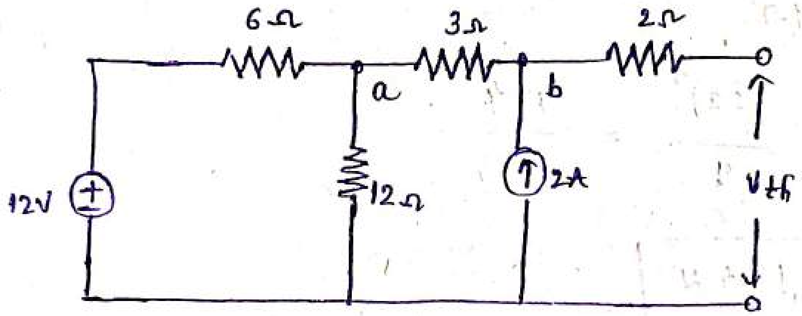
Compensation theorem:

Any network Consisting of linear or bilateral impedences and independent Sources if a branch having Current "I" and impedence "Z" that increases by ΔZ Then the change of voltage and Current in other branches of the network is Same as the voltage are Current produced by an opposing voltage Source of value $I\Delta Z$ placed in that branch after replacing original sources by their internal sources.

b) <u>problem</u>



To find V_{th}



Applying kcl at node a.

$$\frac{V_1 - 12}{6} + \frac{V_1}{12} + \frac{V_1 - V_{th}}{3} = 0.$$

$$\frac{2V_1 - 24 + V_1 + 4V_1 - 4V_{th}}{12} = 0$$

$$7V_1 - 4V_{th} = 24 \rightarrow (1)$$

Apply kcl at node b

$$\frac{V_{th} - V_1}{3} = 2 \left(\frac{22-16}{3} = 2 \right).$$

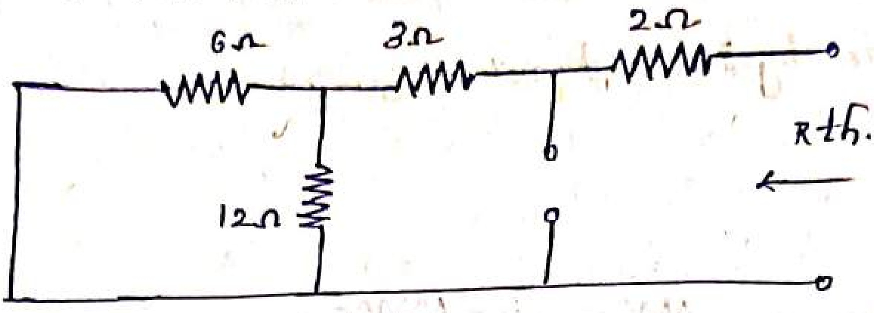
$$V_{th} - V_1 = 6 \rightarrow (2)$$

By solving (1) & (2) Equ we get

$$\boxed{V_1 = 16V}$$

$$\boxed{V_{th} = 22V}$$

To find Rth



$$R_{th} = \frac{6\times12}{6+12} + 3 + 2$$

$$R_{th} = 4+3+2$$

$$\boxed{R_{th} = 9\Omega}$$

∴ The value of R_L at max power theorem.

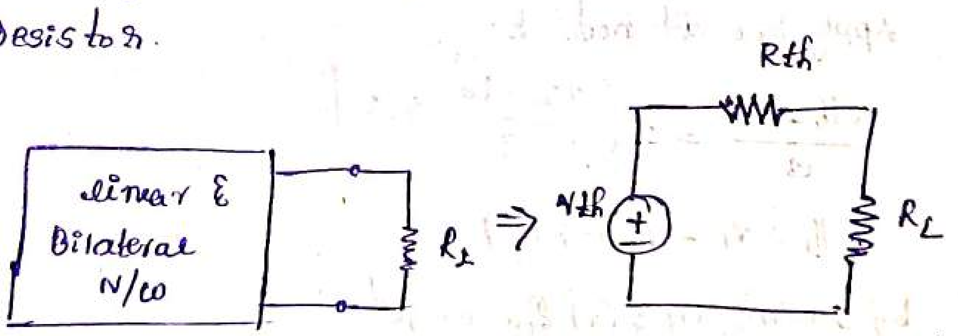
$$R_L = R_{th} = 9\Omega$$

$$P_{max} = \frac{V_{th}^2}{R_{th}} = \frac{(22)^2}{9} = \frac{484}{36}$$

$$\boxed{P_{max} = 13.4\ W}$$

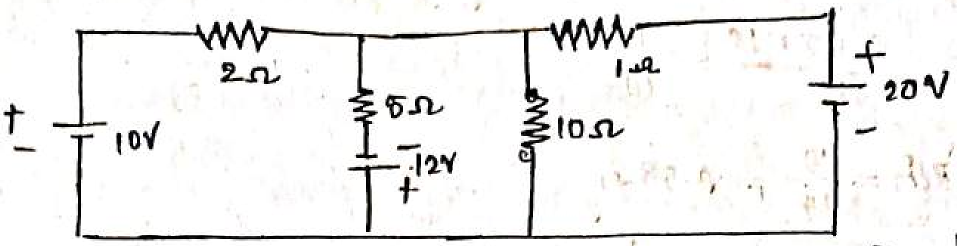
4. State and Explain Thevenin's theorem with an Example.

A. In a linear two terminal, bilateral network consisting of several voltage sources, current sources, resistors it can be replaced by an Equivalent circuit consisting of a voltage source in series with a resistor.



Thevenin's Equivalent circuit.

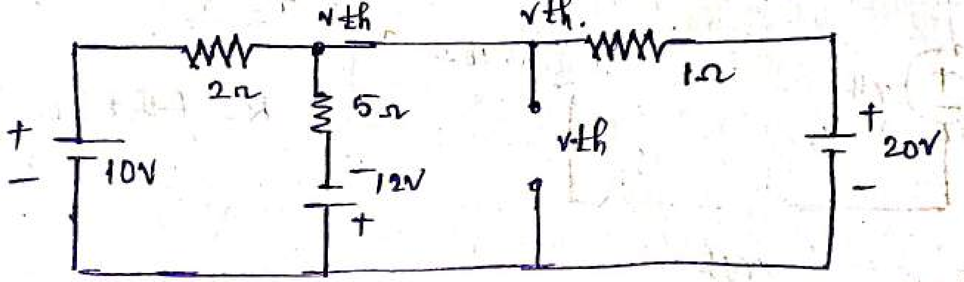
Example -



finding current through 10Ω resistor.

Take. for finding V_{th}.

Remove the resistance and find out the voltage across it using any analysis.



$$\frac{V_{th}-10}{2}+\frac{V_{th}+12}{5}+\frac{V_{th}-20}{1}=0.$$

$$\frac{5V_{th}-50+2V_{th}+24+10V_{th}-200}{10}=0.$$

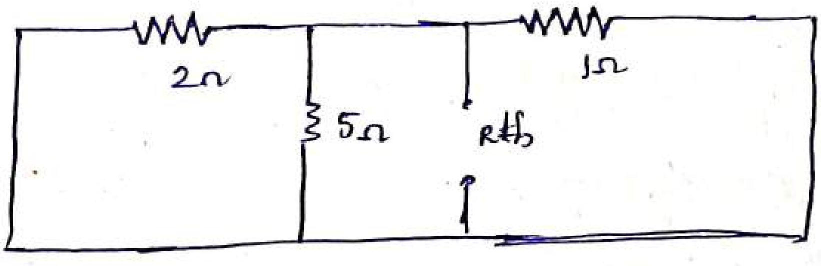
$$5V_{th}+2V_{th}+10V_{th}-50-200+24=0.$$

$$17V_{th}=226$$

$$V_{th}=\frac{226}{17}$$

$$\boxed{V_{th}=13.29}$$

for finding R_{th}



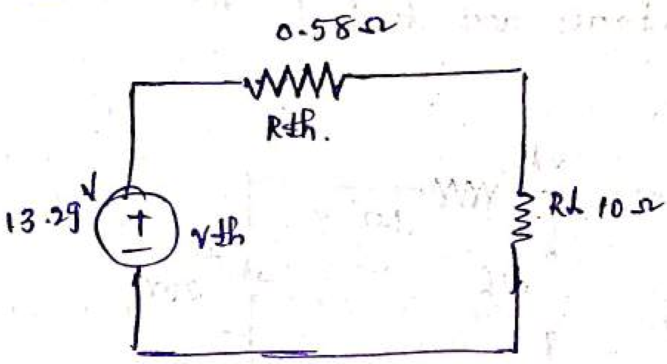
$$\frac{1}{R_{th}} = \frac{1}{2} + \frac{1}{5} + 1$$

$$= \frac{5+2+10}{10} = \frac{17}{10}$$

$$R_{th} = \frac{10}{17} = 0.58\,\Omega$$

$$\boxed{R_{th} = 0.58\,\Omega}$$

<u>Thevinins Equivalent circuit.</u>



$$\left(\begin{array}{l} V = IR \\ I = \frac{V}{R} \\ R = R_{th} + R_L \end{array} \right)$$

$$I_{10\Omega} = \frac{V_{th}}{R_{th} + R_L}$$

$$= \frac{13.29}{0.58 + 10}$$

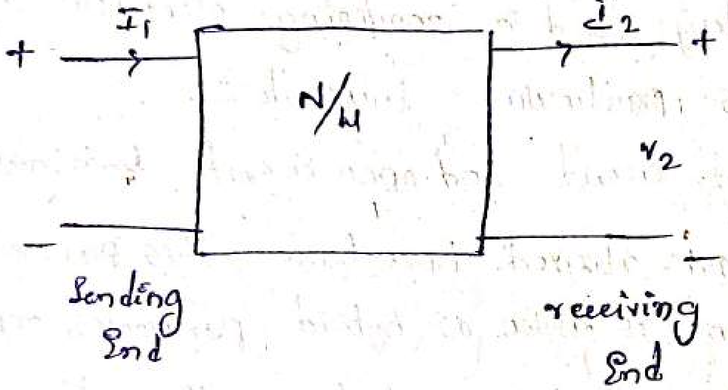
$$= \frac{13.29}{10.58}$$

$$\boxed{I_{10\Omega} = 1.26\,A}$$

5 - Explain ABCD paramiters and H+ Parameters ?

A. ABCD Parameters are widely used in power transmission Engeneering where they are termed as generalized circuit parameters.

* It is Conventional and designate the input port as Sending End and the output port is receiving End while represented ABCD Parameters more over the output Current direction is taken as reverse.



In the matrix form the representation is

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$V_1 = AV_2 - BI_2$

$I_1 = CV_2 - DI_2$.

Assume the output port is opened. (Current = 0)

$\therefore I_2 = 0$.

$V_1 = AV_2$

$A = \left.\dfrac{V_1}{V_2}\right|_{I_2=0}$; $C = \left.\dfrac{I_1}{V_2}\right|_{I_2=0}$.

Assume the output port is Short Circuited (voltage = 0)

$\therefore V_2 = 0$

$$\therefore B = \frac{-V_1}{I_2}\Big|_{V_2=0} \quad ; \quad D = \frac{-I_1}{I_2}\Big|_{V_2=0}.$$

A = reverse voltage ratio.

B = Transfer impedence.

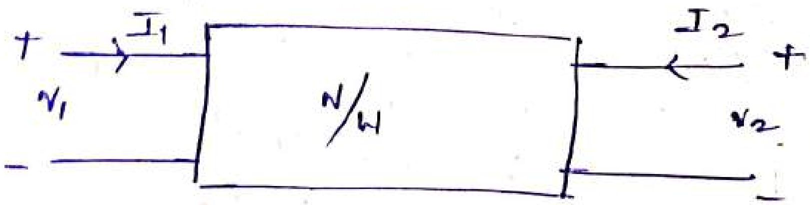
C = Transfer admitence.

D = reverse current ratio.

H + Parameters

* These are widely used in modeling electronic components and circuits particular transistors.
* As Both short circuit and open circuit terminal conditions are utilized here hence this parameter representation is known as hybrid parameter representation.
* In this form of representation the voltage of input port and the current of output port are expressed in the terms of current of input port and voltage of output port.

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$



$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

circuit is shorted ($V_2 = 0$.)

$$h_{11} = \frac{V_1}{I_1} \Big/ _{V_2 = 0.}$$

$$h_{21} = \frac{I_2}{I_1} \Big/ _{V_2 = 0.}$$

circuit is opened ($I_1 = 0$.)

$$h_{12} = \frac{V_1}{V_2} \Big| _{I_1 = 0}$$

$$h_{22} = \frac{I_2}{V_2} \Big/ _{I_1 = 0.}$$

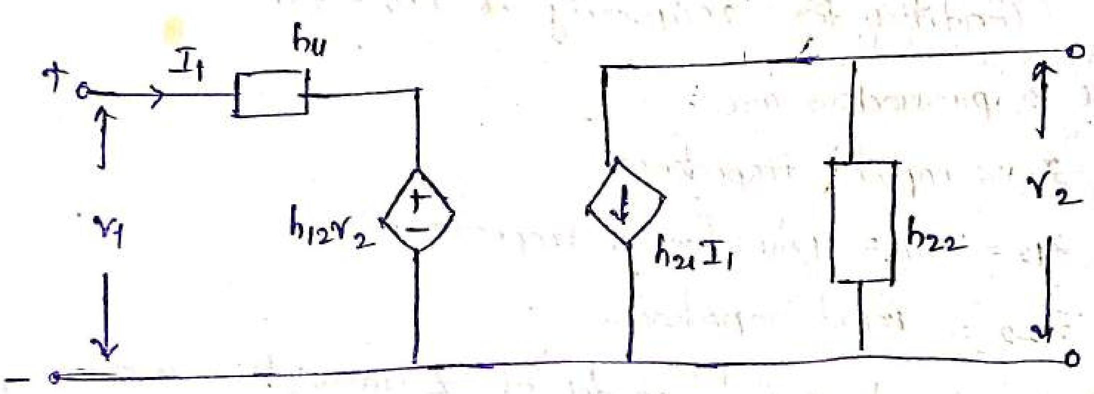
h_{11} = input inpedences (Ω)

h_{12} = reverse voltage gain

h_{21} = forward current gain

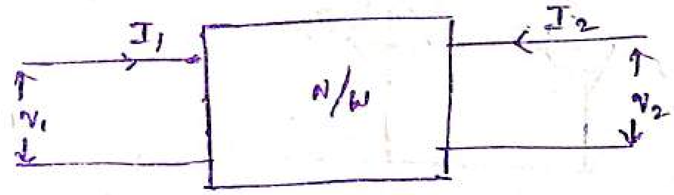
h_{22} = output admitance ($\mho$)

The Equivalent circuit of h-parameters is represented by



6. Explain about Z & Y - parameters.

A. Z Parameters



$$[V] = [Z][I]$$

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \rightarrow (1)$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \rightarrow (2).$$

where

$$Z_{11} = V_1/I_1 \Big/ _{I_2 = 0}$$

$$Z_{21} = V_2/I_1 \Big/ _{I_2 = 0.}$$

$$Z_{12} = V_1/I_2 \Big/ _{I_1 = 0}$$

$$Z_{22} = \frac{V_2}{I_2} \Big/ _{I_1 = 0.}$$

$Z_{11}, Z_{12}, Z_{21}, Z_{22}$ are the network functions and are Called as impedence of Z - Perimeters This Z perimeters Can be represented by matrix notation

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Always the voltage & Current matrix are Column matrix and Z Perimeter matrix is a Square matrix.

Note Condition for Semmitry is $Z_{11} = Z_{22}$

Condition for reciprocity is $Z_{12} = Z_{21}$

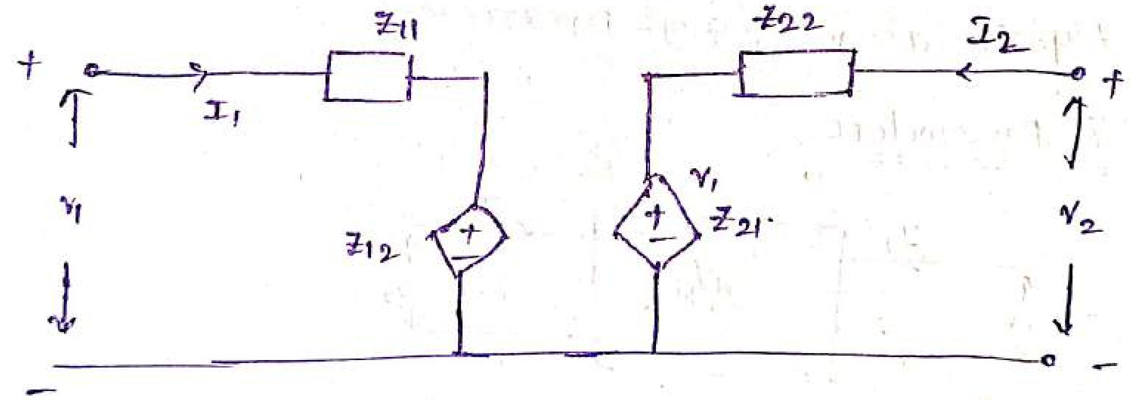
The Z parameters are

Z_{11} = input impedence

$Z_{12} = Z_{21}$ = Transformer impedence

Z_{22} = input impedence.

* the Equivalent circuit model of Z Parameters are



Y- parameters / short circuit parameters

Admittance: resiprocal of impedence is Called Admitance

* units for admittance is Mho's (℧).

* In a two port network the input Current I_1, I_2 be Expressed in terms of input and output voltages V_1 & V_2 respectively as.

$[I] = [y][V]$.

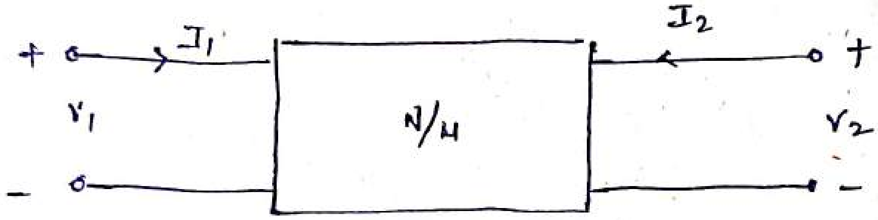
where $[y]$ is a square type Admittance matrix.

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Equation representation.

$$I_1 = y_{11} V_1 + y_{12} V_2$$

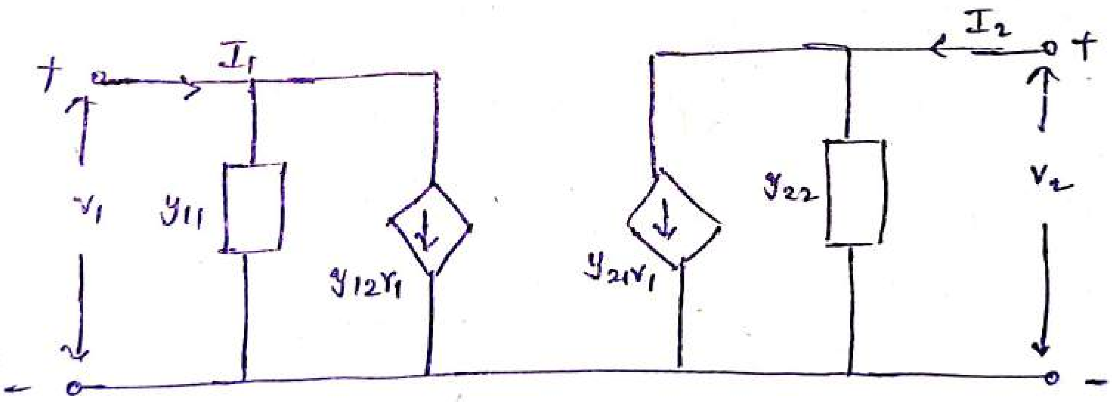
$$I_2 = y_{21} V_1 + y_{22} V_2$$



$$y_{11} = \frac{I_1}{V_1}\Big|_{V_2=0} \qquad y_{12} = \frac{I_1}{V_2}\Big|_{V_1=0}$$

$$y_{21} = \frac{I_2}{V_1}\Big|_{V_2=0} \qquad y_{22} = \frac{I_2}{V_2}\Big|_{V_1=0}$$

The Equivalent circuit representation of y parameters.



Unit - V

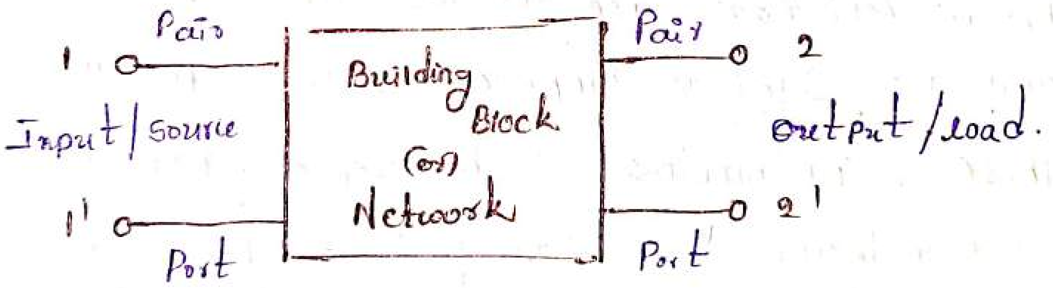
Two-port Networks.

Port : A pair of terminals at which a signal may enter or leave a network is called a port.

(or)

A port is defined as pair of terminals on to which the Energy is supplied or from which the Energy is withdrawn.

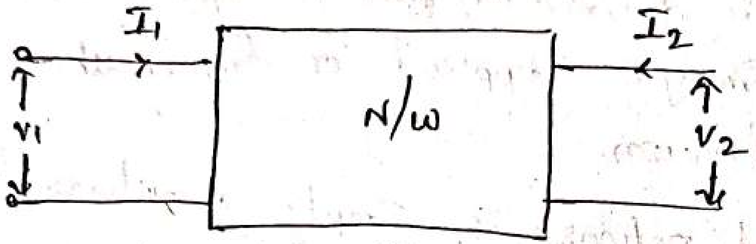
* A two port network is simply a network inside a building block and the network has only two pairs of accesible terminals. usually one pair represents input and the other pair represents the output.
* Such a building block is very common in Electronic System, Communication System, transmition and distribution Systems.



In the below figure shows two port network in which the four terminals have been paired into two ports 1-1' & 2-2'. the terminals 1-1' together constituts a port and the terminals 2-2' Constitutes another Port.

* If the port Containing no sources in their Brances. are Called passive ports If the ports Containing Sources in their branches or Called active ports.

<u>open circuit (or) Z parameters. (or) impedence parameters</u>



$$[V] = [Z][I]$$

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \longrightarrow ①$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \longrightarrow ②$$

where

$$Z_{11} = \frac{V_1}{I_1}\bigg|_{I_2=0} \qquad Z_{12} = \frac{V_1}{I_2}\bigg|_{I_1=0}.$$

$$Z_{21} = \frac{V_2}{I_1}\bigg|_{I_2=0} \qquad Z_{22} = \frac{V_2}{I_2}\bigg|_{I_1=0}.$$

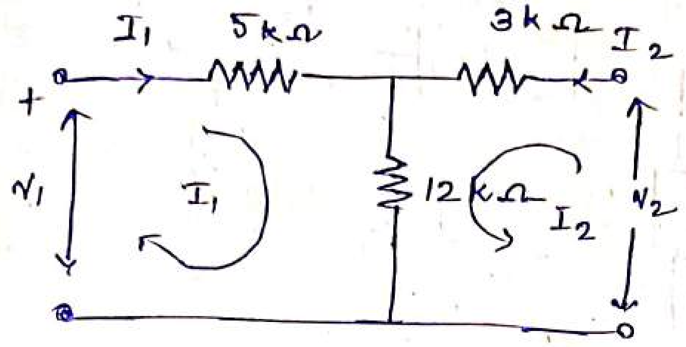
$Z_{11}, Z_{12}, Z_{21}, Z_{22}$ are the network functions and are Called as impedences or Z parameters.

These Z parameters Can be represented by matrix notation i.e.

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Always the voltage and Current matrices or Column matrices and Z Parameters matrix is a square matrix.

1. Find the z parameters for the network shown in below figure.



Solution

Apply kvl in loop-1

$$-V_1 + 5kI_1 + 12k(I_1 + I_2) = 0.$$

$$V_1 = 5k\,I_1 + 12k\,I_1 + 12k\,I_2$$

$$V_1 = 17k\,I_1 + 12k\,I_2 \rightarrow (1)$$

Apply kvl in loop-2

$$-V_2 + 3k\,I_2 + 12k\,(I_2 + I_1) = 0.$$

$$V_2 = 3k\,I_2 + 12k\,I_2 + 12k\,I_1$$

$$V_2 = 15k\,I_2 + 12k\,I_1 \rightarrow (2)$$

Represent Eqn (1) & (2) in ~~matrix~~ standard matrix representation of z parameters we get.

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 17k & 12k \\ 12k & 15k \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

By comparing with standard representation the z parameters are

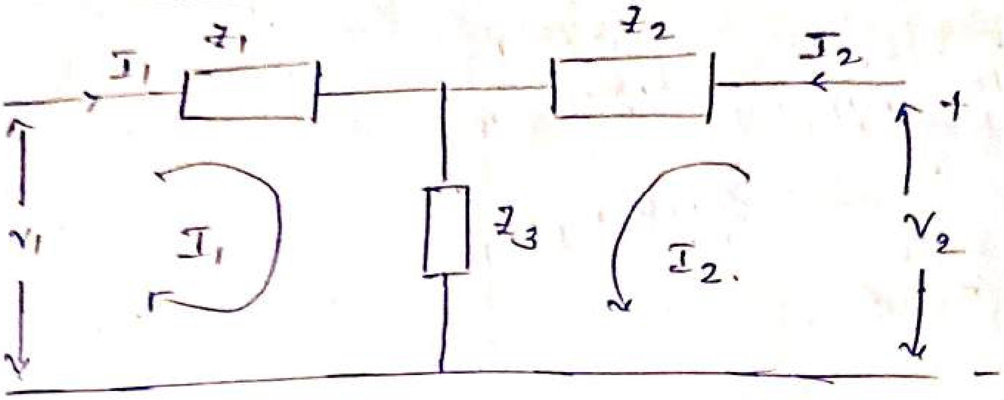
$Z_{11} = 17\,k\Omega$

$Z_{12} = 12\,k\Omega$

$Z_{21} = 12\,k\Omega$

$Z_{22} = 15\,k\Omega$

In a T - Network of given Circuit $Z_1 = 2\angle 0^\circ \Omega$ $Z_2 = 5\angle -90^\circ \Omega$, $Z_3 = 3\angle 90^\circ \Omega$ find the Z parameters.



Apply kvl in loop (1).

$$-V_1 + Z_1 I_1 + (I_1 + I_2) Z_3 = 0.$$

$$V_1 = Z_1 I_1 + Z_3 I_1 + Z_3 I_2$$

$$V_1 = (Z_1 + Z_3) I_1 + Z_3 I_2 \quad - (1)$$

Apply kvl in loop (2).

$$-V_2 + I_2 Z_2 + Z_3 (I_2 + I_1)$$

$$V_2 = (Z_2 + Z_3) I_2 + Z_3 I_1 \longrightarrow (2)$$

Repretion Eq (1) & (2) in standard matrix representation of Z parameters we get.

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_1 + Z_3 & Z_3 \\ Z_3 & Z_2 + Z_3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Compare the above representation with standard form the Z Parameters are

$$Z_{11} = Z_1 + Z_3.$$

$$Z_{11} = 2\angle 0 + 3\angle 90^\circ$$

$$= 2 + 3j$$

$$= 3.605 \angle 56\ \Omega$$

The following readings are obtained Experimental in an unknown two port network.

	V_1	V_2	I_1	I_2
Output open	100V	60V	10A	0
Input open	30V	40V	0	3A

Sol

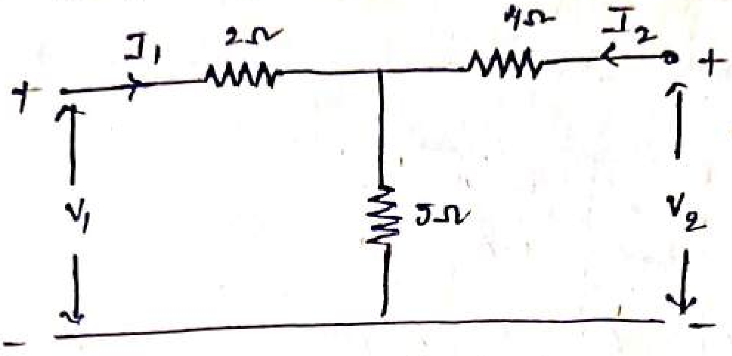
$$Z_{11} = \frac{V_1}{I_1}\Big|_{I_2=0} = \frac{100}{10} = 10\,\Omega$$

$$Z_{12} = \frac{V_1}{I_2}\Big|_{I_1=0} = \frac{30}{3} = 10\,\Omega$$

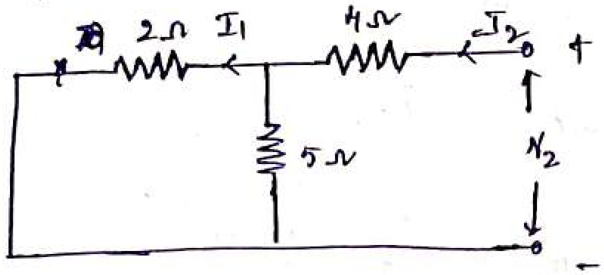
$$Z_{21} = \frac{V_2}{I_1}\Big|_{I_2=0} = \frac{60}{10} = 6\,\Omega$$

$$Z_{22} = \frac{V_2}{I_2}\Big|_{I_1=0} = \frac{40}{3} = 13.33\,\Omega$$

1. Find y parameters for the circuit shown below.



A. Port 1 short circuited $(V_1 = 0)$



$I_1 = y_{11} V_1 + y_{12} V_2$

$I_2 = y_{21} V_1 + y_{22} V_2$

$$y_{12} = \frac{I_1}{V_2}$$

$$y_{22} = \frac{I_2}{V_2}$$

$$R_{eq} = \frac{2 \times 5}{2+5} + 4.$$

$R_{eq} = 5.42\,\Omega$

$\therefore$ Total Current $I_2 = \dfrac{V_2}{R_{eq}}$

$$\therefore y_{22} = \frac{I_2}{V_2} = \frac{1}{R_{eq}} = \frac{1}{5.42}$$

$$\boxed{y_{22} = 0.18\,\Omega}$$

By using Current division rule.

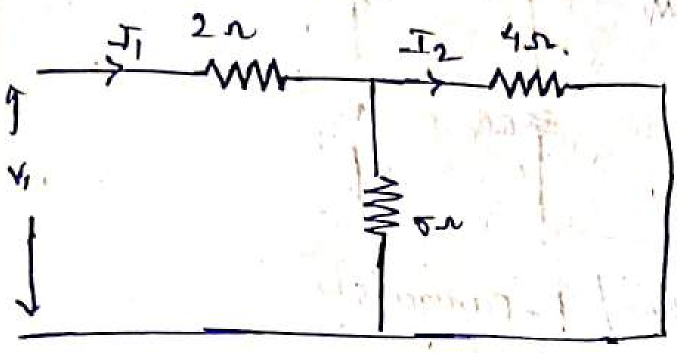
$$I_1 = I_2\left(\frac{5}{5+2}\right)$$

$$I_1 = \frac{V_2}{R_{eq}}\left(\frac{5}{7}\right)$$

$$\therefore y_{12} = \frac{I_1}{V_2} = \frac{5}{7 R_{eq}} = \frac{\cancel{25}\,5}{7 \times 5.42}$$

$y_{12} = -0.131\,\mho$

Port-2- Short circuited



$$R_{equ} = \frac{4\times5}{4+5} + 2$$

$$= \frac{20}{9} + 2$$

$$= 4.222\,\Omega$$

$\therefore$ Total current $I_1 = \dfrac{V_1}{R_{eq}}$

$$y_{11} = \frac{I_1}{V_1} = \frac{1}{R_{eq}} = \frac{1}{4.22} = 0.23\,\mho$$

$$\boxed{y_{11} = 0.23\,\mho}$$

By using Current division rule.

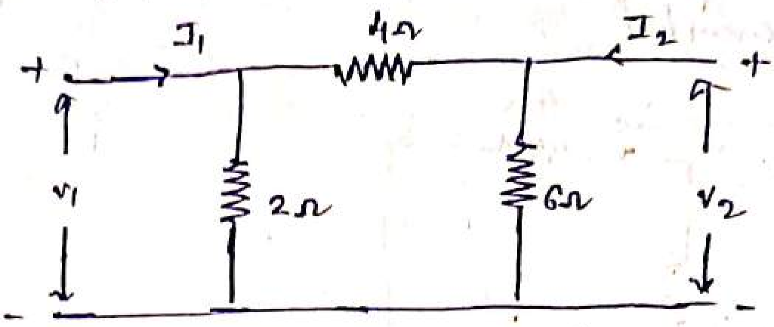
$$I_2 = I_1\left(\frac{5}{4+5}\right)$$

$$I_2 = \frac{-V_1}{R_{eq}}\left(\frac{5}{4+5}\right)$$

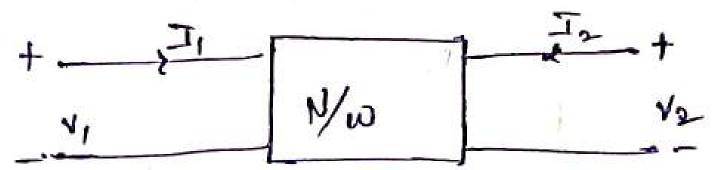
$$\therefore\ y_{21} = \frac{-I_2}{V_1} = \frac{-5}{9\times R_{eq}} = \frac{-5}{9\times4.22}$$

$$\boxed{y_{21} = -0.131\,\mho}$$

1 Find y parameters for the circuit shown below.



Hybrid parameters / h-parameters



Hybrid parameter representation is widely used in modeling of electronic components and circuits particularly transistors.

As both short circuit & open circuit condition terminals are utilised here hence this parameter representation is known as hybrid parameter representation in this form of representation the voltage of the input port and the current of the output port are expressed in terms of current of input port and voltage of output port.

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2 .$$

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0} \ \Omega \qquad h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0} \ \mho$$

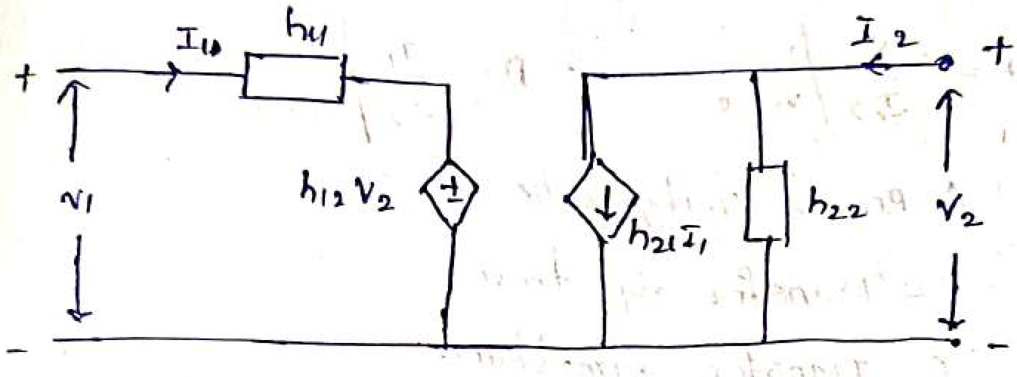
$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0} \ \Omega \qquad h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0} \ \mho$$

short

h_{11} = Input Impedance (Ω) (h_i) | h_{12} = Reverse Voltage gain (H_R)

h_{22} = Output admittive ($\mho$) (h_o) | h_{21} = Forward Current gain (h_f)

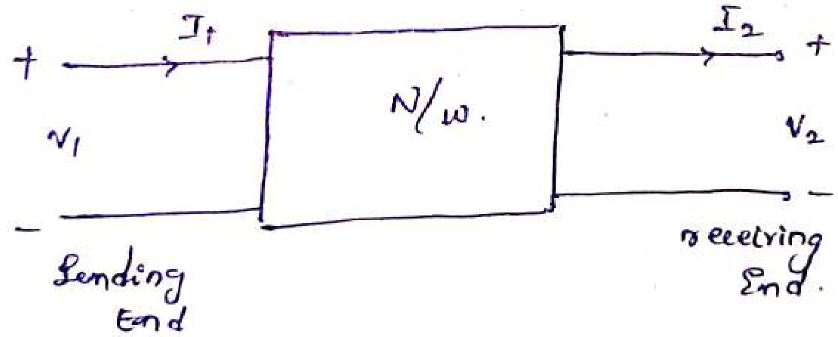
The Equivalent circuit model of h parameter representation is



ABCD parameters / Transmission parameters

These are widely used in power transmission Engenoring where they are termed as generalized circuit parameters.

It is Conventional to disignate the input port as Sending End and the output port as receiving End while representing ABCD parameters. More over the output Current direction is taken as reverse



In matrix form the representation is

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

Assume the output port is open circuited &

$\therefore A = \frac{V_1}{V_2}\Big|_{I_2=0}$; $C = \frac{I_1}{V_2}\Big|_{I_2=0}$.

Assume the output port is ~~open~~ short circuited

$\therefore B = \frac{-V_1}{I_2}\Big|_{V_2=0}$; $D = \frac{I_1}{I_2}\Big|_{V_2=0}$

A = Reverse Voltage ratio.

B = Transfer impedance

C = Transfer admittance.

D = Reverse Current ratio.

1. Photo diode
2. SCR
3. Solar Cell
4.